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Cooperative Game Theory for Explainable Artificial Intelligence (XAI) in Recommendation Systems: A Shapley Framework for Actionable Insight

*A thesis submitted in partial fulfillment of the requirements of the degree of
Doctor of Philosophy in Computer Science*

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Abstract

Machine learning models are increasingly used in settings where transparency, accountability, and trust are essential. Recommender systems and clustering pipelines, which shape user choices and support decision-making in domains such as digital commerce, media platforms, and environmental monitoring, are particularly affected by the opacity of their underlying algorithms. This thesis develops principled Explainable Artificial Intelligence (XAI) frameworks for clustering and recommender systems by integrating cooperative game theory with graph-based and hypergraph-based learning architectures.

The first contribution introduces a model-agnostic framework for explaining black-box clustering through a PCA–K-Means–LightGBM–SHAP pipeline. This approach produces interpretable cluster explanations and yields more attribution-consistent results than LIME-based alternatives, while motivating cluster selection on interpretability grounds rather than geometric separation alone.

The second contribution extends this framework to a multi-level clustering architecture handling more than 380,000 observations. A hierarchical attribution aggregation mechanism, supported by a formal consistency argument, reveals cross-level feature relationships and demonstrates generalisation across structurally distinct datasets, including the Beijing Multi-Site Air Quality dataset.

The third and principal contribution introduces DyHuCoG, a Dynamic Hypergraph Cooperative Game model that embeds preference-aware Monte Carlo Shapley estimation into hypergraph message passing. Evaluated on MovieLens-1M and Amazon-Book, DyHuCoG improves ranking accuracy, catalogue coverage, and intra-list diversity relative to strong graph and hypergraph baselines. Statistical significance is established on MovieLens-1M, while Amazon-Book results show strong descriptive gains.

Taken together, these contributions show that Shapley-value-based attribution provides a transferable explanatory mechanism spanning static clustering and dynamic recommendation. The thesis therefore advances cooperative-game explainability as a unified framework for interpretable unsupervised learning and recommendation.

Keywords: Explainable Artificial Intelligence, Cooperative Game Theory, Shapley Values, Recommender Systems, Hypergraph Neural Networks, Interpretable Clustering, Feature Attribution, Trustworthy AI

Résumé

Les modèles d'apprentissage automatique sont de plus en plus utilisés dans des contextes où la transparence, la responsabilité et la confiance sont essentielles. Les systèmes de recommandation et les pipelines de clustering, qui orientent les choix des utilisateurs et soutiennent la prise de décision dans des domaines tels que le commerce numérique, les plateformes médiatiques et la surveillance environnementale, sont particulièrement affectés par l'opacité de leurs algorithmes sous-jacents. Cette thèse développe des cadres méthodologiques d'intelligence artificielle explicable (XAI) pour le clustering et les systèmes de recommandation en intégrant la théorie des jeux coopératifs à des architectures d'apprentissage fondées sur des graphes et des hypergraphes.

La première contribution introduit un cadre agnostique au modèle pour expliquer le clustering boîte noire au moyen d'un pipeline PCA–K-Means–LightGBM–SHAP. Cette approche produit des explications de clusters interprétables et fournit des résultats d'attribution plus cohérents que les alternatives fondées sur LIME, tout en justifiant la sélection des clusters sur des critères d'interprétabilité plutôt que sur la seule séparation géométrique.

La deuxième contribution étend ce cadre à une architecture de clustering multi-niveaux traitant plus de 380,000 observations. Un mécanisme hiérarchique d'agrégation des attributions, étayé par un argument formel de cohérence, met en évidence des relations entre variables à différents niveaux et démontre une capacité de généralisation sur des jeux de données structurellement distincts, notamment le jeu de données Beijing Multi-Site Air Quality.

La troisième contribution, qui constitue la contribution principale, introduit DyHuCoG, un modèle coopératif dynamique sur hypergraphe qui intègre une estimation Monte Carlo des valeurs de Shapley sensible aux préférences dans le passage de messages sur hypergraphe. Évalué sur MovieLens-1M et Amazon-Book, DyHuCoG améliore conjointement la précision du classement, la couverture du catalogue et la diversité intra-liste par rapport à de solides approches de référence fondées sur des graphes et des hypergraphes. La significativité statistique est établie sur MovieLens-1M, tandis que les résultats obtenus sur Amazon-Book montrent de forts gains descriptifs.

Pris ensemble, ces apports montrent qu'une attribution fondée sur les valeurs de Shapley constitue un mécanisme explicatif transférable entre le clustering statique et la

recommandation dynamique. La thèse fait ainsi progresser l’explicabilité fondée sur la théorie des jeux coopératifs en tant que cadre unifié pour l’apprentissage non supervisé interprétable et la recommandation.

Mots-clés : Intelligence artificielle explicable, Théorie des jeux coopératifs, Valeurs de Shapley, Systèmes de recommandation, Réseaux de neurones sur hypergraphes, Clustering interprétable, Attribution de variables, IA digne de confiance

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List of Abbreviations

Abbreviation	Meaning
Artificial Intelligence and Learning	
AI	Artificial Intelligence
GNN	Graph Neural Network
HGNN	Hypergraph Neural Network
LightGBM	Light Gradient Boosting Machine
ML	Machine Learning
PCA	Principal Component Analysis
XAI	Explainable Artificial Intelligence
Cooperative Game Theory	
SHAP	SHapley Additive exPlanations
TreeSHAP	Tree-Specific SHAP Algorithm
TU	Transferable Utility Game
Recommender Systems	
CF	Collaborative Filtering
DyHuCoG	Dynamic Hypergraph Cooperative Game
IBCF	Item-Based Collaborative Filtering
NCF	Neural Collaborative Filtering
Top- N	Top- N Recommendation
UBCF	User-Based Collaborative Filtering
Graph-Based Models	
GAT	Graph Attention Network
GCN	Graph Convolutional Network
HCCF	Hypergraph Contrastive Collaborative Filtering
HPCF	Hypergraph Projection Enhanced Collaborative Filtering
KG	Knowledge Graph
LightGCN	Lightweight Graph Convolutional Network
RecDCL	Dual Contrastive Learning for Recommendation

Abbreviation	Meaning
Clustering and Dimensionality Reduction	
CH	Calinski–Harabasz Index
DB	Davies–Bouldin Index
K-Means	K-Means Clustering Algorithm
PC	Principal Component
SC	Silhouette Coefficient
Explainability Methods	
LIME	Local Interpretable Model-Agnostic Explanations
Matrix and Neural Baselines	
BPR	Bayesian Personalised Ranking
MF	Matrix Factorisation
Evaluation Metrics and Optimisation	
F1	Harmonic Mean of Precision and Recall
ILD	Intra-List Diversity
MSE	Mean Squared Error
NDCG	Normalised Discounted Cumulative Gain
PM2.5	Particulate Matter Below 2.5 Micrometres
PM10	Particulate Matter Below 10 Micrometres
RQ	Research Question

List of Notations

Symbol	Meaning
α, β, γ	Coalition utility weighting coefficients for ranking quality, diversity, and contextual relevance
δ	Fairness-penalty coefficient in an augmented coalition utility
ε	Residual term in Appendix A.2; in density-based clustering discussion, ε may also denote the DBSCAN neighbourhood radius, with context making the role explicit
$\hat{\phi}_j$	Monte Carlo estimate of the Shapley value for player j
Δ	Shapley refresh period in DyHuCoG training
λ	Generic regularisation or weighting hyperparameter
λ_{pref}	Weight assigned to preference consistency in the coalition utility
μ_1, μ_2	Weights of the diversity and context-alignment regularisers in the composite loss
ρ	Exponential smoothing coefficient for dynamic edge weights
$\sigma(\cdot)$	Non-linear activation function
Θ	Full parameter set of the recommendation model
$a(x)$	Mean intra-cluster distance for instance x in the Silhouette formulation
$b(x)$	Minimum mean distance from instance x to another cluster in the Silhouette formulation
$\mathcal{C}$	Set of contextual entities or contextual states
D	Degree matrix or embedding depth, depending on context
e_u, e_i	Learned embedding vectors of user u and item i
$\mathcal{F}$	Set of features in a clustering game
G	Interaction graph
H	Hypergraph
I	Set of candidate items
K	Cut-off threshold in top- K evaluation
k^*	Selected number of clusters
L	Composite loss function

Symbol	Meaning
M	Number of Monte Carlo samples used for Shapley approximation
N	Set of players in a cooperative game; in recommendation settings, top- N refers to the ranked-list cut-off and is interpreted from local context
p_j	Preference weight or normalised interaction frequency of player j
R	User-item interaction matrix
S	Coalition, with $S \subseteq N$
$v(S)$	Characteristic value of coalition S
$v_{\text{pref}}(S)$	Preference-weighted coalition utility
W	Weight matrix in neural message passing
X	Original data matrix
Y	PCA-transformed data matrix

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List of Publications

This thesis synthesises three peer-reviewed publications into a single monograph-style dissertation. In the main text, these publications are cited by their chapter placement rather than reproduced as stand-alone papers.

- [1] M. Louhichi, R. Nesmaoui, M. Mbarek, and M. Lazaar, “Shapley Values for Explaining the Black Box Nature of Machine Learning Model Clustering,” *Procedia Computer Science*, vol. 220, pp. 806–811, 2023, doi: 10.1016/j.procs.2023.03.107.
- [2] M. Louhichi, R. Nesmaoui, and M. Lazaar, “Game Theory Meets Explainable AI: An Enhanced Approach to Understanding Black Box Models Through Shapley Values,” *International Journal of Advanced Computer Science and Applications*, vol. 16, no. 7, pp. 716–725, 2025, doi: 10.14569/IJACSA.2025.0160780.
- [3] M. Louhichi, R. Nesmaoui, and M. Lazaar, “DyHuCoG: A Dynamic Hypergraph Cooperative Game for Preference-aware Recommendation,” *International Journal of Intelligent Engineering and Systems*, vol. 19, no. 2, 2026, doi: 10.22266/ijies2026.0228.54.

The three publications above are the primary source of truth for Chapters 5–7. Unless a passage is explicitly identified as thesis-level synthesis, proof, or cross-chapter comparison, the reported protocols, numerical results, and source figures in those chapters follow the published papers directly.

Chapter 1

Introduction

Opaque artificial intelligence systems increasingly mediate what people see, what risks they are exposed to, and which options become actionable in everyday digital environments, a concern now central to the trustworthy-AI and explainability literature [1]. When a recommendation pipeline steers a user toward a product, when a clustering workflow identifies an environmental risk regime, or when an analytical dashboard foregrounds one latent pattern over another, the question is no longer whether the model predicts well enough; it is whether its behaviour can be interrogated, justified, and revised responsibly.

This thesis investigates how cooperative game theory can be turned into a practical explanatory and optimisation perspective for contemporary machine learning systems. The empirical research arc is developed through three complementary research contributions presented in Chapters 5–7. Its central claim is that Shapley values matter not only because they yield mathematically principled post-hoc explanations, but also because they provide a common cooperative-game language through which clustering models, graph-based recommenders, and dynamic hypergraph architectures can be analysed in comparable attributional terms. The thesis therefore treats attribution not as an accessory attached after prediction, but as part of the modelling logic itself.

The thesis is motivated by a structural tension that has become increasingly visible across the machine-learning literature. As models gain expressive power, they frequently lose the degree of transparency needed for trustworthy deployment in settings where outputs shape human choices, organisational decisions, or regulatory action. Recommender systems illustrate this tension particularly clearly. They are embedded in digital commerce, entertainment platforms, health monitoring tools, and decision-support applications, yet the strongest models often expose little about why a specific item was recommended, why a cluster was formed, or which factors most strongly influenced the result [2]. The issue is therefore not simply technical opacity. It is the absence of explanations that can guide intervention.

Research Context and Motivation

The last decade has seen recommender systems evolve from relatively simple similarity-based filters into increasingly complex representation-learning systems that operate on sparse, high-dimensional, and dynamic interaction data. Matrix factorisation made it possible to model large user–item spaces with latent factors [3], while subsequent neural and graph-based recommenders widened the modelling space further. Neural collaborative filtering introduced richer non-linear interaction functions [4]. Graph neural recommendation then extended this shift through neighbourhood propagation on interaction graphs [5]. Hypergraph-based recommenders add another layer of sophistication by modelling higher-order relations among users, items, and contextual entities, thereby moving beyond pairwise interaction assumptions [6]. These developments improved ranking quality, but they also intensified the interpretability deficit.

That deficit matters for three reasons. First, opacity undermines user trust. If a recommendation changes consumption behaviour, or a cluster assignment informs intervention, the affected actor should be able to understand the main drivers of the output. Second, opacity constrains debugging and scientific learning. System designers who cannot trace influential features or interaction patterns are forced to optimise models as black boxes, which slows diagnosis of bias, instability, and distribution shift. Third, opacity increasingly collides with regulatory expectations. Emerging governance frameworks for high-impact AI systems emphasise transparency, traceability, and the capacity for meaningful human oversight, particularly where decisions shape access, risk, or exposure [7]. In such settings, explanation is not merely a usability enhancement; it is an accountability mechanism.

The motivation for this thesis emerges at the intersection of these pressures. We are interested in models that do not merely predict well, but also provide explanations that point toward action: a domain expert identifying which pollutants define a high-risk air-quality regime, a product designer understanding why a recommendation pipeline concentrates attention on a narrow item subset, or a system operator tracing whether a recommendation was driven by genuine preference structure or by popularity bias. This requirement for *actionable insight* echoes the stronger intervention-oriented view of explanation developed in the XAI literature [8]. In the present thesis, however, actionable insight is used primarily as a framing concept rather than as a separately measured endpoint: the empirical chapters illustrate it through domain-grounded explanatory cases rather than through an independent user study or intervention experiment.

Definition 1.1 (Actionable insight). *An explanation is actionable when it identifies at least one modifiable factor whose change is associated with a specifiable change in model output, and when that factor is accessible to the relevant decision-maker. In this thesis, accessibility means that the factor is expressed in the semantic vocabulary of the task*

domain, a physicochemical variable, a pollution indicator, or a preference signal, rather than in an opaque latent code.

Problem Statement

The thesis addresses two intertwined research problems. The first concerns unsupervised learning. Although Shapley-based explanation has become influential in supervised tasks, clustering remains comparatively under-explained. Existing clustering-interpretability approaches tend to privilege either local explanation or global characterisation, but not both, and they often struggle to scale beyond small tabular settings or to preserve coherence when explanations are generated at multiple resolutions [9]. Methods such as LIME provide local surrogates, yet their dependence on perturbation design makes their attributions unstable and difficult to reconcile across clusters [10]. Consequently, practitioners may obtain an explanation of a cluster assignment without gaining a reliable account of how the cluster structure itself emerged.

The second problem concerns graph-based recommendation. State-of-the-art recommenders routinely optimise ranking quality, but often treat diversity, contextual relevance, and interpretability as secondary or external concerns. Hypergraph collaborative filtering improves the modelling of higher-order relations, yet prevailing models still assume static or uniform message importance across interaction structures. This assumption is problematic because not every user–item–context coalition contributes equally to recommendation quality. Some interactions encode strong preference evidence, while others are noisy, outdated, or popularity-driven. If a model cannot distinguish among these interaction types in a principled way, then both performance and explanation suffer.

Taken together, these problems reveal a deeper gap: there is still no sufficiently coherent cooperative-attribution perspective that explains unsupervised structure, scales to realistic datasets, and then carries the same attribution logic into dynamic recommendation models. In other words, the literature lacks a coherent research programme in which cooperative attribution is not tied to a single algorithmic niche, but also serves as the invisible spine of a broader explainability agenda. This thesis develops such a programme by integrating three complementary studies into one methodological trajectory. In one sentence, the thesis gap is this: the literature still lacks a single cooperative-attribution framework that can explain clustering faithfully, remain coherent under hierarchy, and then operate as an in-training signal in recommendation rather than remaining a purely post-hoc device.

Research Aim and Objectives

The overall aim of this thesis is to develop, justify, and evaluate a cooperative-game-theoretic perspective for explainable artificial intelligence in clustering and recommender systems, using Shapley-value attribution as both an explanatory mechanism and, where appropriate, an in-training optimisation signal.

That overall aim is pursued through five tightly connected objectives. The first objective is to develop a partially model-agnostic clustering explanation pipeline, model-agnostic at the clustering stage but tree-specific at the explanation stage, capable of yielding both instance-level and cluster-level feature attribution. The second is to extend that framework to large-scale and hierarchical clustering settings without losing attribution coherence across levels of analysis. The third is to formulate recommendation as a cooperative game over users, items, and contextual entities so that interaction importance is quantified rather than merely assumed. The fourth is to embed preference-aware Shapley estimation into hypergraph message passing in order to improve ranking quality, catalogue coverage, and intra-list diversity jointly. The fifth is to synthesise the clustering and recommendation studies into a shared attribution perspective on actionable insight for transparent and accountable AI analysis. These objectives are operationalised in the empirical chapters that anchor the thesis contributions.

Research Questions

The thesis is organised around five research questions that define its argumentative spine and provide the traceability framework used by the contribution chapters and by the synthetic discussion in chapter 8.

RQ1: How can cooperative-game-theoretic Shapley values be adapted to explain black-box clustering models so that explanation remains faithful at the instance level and coherent at the cluster level? This question motivates the first contribution by focusing on the foundational explanatory bridge between unsupervised partitioning and semantically meaningful feature attribution. It is answered primarily in chapter 5.

RQ2: How can Shapley-based clustering explanation be extended from single-level analysis to large-scale and hierarchical settings without losing computational tractability or attribution consistency? This question matters because explanation that collapses under scale is of limited practical value. It is addressed primarily in chapter 6.

RQ3: Can cooperative attribution move beyond post-hoc analysis and be incorporated directly into the learning dynamics of graph-based recommender systems? This question marks the thesis’s transition from explanation of model behaviour to explanation-informed optimisation. It is answered primarily in chapter 7.

RQ4: How can a recommendation model jointly optimise ranking accuracy, con-

textual relevance, and diversity when interaction importance is estimated through a cooperative-game utility rather than assumed to be uniform? This question probes whether the accuracy–diversity trade-off can be managed through principled contribution weighting rather than heuristic balancing alone. It is answered primarily in chapter 7 and synthesised in chapter 8.

RQ5: What theoretical and practical insights emerge when clustering explanation and recommendation learning are interpreted as two stages of a shared cooperative-game perspective on explainability? This final question is cumulative rather than isolated. It is answered at thesis level in chapter 8, where the three studies are read as one research arc rather than as separate papers.

Scope and Delimitations

The thesis is deliberately bounded in four ways. First, the clustering studies focus on tabular real-world datasets rather than image, text, or audio modalities. This delimitation is principled: it allows the thesis to examine cooperative attribution in settings where semantic interpretability of features can be evaluated directly. Second, the recommendation studies emphasise top- N item recommendation under implicit or converted-implicit feedback, rather than sequence generation, reinforcement learning, or large-language-model recommenders. Third, the work focuses on Shapley-value-based attribution rather than a broad comparison of all possible explanation families. This narrower focus is methodologically necessary because the thesis aims to examine the internal development of one mathematically grounded explanatory mechanism across multiple model classes. Fourth, although fairness and regulation are discussed as implications, the empirical studies do not constitute a dedicated fairness audit or legal-compliance evaluation. That bounded scope follows the empirical design of the three empirical contributions and the broader methodological emphasis of current Shapley-based XAI research.

These delimitations do not weaken the thesis; they sharpen its claims. The contribution is not that Shapley values solve all interpretability problems everywhere. It is that they provide a uniquely coherent framework for linking explanation, optimisation, and intervention across specific but important classes of machine-learning systems, which is also the measured position taken in the wider Shapley literature [11].

Methodological Approach

Methodologically, the thesis adopts a multi-study design that progresses from explanation of static unsupervised structure to learning with dynamic cooperative attribution. The first study develops a clustering explanation pipeline based on PCA, K-Means,

LightGBM, and TreeSHAP. The second study extends the same logic to a multi-level architecture for large-scale clustering, introducing hierarchical attribution aggregation and consistency analysis. The third study develops DyHuCoG, a hypergraph recommender in which preference-aware Monte Carlo Shapley estimates are refreshed during training and injected into message passing. Across all studies, evaluation combines quantitative performance metrics with interpretability-oriented analysis, ablation, and cross-study synthesis. The full protocol is detailed in chapter 4, while the study designs remain aligned across the empirical chapters.

Research Contributions

The thesis makes three original contributions, stated here in the same form in which they are revisited in chapter 9.

C1. The thesis develops a partially model-agnostic Shapley framework for explainable black-box clustering, operationalised through a PCA–K-Means–LightGBM–TreeSHAP pipeline and validated on the Portuguese Wine Quality dataset. The clustering stage remains flexible, whereas the explanation stage is tree-specific because it relies on a LightGBM surrogate together with TreeSHAP.

C2. The thesis extends that framework to large-scale, multi-level clustering by introducing hierarchical attribution aggregation, cross-level consistency analysis, and scalable explanatory reporting on the Beijing Multi-Site Air Quality dataset. This contribution deepens the methodological argument by showing that Shapley-based clustering explanation can remain coherent under scale and hierarchy.

C3. The thesis introduces DyHuCoG, a dynamic hypergraph recommender in which preference-aware Monte Carlo Shapley estimation is injected directly into message passing and training, thereby improving ranking quality, coverage, and intra-list diversity simultaneously on MovieLens-1M and Amazon-Book.

Taken together, these three contributions support a single thesis-level claim: cooperative game theory can function as a shared methodological perspective for actionable explanation across clustering and recommendation problems. The thesis contribution is therefore cumulative rather than additive; it is not a binder of three papers, but a progressively developed argument about what cooperative attribution can do.

Thesis Organisation

Chapter 2 establishes the conceptual foundation of the thesis by reviewing recommender systems, graph learning, explainable AI, cooperative game theory, and the clustering concepts needed later in the document. Rather than serving as a generic textbook

chapter, it introduces only those concepts that become analytically necessary in the contribution chapters.

Chapter 3 develops a critical, theme-based literature review that positions the thesis relative to interpretable clustering, explainable recommendation, Shapley-based attribution, and adaptive recommender systems. It is in this chapter that the thesis gap is articulated most explicitly, and where the three contribution chapters are positioned as answers to that gap.

Chapter 4 defines the common experimental protocol used throughout the thesis. It describes the datasets, splitting strategies, baselines, evaluation metrics, optimisation procedures, statistical testing, and implementation details that make the empirical claims comparable across studies.

Chapter 5 presents the first contribution: a Shapley-based framework for explaining black-box clustering via a PCA–K-Means–LightGBM–SHAP pipeline. It establishes the feasibility of using cooperative attribution for unsupervised explanation and answers **RQ1** directly.

Chapter 6 extends the clustering framework to multi-level and large-scale settings. It introduces hierarchical attribution aggregation, analyses cross-level consistency, and demonstrates that scalable, multi-granular explanation is possible without collapsing interpretability into a single flat summary. This chapter addresses **RQ2** and part of **RQ5**.

Chapter 7 presents DyHuCoG, the thesis’s flagship contribution. It embeds preference-aware Monte Carlo Shapley estimation into hypergraph message passing and evaluates the resulting model on MovieLens-1M and Amazon-Book. The chapter answers **RQ3** and **RQ4**, while also creating the main empirical basis for the thesis-level synthesis.

Chapter 8 synthesises the three studies, answers all five research questions explicitly, discusses theoretical and practical implications, consolidates limitations, restates the thesis’s original contributions, and closes with future research directions together with a broader reflection on the significance of cooperative-game explainability for transparent and accountable AI analysis.

Summary

This introductory chapter established the thesis problem, scope, research questions, and claimed contributions. It argued that clustering explanation and recommender-system optimisation can be read as related importance-allocation problems and positioned Shapley-value attribution as the common formal language used to investigate that claim. The next chapter builds the theoretical foundation required for that programme by reviewing recommender systems, graph learning, explainable AI, cooperative game theory, and the clustering concepts on which the contribution chapters depend.

Chapter 2

Background and Theoretical Foundations

Introduction

Building on the problem framing established in the Introduction, this chapter provides the theoretical foundation required for the thesis’s three empirical contributions. The sequence is deliberate. We begin with classical recommendation because the limitations of those models motivate the later move to graph and hypergraph structures. We then turn to explainable AI and cooperative game theory because the thesis argues that predictive strength without intelligibility is inadequate in the domains under consideration. The chapter closes with clustering and dimensionality-reduction concepts, which are necessary for the first two contribution chapters.

The chapter is not intended as an encyclopaedic survey. Every concept introduced here is used analytically in a later chapter. In that sense, the chapter functions as a scaffold for the thesis argument rather than as detached background material, which is also how the later empirical chapters mobilise background material rather than treating it as generic textbook review. Readers already familiar with core recommender-system concepts may skim sections [2.2](#) and [2.3](#), while those interested primarily in the thesis’s explanatory logic should attend especially to sections [2.7](#) and [2.8](#).

2.1 Recommender Systems: Overview and Motivation

This section defines recommender systems as the thesis’s principal application domain and explains why recommendation remains an analytically rich setting for questions of explanation, trust, and intervention.

Recommender systems are information-filtering mechanisms that estimate which items are likely to be relevant, useful, or desirable for a particular user under a particular decision context. Knowledge-graph surveys reinforce the same formulation while emphasising richer relational context [12]. Their practical importance stems from the same condition that makes them technically difficult: digital platforms expose users to an item space far larger than any individual can evaluate directly. Recommendation therefore acts as a ranking, exposure, and relevance-allocation mechanism rather than as a neutral search convenience. It determines which books are surfaced, which films are foregrounded, which products are repeatedly shown, and which options become behaviourally available.

Formally, let U denote the set of users, I the set of items, and $f(u, i)$ a scoring function that maps user-item pairs to relevance scores. The top- N recommendation task seeks a ranked list $\hat{L}_u \subset I$ that maximises utility for user u , where utility may refer to immediate click likelihood, long-term engagement, satisfaction, novelty, or some weighted combination thereof. The apparent simplicity of this formulation conceals several layers of modelling difficulty: the interaction matrix is sparse, preferences drift over time, and item popularity is highly skewed. The relevance of an item may also depend on novelty, exposure diversity, and fairness constraints rather than on user and item alone [13]. Recent work on explainability and fairness makes the same point more explicitly by showing that exposure allocation carries distributional consequences [14].

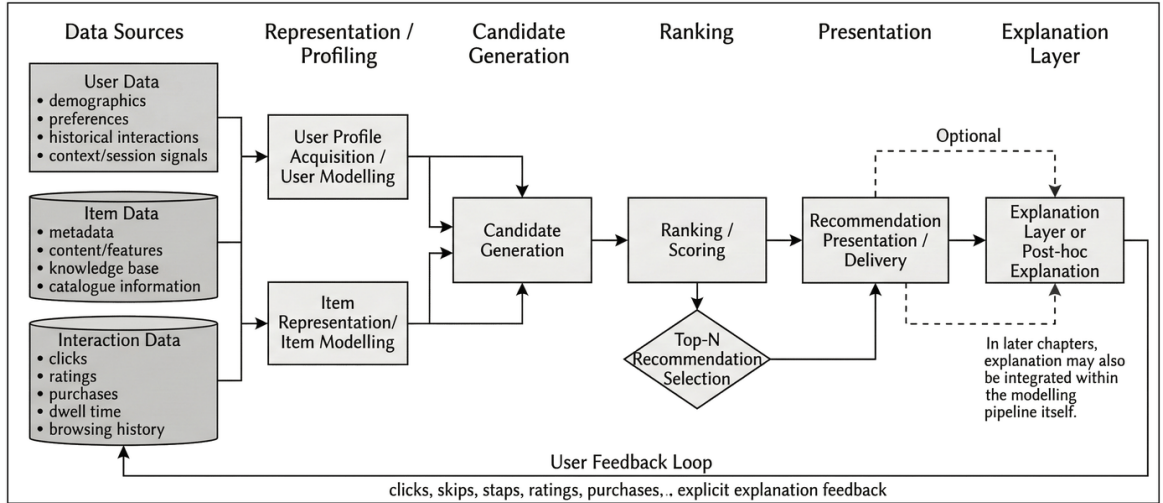


Figure 2.1: Conceptual architecture of a recommender system

Recommendation is therefore not just a predictive problem. It is also a problem of representation, exposure, and responsibility. A system that ranks accurately but collapses diversity, reinforces popularity bias, or cannot explain its own choices may still be technically impressive while remaining operationally deficient. This is the central reason the recommender-systems domain provides an especially productive setting for the thesis’s cooperative-game perspective.

Recent AI ethics frameworks already frame recommendation as a governance problem rather than a purely technical one [15]. Legal and policy work then connects opaque ranking systems to discrimination, bias auditing, and formal risk management [16]. International governance principles make transparency and accountability explicit design expectations, and emerging regulation turns some of those expectations into compliance obligations [17]. This broader legal background also interacts with data-protection arguments around automated decision-making [18].

2.2 Traditional Recommendation Techniques

This section reviews classical recommendation families and clarifies why they remain conceptually important even when modern graph-based models outperform them empirically.

2.2.1 Collaborative Filtering

Collaborative filtering rests on a relational assumption: users who behaved similarly in the past will tend to value similar items in the future. In user-based collaborative filtering, predictions are derived from neighbouring users, typically using cosine or Pearson similarity over observed interaction histories. Item-based collaborative filtering inverts the logic, recommending items similar to those the user has previously consumed. If $R \in \mathbb{R}^{|U| \times |I|}$ denotes the interaction matrix, a standard user-based estimate can be written as:

$$\hat{r}_{ui}^{\text{UBCF}} = \bar{r}_u + \frac{\sum_{v \in N_u(i)} s(u, v) (r_{vi} - \bar{r}_v)}{\sum_{v \in N_u(i)} |s(u, v)|}, \quad (2.1)$$

where $N_u(i)$ denotes the neighbouring users who rated item i , $s(u, v)$ is a similarity score, and $\bar{r}_u$ is the mean rating of user u . The corresponding item-based variant is:

$$\hat{r}_{ui}^{\text{IBCF}} = \frac{\sum_{j \in N_i(u)} s(i, j) r_{uj}}{\sum_{j \in N_i(u)} |s(i, j)|}, \quad (2.2)$$

with $N_i(u)$ denoting items similar to i that user u has already consumed. The key strength of collaborative filtering is that it does not require explicit content features. Its weakness is equally clear: if the interaction matrix is sparse or unstable, similarity estimates become unreliable.

From the perspective of this thesis, collaborative filtering is important for two reasons. First, it establishes the relational ontology that later graph-based methods refine rather than abandon. Second, it reveals an early version of the interpretability problem. Neighbourhood methods can often say *which* users or items drove a recommendation, but they rarely offer a principled account of how multiple relational signals combine, or why one latent influence should count as more important than another. Cooperative-game

attribution can therefore be understood as a mathematically disciplined continuation of a question classical collaborative filtering already exposed.

2.2.2 Content-Based Filtering

Content-based filtering predicts preference from item attributes and user profiles [19]. If a user tends to prefer items with particular textual, visual, or categorical characteristics, the model recommends items sharing those characteristics. Content-based systems are attractive when rich metadata are available or when the user history is too sparse for collaborative inference alone. They also provide a more direct route to explanation because the model can, at least in principle, point to descriptive features rather than latent interaction patterns.

However, content-based systems inherit their own limitations. They are strongly dependent on the availability and quality of side information, they frequently overspecialise, and they may fail to recommend genuinely novel items outside the descriptive manifold of the user’s history. In practice, therefore, content-based methods are often most useful as components within larger hybrid systems rather than as complete solutions.

2.2.3 Hybrid Approaches

Hybrid recommendation combines collaborative and content-based signals in order to mitigate the weaknesses of each component [20]. The literature includes weighted hybrids, switching models, cascaded architectures, and feature-augmentation strategies. The conceptual value of hybridisation is clear: collaborative signals capture collective behavioural structure, while content features stabilise recommendations for sparse or cold-start entities.

The limitation, however, is that hybridisation alone does not resolve the explanatory problem. Combining multiple signal sources can improve predictive robustness while simultaneously making causal interpretation harder. A recommendation may depend partly on user similarity, partly on item content, and partly on downstream ranking heuristics. Without an attribution mechanism that quantifies marginal contribution, the hybrid model gains flexibility at the cost of analytical transparency.

2.2.4 Matrix Factorisation

Matrix factorisation was a decisive step in recommendation research because it replaced direct similarity lookup with latent-factor learning. By decomposing the interaction matrix into user and item embeddings, the model estimates hidden dimensions of taste and item structure that are not directly observed but are inferred from data. In its

canonical form:

$$R \approx PQ^\top, \quad (2.3)$$

where $P \in \mathbb{R}^{|U| \times d}$ and $Q \in \mathbb{R}^{|I| \times d}$ are latent user and item factors. The predicted relevance is then:

$$\hat{r}_{ui} = p_u^\top q_i. \quad (2.4)$$

This formulation improves generalisation under sparsity and provides a compact representation space in which ranking can be computed efficiently.

Despite its enduring importance, matrix factorisation makes interpretability more difficult rather than less. Latent factors are powerful precisely because they do not correspond straightforwardly to human-readable variables. As a result, matrix factorisation sharpened the trade-off that motivates the present thesis: representational power increased, but the semantic accessibility of recommendations declined. Table 2.1 summarises how the major recommendation paradigms differ in signal source, strength, and interpretability.

Table 2.1: Comparison of recommendation paradigms

Paradigm	Core signal	Main strength	Main weakness	Interpretability implication
Collaborative filtering	User–item interaction overlap	Strong behavioural relevance	Sparse and cold-start sensitive	Explanations depend on neighbour traceability
Content-based filtering	Item attributes and profiles	Strong item-level semantics	overfitting	Explanations inherit feature semantics directly
Hybrid recommendation	Multiple combined signals	Robustness across regimes	System complexity	Explanation must reconcile heterogeneous evidence
Matrix factorisation	Latent preference factors	Compact and efficient	Latent dimensions are opaque	Requires additional interpretive bridge

2.3 Limitations of Classical Recommender Systems

This section identifies the limitations that made graph-based and explainable recommendation necessary, not fashionable.

2.3.1 Data Sparsity and Scalability

In realistic recommendation settings, the user–item interaction matrix is overwhelmingly empty. Even benchmark datasets that appear substantial in absolute terms are extremely sparse when measured relative to the full combinatorial user–item space. Sparse matrices undermine neighbour estimation, reduce the reliability of latent-factor updates, and amplify sensitivity to random exposure effects. Scalability compounds the problem because the computational burden of storing, updating, and searching large interaction spaces grows with platform size.

2.3.2 Cold-Start Problem

Cold-start refers to the difficulty of recommending effectively for new users, new items, or both. Classical collaborative methods depend on prior interaction evidence, so entities with few observations are structurally disadvantaged. Content-based and hybrid methods partially alleviate the problem, but only when auxiliary data are informative and well aligned with user behaviour. The persistence of cold start is one reason later chapters attend carefully to context, higher-order structure, and coalition-based utility.

2.3.3 Popularity Bias and Lack of Diversity

Classical recommenders frequently overweight popular items because popular items generate more observations, stronger similarity signals, and lower perceived ranking risk. The result is a feedback loop in which exposure begets interaction and interaction begets further exposure [21]. The system can look accurate on conventional top- K metrics while narrowing the user’s effective choice set. This problem is central to chapter 7, where diversity is treated as a first-class objective rather than a post-processing preference.

2.3.4 Absence of Interpretability

The most fundamental limitation, for the purposes of this thesis, is that classical recommenders provide weak explanatory traction. Even when they can identify neighbouring users or factor similarities, they do not provide a normative account of why one signal should count more than another, nor do they offer a general method for distributing credit fairly among multiple interacting features or entities. Shapley-based attribution

becomes relevant precisely here: it addresses the credit-assignment problem in a way that is mathematically explicit rather than heuristically asserted.

2.4 Graph-Based Recommender Systems

This section explains why graph representations are a qualitative advance over matrix views when interactions are relational, heterogeneous, and context dependent.

2.4.1 Graph Representation of User–Item Interactions

Graph-based recommendation represents users and items as nodes and observed interactions as edges. This shift is conceptually important because it treats recommendation as structure learning over relations rather than as algebraic completion of a matrix. In the simplest case the interaction structure is represented as a bipartite graph:

$$G = (V, E), \quad V = U \cup I, \quad E \subseteq U \times I, \quad (2.5)$$

where each edge $(u, i) \in E$ records an observed interaction. Graphs make it natural to encode neighbourhood effects, higher-order dependencies, and heterogeneous entity types. They also expose a more faithful substrate for explanation, since influence can be traced across connection patterns rather than inferred indirectly from factor proximity. Figure 2.2 makes that representational shift concrete.

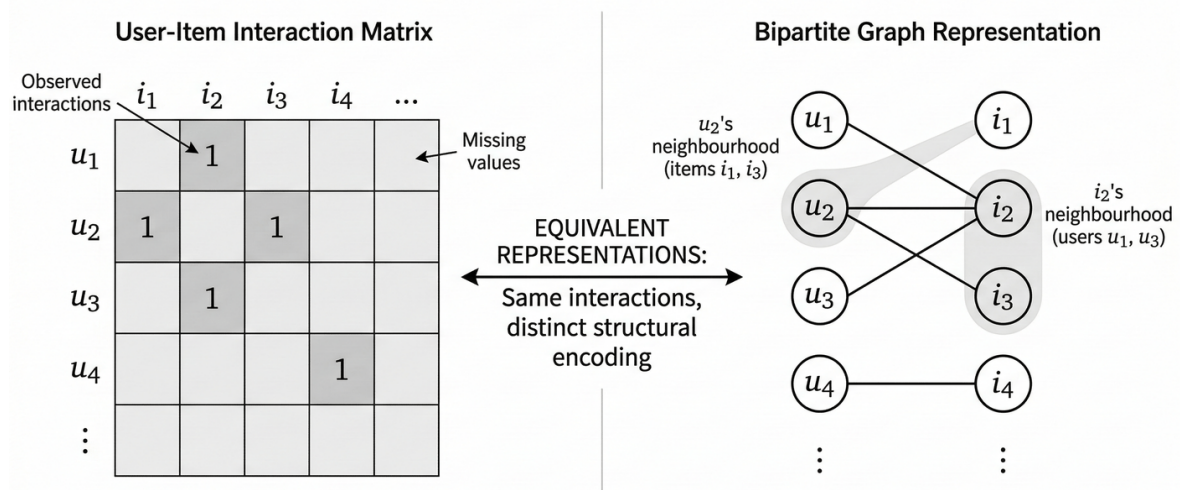


Figure 2.2: Matrix and graph views of interaction data

2.4.2 Bipartite Graph Collaborative Filtering

In the simplest graph formulation, recommendation is modelled on a bipartite graph connecting users and items. Message passing or propagation algorithms aggregate neighbourhood information to refine user and item representations iteratively. Compared with classical collaborative filtering, the graph view captures multi-hop relational structure more explicitly. It is therefore unsurprising that graph-based methods often outperform matrix methods on sparse implicit-feedback tasks. Figure 2.3 shows the neighbourhood-expansion logic that later underpins graph propagation.

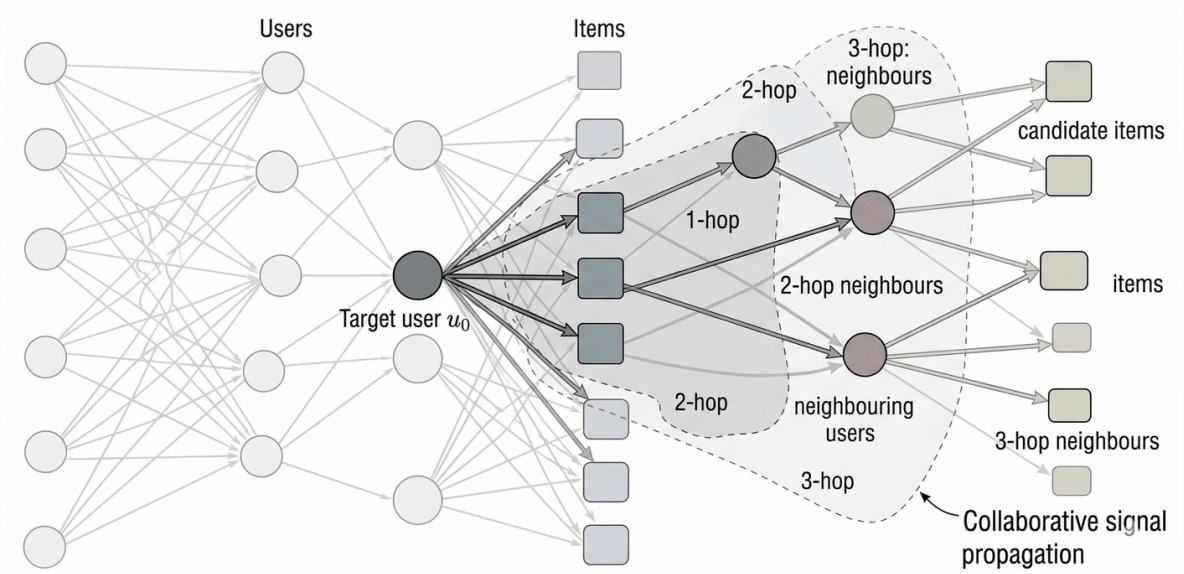


Figure 2.3: Neighbourhood hops in a bipartite recommendation graph

2.4.3 Hypergraph Extensions for Higher-Order Relations

The jump from pairwise graphs to hypergraphs is illustrated in Figure 2.4.

Standard graphs still privilege pairwise relations. Hypergraphs generalise this structure by allowing a single hyperedge to connect more than two entities at once, which is particularly attractive when recommendation depends on user, item, and context simultaneously. A hypergraph is written as:

$$\mathcal{H} = (V, \mathcal{E}), \quad \mathcal{E} \subseteq 2^V, \quad (2.6)$$

where each hyperedge $e \in \mathcal{E}$ may connect an arbitrary subset of entities. Hypergraph collaborative filtering therefore offers a principled way to encode session, genre, temporal, or contextual co-occurrence that would be fragmented under an ordinary graph representation. This representational move is foundational for DyHuCoG, where the unit of importance is not the isolated node but the coalition embedded in a hyperedge.

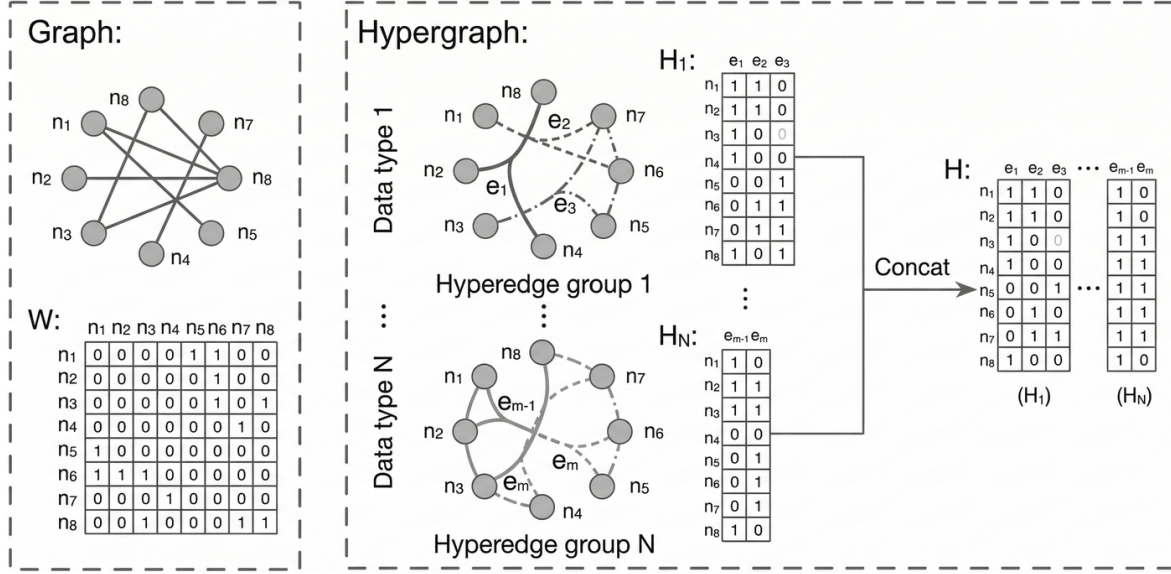


Figure 2.4: Graph versus hypergraph representation

2.4.4 Knowledge Graph-Augmented Recommendation

Knowledge-graph recommendation extends graph representation further by linking users and items to structured semantic entities, relations, or ontological concepts. The benefit is explanatory richness and improved cold-start performance, but the cost is additional modelling complexity and dependence on external knowledge resources. The thesis does not implement a knowledge-graph system directly, but the literature is relevant because it shows that richer relational structure often improves both performance and explanation, but only when there is a principled way to distribute importance across multiple relation types.

2.5 Graph Neural Networks

This section introduces graph neural networks as the dominant machinery for learning from relational structure in recommendation and related tasks.

2.5.1 The Message-Passing Framework

Graph neural networks learn node representations by aggregating information from neighbouring nodes in repeated propagation steps, extending core representation-learning ideas from deep learning into relational settings [22]. Contemporary attention-based architectures further influenced how messages can be weighted and aggregated in graph models [23]. Figure 2.5 illustrates the generic message-passing template used throughout graph learning. Figure 2.6, discussed later in section 2.5.3, then shows the simplified

propagation design that is most relevant for recommendation. In its generic form, message passing alternates between message construction, neighbourhood aggregation, and state update:

$$h_v^{(\ell+1)} = \phi^{(\ell)}\left(h_v^{(\ell)}, \text{AGG}_{u \in \mathcal{N}(v)} \psi^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, e_{uv})\right), \quad (2.7)$$

where $\psi^{(\ell)}$ is the message function and $\phi^{(\ell)}$ is the update rule. The result is a representation that encodes both the node's own features and the structure of its local environment. This mechanism is especially appealing for recommendation because preference often emerges from relational context rather than from isolated attributes alone. Comprehensive surveys confirm that message passing remains the common denominator of modern GNN design [24]. Dedicated review work reaches the same conclusion from a model-taxonomy perspective [25]. Benchmarking studies further show that dataset choice and evaluation protocol strongly shape apparent architectural gains [26].

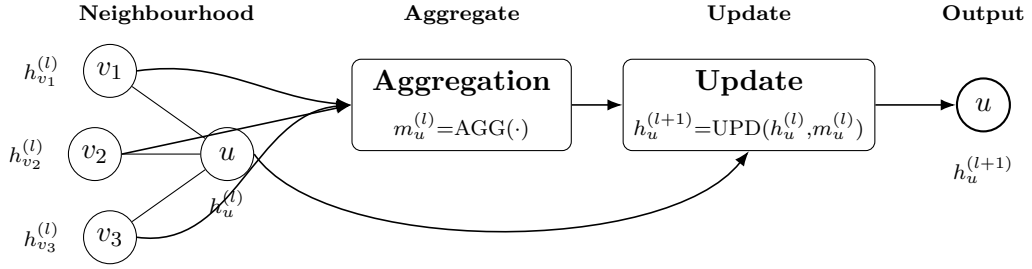


Figure 2.5: Generic message-passing framework

2.5.2 Graph Convolutional Networks (GCN)

Graph Convolutional Networks formalised the idea that neighbourhood aggregation can be interpreted as a graph analogue of convolution in Euclidean data. Their power lies in smoothing, information diffusion, and representation enrichment. Their weakness, for recommendation, is that feature transformation and repeated non-linearities are not always necessary when the graph itself already carries most of the signal. This insight informs the development of simplified architectures such as LightGCN. Subsequent theoretical work sharpened the same point by analysing oversmoothing directly [27]. Stabilisation schemes such as DropEdge then offered one practical response [28]. Simple and deep residual GCN variants pushed the architectural side of the same argument [29].

2.5.3 LightGCN and Simplified Propagation

LightGCN removes non-essential transformations and retains only the propagation mechanism most relevant for collaborative recommendation. For a user node u the propagation step is:

$$e_u^{(\ell+1)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)| |\mathcal{N}(i)|}} e_i^{(\ell)}, \quad (2.8)$$

and the final representation is obtained by layer fusion:

$$e_u = \sum_{\ell=0}^L \alpha_\ell e_u^{(\ell)}, \quad (2.9)$$

with an analogous update for items. The model is important for this thesis because it shows that, in recommender settings, structural propagation may matter more than generic neural complexity. DyHuCoG inherits this lesson: the goal is not to make hypergraph learning maximally deep, but to make propagation selectively sensitive to coalition importance.

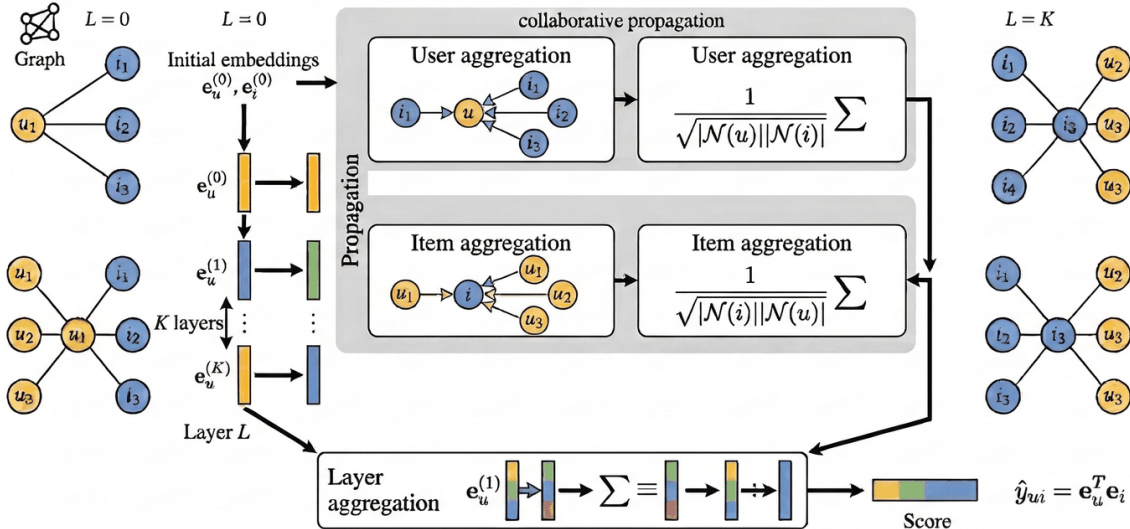


Figure 2.6: LightGCN propagation mechanism

2.5.4 Hypergraph Neural Networks

Hypergraph neural networks extend graph propagation to higher-order relations. Instead of aggregating only along pairwise edges, they propagate information through hyperedges that encode multiway interactions. Using an incidence matrix H , node-degree matrix D_v , and hyperedge-degree matrix D_e , a standard hypergraph propagation operator can be written as:

$$E^{(\ell+1)} = \sigma \left(D_v^{-1/2} H W D_e^{-1} H^\top D_v^{-1/2} E^{(\ell)} \right), \quad (2.10)$$

where W contains hyperedge weights. This makes hypergraph models particularly apt for recommendation settings in which user preferences are shaped by context, topic, or co-consumption structure that cannot be reduced cleanly to dyads [30]. Hypergraph learning is therefore not merely a representational luxury in this thesis; it is the architectural condition that makes coalition-based attribution meaningful. Later variants also model signed relations [31]. Other work introduces sequential hyperedges for recommendation [32]. Direct alignment objectives provide another branch of the literature [33]. Multi-channel higher-order reasoning further illustrates how hypergraph recommendation has diversified beyond a single incidence design [34].

2.5.5 Attention Mechanisms in GNNs

Table 2.2: GNN architectures relevant to recommendation

Model	Aggregation principle	Distinctive feature	Recommendation relevance	Limitation
GCN	Normalised neighbourhood averaging	Spectral smoothing	Foundational graph representation learning	Can over-smooth under depth
GAT	Learned attention weights	Edge-specific weighting	Captures heterogeneous neighbour importance	Higher computational cost
LightGCN	Linear propagation without transforms	Minimalist CF design	Strong collaborative-filtering baseline	Uniform edge treatment
HGNN	Hyperedge incidence propagation	Higher-order relation modelling	Captures multi-way user-item structure	More complex construction and tuning

Attention mechanisms allow graph models to weight incoming messages unequally. In recommendation, this is attractive because not all neighbours provide equally informative evidence. However, attention weights alone do not solve the interpretability problem. They indicate relative weighting internal to the model, but they do not generally satisfy fairness, completeness, or symmetry properties. One of the thesis’s central arguments is that Shapley-weighted propagation offers a complementary and more principled account of importance than attention alone. Empirical analyses of graph attention also caution that learned coefficients are not automatically reliable explanations of model behaviour

[35]. Table 2.2 summarises the architectural differences that matter most for the later comparison chapters.

2.6 Graph Neural Networks for Recommendation

This section connects generic graph learning to the particular modelling demands of recommendation.

2.6.1 Embedding Propagation on Interaction Graphs

The main attraction of GNN recommendation lies in the ability to propagate behavioural evidence through interaction graphs so that a user’s representation reflects not only direct history but also relational structure beyond immediate observations. This is especially useful in sparse settings, where multi-hop signals can compensate for missing direct evidence. The downside is that propagation can also amplify noise, outdated preferences, and popularity bias if message importance is not controlled.

2.6.2 Contrastive Self-Supervised Learning for Recommendation

Contrastive recommendation methods generate augmented graph views and encourage agreement between compatible representations. Such methods have proved effective for sparse feedback because they regularise embeddings and make better use of structure beyond explicit labels [36]. Their relevance to this thesis is twofold. They demonstrate the field’s interest in auxiliary learning signals, and they provide strong baselines against which the cooperative-game perspective must be assessed. Within recommendation specifically, this family now includes mixup-based hard-negative generation [37]. Self-supervised graph learning provides a closely related baseline family [38]. Simplified contrastive regularisation appears in SimGCL [39]. Light graph contrastive learning is represented by LightGCL [40]. Dual-graph consistency objectives continue that line [41]. Layer-to-layer consistency offers another recent variant [42]. Disentangled propagation objectives broaden the same contrastive design space [43]. Its broader methodological roots lie in graph-contrastive work on augmentation design [44]. Graph augmentation design has also been studied explicitly as a problem in its own right [45]. Multi-view graph contrast provides another root of the paradigm [46]. Comparative surveys now synthesise this wider landscape [47].

2.6.3 Context-Aware Graph Recommendation

Context-aware recommendation attempts to incorporate temporal state, session dynamics, location, genre, or other situational features into ranking. The literature shows that context can improve relevance substantially, but often at the price of additional complexity and lower transparency [48]. DyHuCoG addresses this tension by incorporating context directly into the coalition definition, allowing contextual importance to be estimated rather than appended heuristically.

2.7 Explainable Artificial Intelligence (XAI)

This section situates the thesis within the broader field of explainable AI and clarifies what kind of explanation the thesis values.

2.7.1 Taxonomy of Interpretability Methods

The overall XAI landscape used to position the thesis is summarised in Figure 2.7.

Interpretability methods are often divided into intrinsic versus post-hoc, local versus global, and model-specific versus model-agnostic approaches. These distinctions are useful but incomplete. For the purposes of this thesis, an explanation must also be evaluated in terms of whether it can support intervention. A locally faithful explanation that cannot be aggregated, compared, or operationalised is analytically limited, even if it is visually persuasive.

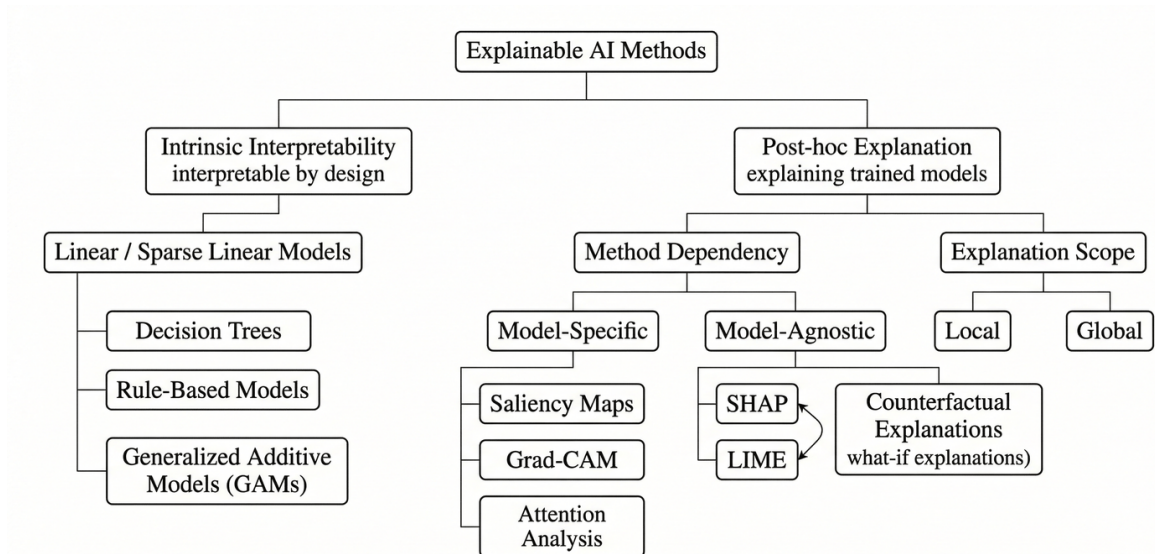


Figure 2.7: XAI taxonomy relevant to the thesis

2.7.2 Post-Hoc Interpretability Methods

Post-hoc methods attempt to explain models after training. LIME approximates local behaviour with an interpretable surrogate; SHAP computes additive feature contributions grounded in Shapley values [49]; other approaches rely on gradients, perturbations, prototypes, counterfactuals, or concept activation. Post-hoc explanation is attractive because it can be applied to strong predictive models without retraining them. Its limitation is that it may remain external to the model’s learning process and therefore insufficient for improving behaviour rather than merely describing it. Later developments extend this family to dependent-feature SHAP [50]. Fast amortised approximations provide another practical direction [51]. Counterfactual variants extend the same attribution logic into intervention-style explanation [52]. Broader XAI surveys explicitly compare attribution with alternative explanation families [53]. Model-oriented interpretability texts reinforce the same comparative perspective [54]. More recent domain-specific surveys continue that synthesis [55]. Figure 2.8 contrasts that post-hoc workflow with a more explanation-aware modelling pipeline.

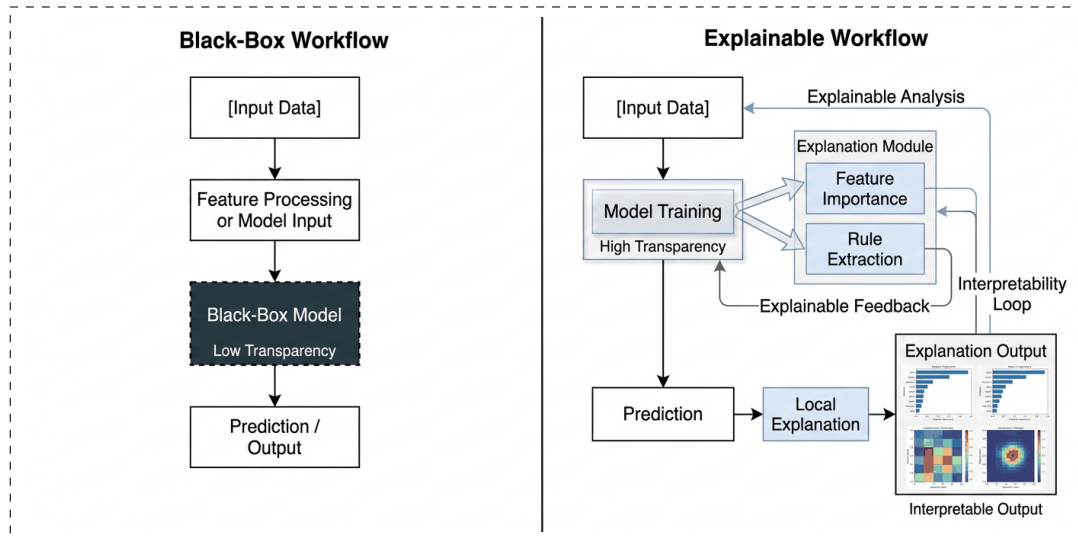


Figure 2.8: Black-box and explainable modelling workflows

2.7.3 Intrinsic Interpretability

Intrinsic interpretability builds transparency into the model architecture itself, for example through linear structures, sparse rules, or inherently readable decision paths. Such models can be highly valuable when explanation is essential by design. However, the interpretability–performance trade-off is not absolute; it is contingent on how intelligently the explanatory mechanism is integrated. The thesis explores this contingency by showing that Shapley values can sometimes migrate from post-hoc explanation toward in-training guidance [56].

2.7.4 Explainability in Recommendation Systems

In recommendation, explanation serves multiple audiences. Users seek reasons to trust or contest an output; designers seek debugging signals; platform operators seek evidence about bias and exposure patterns. Survey work on recommender explanation evaluation likewise shows that transparency, scrutability, trust, and effectiveness are distinct user-centred criteria rather than one interchangeable goal [57]. Textual templates, aspect explanations, attention visualisations, and path-based reasoning over knowledge graphs all attempt to address these needs, but they often optimise communicative plausibility rather than faithful causal attribution. This distinction becomes critical in later chapters, where the thesis insists that explanation should be anchored in marginal contribution rather than rhetorical convenience. Broader explainable-recommender surveys reinforce the same point [58]. Applied reviews likewise show that user-facing plausibility, trust support, and domain transferability are related but non-identical goals [59].

2.7.5 Evaluation of Explanations

Explanation quality is difficult to assess because persuasiveness, faithfulness, stability, and usefulness need not coincide. In a model-based reading, faithfulness asks whether the features identified as important genuinely matter to the predictor; one common intervention-style formulation is deletion or insertion performance, where explanation quality is judged by how quickly model performance deteriorates or recovers as highlighted features are removed or reintroduced [60]. Stability asks whether small perturbations in data or explanation settings produce large attribution changes, a concern made especially visible by adversarial and perturbation-based failures of post-hoc explanation tools [61]. Sparsity and usability matter because an explanation that spreads importance thinly across many variables may be formally valid yet operationally unreadable, while human-centred work shows that causability and usability cannot be read directly from attribution maps alone [62]. A closely related user study reaches the same conclusion for domain comprehension [63]. A method may therefore generate plausible stories that are not causally relevant to the model. The present thesis consequently treats explanation evaluation as multi-criterion: formal coherence, local accuracy, stability under approximation, semantic plausibility, and usefulness for downstream analysis all matter together rather than sequentially. This evaluative stance is a major reason for choosing Shapley-based methods over ad hoc alternatives. It is also why recent trustworthy-AI work links explanation to causability [64]. Governance frameworks make the same move at policy level [65]. Bias-and-fairness surveys show why explanation cannot be reduced to visual plausibility alone [66]. Table 2.3 compares the explanation families most relevant to the thesis in terms of scope, dependence, and

normative coherence.

Table 2.3: Comparison of major XAI families

Method	Local/global scope	Model dependence	Main strength	Main limitation
LIME	Primarily local	Model-agnostic	Simple local surrogate	Sensitive to perturbation design
SHAP	Local and global	Model-agnostic / model-specific variants	Axiomatic attribution	Computational burden
TreeSHAP	Local and global	Tree ensembles	Exact fast tree attribution	Limited to tree-family models
Integrated Gradients	Local	Differentiable models	Gradient-based faithfulness	Requires differentiable pathway
Counterfactual methods	Local and actionable	Broadly model-agnostic	Intervention-oriented explanation	Often optimisation-heavy

2.8 Cooperative Game Theory and Shapley Values

This section introduces the cooperative-game concepts that provide the thesis’s main theoretical backbone.

2.8.1 Transferable Utility Games

A transferable-utility cooperative game is defined by a player set N and a characteristic value function $v(S)$ for every coalition $S \subseteq N$. In compact notation:

$$\mathcal{G} = (N, v), \quad v : 2^N \rightarrow \mathbb{R}, \quad (2.11)$$

where $v(S)$ measures what the coalition can achieve collectively. In the context of explainable AI, the players are typically features, interactions, or entities, and the coalition value is defined in terms of model output, clustering quality, or recommendation utility. This formulation is powerful because it translates the vague question “what

mattered?” into the mathematically precise question “what is the expected marginal contribution of each player across all coalition orders?” [67].

2.8.2 Shapley Value: Definition and Axioms

The Shapley value allocates the total value of a cooperative game among players according to their expected marginal contribution across all orderings. Later expository work summarises the same axiomatic logic for machine-learning audiences [68]. For player $i \in N$, it is defined as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (2.12)$$

Its enduring importance stems from the axioms it satisfies: efficiency, symmetry, null player, and additivity. Formally, efficiency requires $\sum_{j \in N} \phi_j = v(N) - v(\emptyset)$; symmetry requires $\phi_i = \phi_j$ whenever players i and j make identical marginal contributions to all coalitions; the null-player axiom requires $\phi_i = 0$ whenever $v(S \cup \{i\}) = v(S)$ for all $S \subseteq N \setminus \{i\}$; and additivity requires $\phi_i(v + w) = \phi_i(v) + \phi_i(w)$ for any two games v and w . In machine learning, these axioms matter because they turn attribution from a heuristic ranking into a principled allocation rule. This thesis returns repeatedly to these properties because they explain why Shapley values remain attractive even when exact computation must be approximated.

Two common alternatives in the literature are the Banzhaf index and the nucleolus. The Banzhaf index also averages marginal contributions, but in its standard form it does not satisfy efficiency, so its attributions need not sum to the total value being explained. For a thesis that treats attribution completeness as an important design requirement, this is a meaningful limitation. The nucleolus, by contrast, was developed for dissatisfaction minimisation in cooperative bargaining problems rather than for additive feature attribution, and is therefore less naturally suited to the explanatory setting considered here. The Shapley value is preferred here because it is the unique allocation rule satisfying all four axioms simultaneously, as proved in section A.1. Figure 2.9 provides an intuitive coalition-based picture of the formal definition discussed in the cooperative-game literature, while table 2.4 summarises the axioms and their interpretive significance.

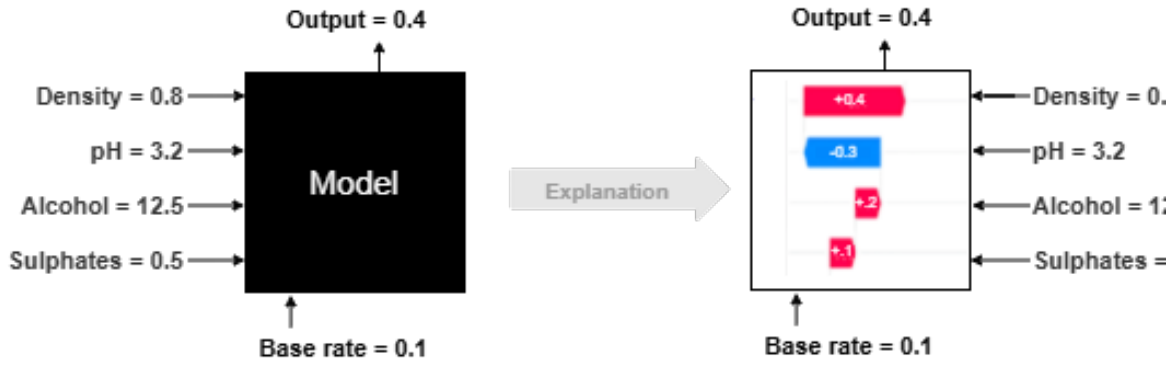


Figure 2.9: Coalition view of the Shapley value

Table 2.4: Shapley axioms and their interpretive meaning

Axiom	Formal statement	Interpretive meaning
Efficiency	$\sum_j \phi_j = v(N) - v(\emptyset)$	All explanatory mass is allocated
Symmetry	Identical contributors receive equal allocation	Equivalent players are treated fairly
Null player	Zero marginal contributor receives zero value	Irrelevant variables are not credited
Additivity	$\phi(v + w) = \phi(v) + \phi(w)$	Explanations compose consistently across tasks

2.8.3 Monte Carlo Approximation of Shapley Values

Exact Shapley computation is exponential in the number of players, which is often impractical for realistic models or interaction spaces [69]. Modern approximation methods therefore emphasise efficient sampling and learned surrogates [70]. Monte Carlo approximation addresses this difficulty by sampling permutations or coalitions and estimating the expected marginal contribution empirically. A generic estimator based on M sampled coalitions is:

$$\hat{\phi}_i = \frac{1}{M} \sum_{m=1}^M [v(S_m \cup \{i\}) - v(S_m)], \quad (2.13)$$

which converges to the exact value as M increases. The trade-off is between computational cost and estimator variance. This issue is peripheral in the clustering chapters and central in DyHuCoG, where approximation quality directly affects training behaviour. Figure 2.10 visualises the exact-versus-sampled trade-off that recurs later in the thesis.

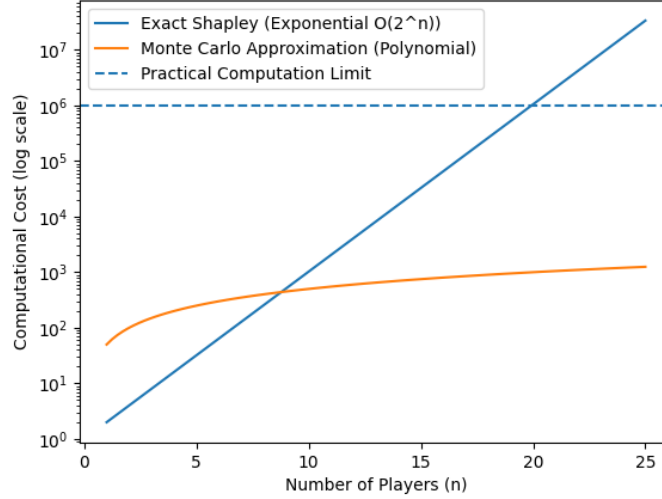


Figure 2.10: Exact versus Monte Carlo Shapley computation

2.8.4 TreeSHAP and KernelSHAP

TreeSHAP exploits the structure of tree ensembles to compute exact or near-exact Shapley attributions efficiently for a large and practically important model family. Its practical relevance is strongest for the gradient-boosted and ensemble tree families used widely in applied machine learning [71]. KernelSHAP generalises the additive explanation framework to broader model classes, but typically at higher computational cost. Follow-on work on tree ensembles has also clarified how path-dependent structure influences practical TreeSHAP behaviour in applied settings [72]. The clustering contributions in this thesis use TreeSHAP because the LightGBM surrogate offers a tractable bridge between unsupervised cluster labels and feature-attribution analysis.

2.8.5 Preference-Aware Shapley Variants

Recent work has explored weighted or modified Shapley formulations that reflect player importance, fairness constraints, or application-specific utility structures [73]. Closely related data-valuation literature makes the same move by redefining utility around training-set usefulness rather than raw prediction output [74]. Other extensions target graph explanation games and counterfactual allocation [75]. Robustness, anomaly detection, and meta-explanatory variants further show how flexible the cooperative-game lens has become [76].

Two distinctions matter for the present thesis. First, FW-Shapley-style estimators modify the weighting or real-time approximation of the Shapley computation itself in order to obtain preference-sensitive attributions. Second, Owen-style coalition-structure variants conceptually redistribute credit under pre-specified unions of players rather than under a flat player set [77]. DyHuCoG does not impose such exogenous coalition

blocks. Its preference-aware utility instead leaves the player set unchanged and alters the coalition value through an additive preference-consistency bonus, so that user–item agreement changes utility directly rather than via a separate union structure. In that sense, the thesis is closer to weighted utility modulation than to coalition-structure redefinition.

This distinction is important because DyHuCoG is not presented as a generic replacement for all modified Shapley games. It is a recommendation-specific formulation in which the utility itself is multi-objective and preference-aware. The contribution therefore lies in embedding that utility inside hypergraph message passing rather than in claiming a universally dominant weighted-Shapley variant.

2.8.6 Shapley Values in Machine Learning: From Theory to Practice

The migration of Shapley values from cooperative game theory into machine learning has been rapid because the method offers what many interpretability approaches lack: a normative answer to the attribution problem. Survey and tutorial work helped transfer the axiomatic language into day-to-day machine-learning practice [78]. That translation is especially visible in graph and model-explanation settings, where Shapley-style allocation is now treated as a general explanatory template and extended to anomaly analysis, counterfactual reasoning, and robustness-oriented explanation games [79]. Yet practice also reveals limitations. Approximation can be expensive, feature independence assumptions can distort interpretation, and explanations can be misread as causal proof [80]. These concerns are echoed by early machine-learning explanation studies that emphasise the gap between attribution and causal inference [81]. Related work on global explanation further shows how widely the Shapley lens now travels across machine learning [82]. Feature-relevance analysis reaches a related conclusion from an information-theoretic angle [83]. Data-valuation surveys extend the same lens to training-data usefulness [84]. Game-theoretic learning work shows that the cooperative perspective also travels beyond explanation into optimisation [85]. The thesis therefore adopts a measured position: Shapley values are not magic, but when the game is formulated carefully they provide a uniquely rigorous starting point for explainable modelling.

2.9 Clustering and Dimensionality Reduction

This final section introduces the concepts most directly used in the first two empirical contributions.

2.9.1 Principal Component Analysis (PCA)

PCA projects data into orthogonal directions that capture maximal variance [86]. If X is the standardised data matrix and W collects the leading eigenvectors of the covariance matrix, the reduced representation is:

$$Y = XW, \quad (2.14)$$

and the explained-variance ratio of principal component k is:

$$r_k = \frac{\lambda_k}{\sum_{i=1}^d \lambda_i}, \quad (2.15)$$

where λ_k is the corresponding eigenvalue. PCA is often used to reduce dimensionality, denoise inputs, and support visualisation. In this thesis, PCA has a deliberately limited but important role. It is used to stabilise clustering and support visual diagnosis of separation, but the explanatory analysis is ultimately returned to the original feature space through the LightGBM surrogate so that the resulting attributions remain semantically interpretable.

2.9.2 K-Means Clustering

K-Means partitions observations into k clusters by minimising within-cluster squared distance to centroids [87]. Its objective is:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2, \quad (2.16)$$

where C_i is cluster i and μ_i is its centroid. The method is attractive for its simplicity, speed, and interpretability of assignments. Its weakness is that it provides cluster membership without explanation. Because K-Means supplies crisp partitions while remaining computationally light, it is well suited to the thesis's strategy of separating cluster discovery from cluster explanation. Two natural alternatives merit explicit consideration. Density-based clustering such as DBSCAN requires calibration of a neighbourhood radius ε and a minimum-points threshold whose optimal values are non-trivial to determine in correlated physicochemical and meteorological spaces; it also designates some observations as noise points, leaving the surrogate classifier without stable training targets for those cases. Gaussian mixture models avoid hard rejection of observations, but they return soft, probabilistic assignments rather than crisp memberships. A LightGBM surrogate trained on fractional cluster labels would then blur the interpretation of SHAP values because the explanation would no longer correspond cleanly to one cluster profile. K-Means++ avoids both problems by delivering

complete, deterministic hard assignments that map directly to a well-defined supervised task, which is the necessary precondition for the surrogate-explanation strategy adopted in this thesis [88].

2.9.3 Cluster Validation Metrics

Clustering quality cannot be assessed through classification accuracy because no ground-truth labels are assumed. Internal validation metrics such as the Silhouette coefficient and Davies–Bouldin index therefore become crucial [89]. The Silhouette coefficient is particularly important in this thesis because it can be interpreted naturally as a coalition value in the clustering game: it rewards compactness and separation simultaneously, while remaining normalised across datasets.

2.9.4 LightGBM as a Surrogate Classifier

A central design move in the clustering chapters is to train a LightGBM classifier on the cluster labels produced by K-Means and then explain the classifier with TreeSHAP. This surrogate strategy is important because it makes otherwise intractable unsupervised explanation computationally manageable while preserving access to the original feature semantics. The method is not innocent, however: its fidelity depends on the surrogate learning the cluster assignments accurately enough to make attribution meaningful. That assumption is therefore scrutinised explicitly in chapters 5 and 6.

Summary

This chapter established the conceptual foundation on which the rest of the thesis depends. We reviewed the evolution from classical to graph-based and hypergraph-based recommendation, clarified why explanation has become a substantive modelling requirement rather than an optional supplement, and introduced the cooperative-game concepts that anchor the thesis’s explanatory framework. The final section prepared the ground for the clustering contributions by explaining why PCA, K-Means, internal validation metrics, and LightGBM surrogates are not arbitrary methodological choices but logically connected components of the thesis argument.

The next chapter moves from conceptual groundwork to critical positioning. If the present chapter explained *what* conceptual tools are available, chapter 3 asks which parts of the existing literature remain inadequate, contradictory, or incomplete, and therefore where the thesis’s original contributions are located. That transition is driven directly by the tensions already visible across the thesis contributions [90].

Chapter 3

Related Work and Research Positioning

Introduction

Building directly on chapter 2, this chapter reviews the literature thematically rather than paper by paper. That choice is deliberate. A paper-by-paper survey would reproduce the fragmentation already present in the field, whereas the present thesis seeks to show that clustering explanation and recommendation learning can be studied under a single explanatory logic. The chapter therefore groups the literature around the conceptual tensions that matter most for the thesis: where interpretability fails in clustering, how graph-based recommendation evolved, how explanation is handled in recommender systems, and why Shapley-based attribution is promising yet still underdeveloped in dynamic learning settings.

The chapter has two purposes. First, it establishes the state-of-the-art critically rather than descriptively. Second, it identifies the research gaps that motivate the three contribution chapters. These gaps are not presented as empty spaces in the literature; they are presented as unresolved tensions among existing approaches, exactly the kind of positioning needed to connect the clustering and recommendation papers into one thesis argument [90].

3.1 Interpretable Clustering: State of the Art

This section reviews the clustering-interpretability literature and shows why existing methods remain fragmented across stages of the clustering workflow.

Interpretable clustering has grown in response to the recognition that unsupervised models are often used in high-stakes settings without accompanying explanations. The literature is now broad enough to distinguish among explanation strategies applied

before clustering, during clustering, and after clustering. This taxonomy is helpful because it reveals that many methods improve only one stage while leaving the overall explanatory pipeline incomplete.

3.1.1 Pre-Clustering Interpretability Methods

Pre-clustering approaches aim to make clustering more interpretable by selecting features, denoising inputs, or constraining the representation space before any partition is learned. Representative examples include minipatch and consensus-style schemes that stabilise feature selection before clustering at scale [91], as well as dimensionality-control pipelines surveyed in recent interpretable-clustering reviews. Their strength is that they simplify the data manifold and can improve semantic clarity before a partition is induced. Their weakness is that they often explain why the data were represented in a certain way rather than why a specific cluster structure emerged. In other words, they improve intelligibility upstream without fully solving downstream explanation. The gap left by this family is therefore cluster-specific attribution after the partition has actually been formed.

3.1.2 In-Clustering Interpretability Methods

In-clustering methods incorporate interpretability into the clustering algorithm itself, for example through explanation-space clustering around learned surrogate features [92]. Broader survey work also covers tree-like partitions, prototype assignments, and sparse centre descriptions. These approaches are valuable when model transparency must be intrinsic. However, they frequently trade flexibility for readability and may underperform on data whose structure is not well captured by simple partition logics, especially when the analyst wants both strong empirical clustering and a post-hoc account in the original feature space. The thesis takes a different route: rather than replacing strong clustering methods with simpler ones, it seeks to explain their outcomes faithfully. The unresolved gap is thus not interpretability in general, but faithful explanation of flexible clustering pipelines without forcing them into overly restrictive intrinsic forms.

3.1.3 Post-Clustering Explanation via SHAP

Post-clustering SHAP methods are the most directly relevant antecedents of this thesis because they attempt to explain cluster formation after clusters have been identified [90]. Their main virtue is that they preserve access to flexible clustering pipelines while still producing feature-attribution analyses. Their main limitation is fragmentation. Many studies report global importance without stable local explanation, or local explanation without a principled account of how cluster-level summaries should be constructed.

Very few address the problem of hierarchical consistency across clustering levels. That unresolved tension directly motivates chapters 5 and 6.

Table 3.1: Representative interpretable clustering methods

Method family	Explanation stage	Strength	Limitation
Pre-clustering feature selection	Before partitioning	Simpler cluster geometry	May discard useful explanatory interactions
Intrinsic interpretable clustering	During partitioning	Transparent objective structure	Limited flexibility on complex data
Post-clustering surrogate explanation	After partitioning	Compatible with black-box clustering	Depends on surrogate fidelity
Explanation-space clustering	Cluster after explanation	Reveals explanation similarity	Harder to relate back to original partition objective

3.2 Graph-Based and Hypergraph Recommender Systems

This section reviews the graph-based recommendation literature with particular attention to what hypergraph models add and what they still fail to explain.

3.2.1 Bipartite Graph Collaborative Filtering

Graph collaborative filtering demonstrated that user–item relations can be modelled more faithfully through propagation on interaction graphs than through matrix completion alone. The literature shows consistent gains under sparse implicit feedback, especially when neighbourhood structure carries information not visible in direct interactions. However, these models usually treat edges as uniformly meaningful once they exist. This assumption is analytically convenient but explanatorily weak.

3.2.2 Hypergraph Collaborative Filtering

Hypergraph recommendation extends graph methods by modelling multiway relations among users, items, and auxiliary entities. This is particularly useful when user

preference depends on sets of co-occurring signals rather than on pairwise edges. The literature demonstrates that hypergraph structure improves expressive capacity, but it rarely asks whether all hyperedges should contribute equally or how the importance of different coalitions should be measured. DyHuCoG addresses precisely this oversight.

3.2.3 Self-Supervised and Contrastive Hypergraph Models

Recent work combines hypergraph learning with contrastive or self-supervised objectives, seeking to stabilise embeddings in sparse settings and improve generalisation. These models are strong baselines because they show that recommendation can benefit from auxiliary signals beyond ordinary ranking losses. At the same time, they leave unexplained why one relational pattern should be privileged over another in the propagation process itself. Their auxiliary objectives are useful, but they are not attributional. Table 3.2 shows exactly where DyHuCoG departs from those representative hypergraph baselines. The gap at the end of this stream is therefore not expressive power alone, but the absence of a principled contribution-allocation mechanism inside the hypergraph learner.

3.3 Graph Neural Network-Based Recommendation

This section examines key strands of the GNN recommendation literature and identifies where explanation remains under-specified.

3.3.1 Knowledge Graph-Augmented GNN Recommendation

Knowledge-graph-augmented recommenders use structured semantic relations to enrich item and user representations. They can improve cold-start behaviour and produce path-based explanation narratives, but the quality of those narratives depends heavily on the completeness and interpretability of the underlying knowledge graph. Moreover, path explanation does not automatically imply fair attribution: a path may exist without representing the most influential causal mechanism.

3.3.2 Session-Based Graph Recommendation

Session-based recommenders model short-term sequences or session graphs rather than stable preference histories. They demonstrate the importance of local context and temporal dependence, but they also reveal a challenge relevant to this thesis: explanation in dynamic recommendation must account for interactions whose importance varies over time. Static post-hoc explanation is often too blunt for this setting.

3.3.3 Multi-Relational Graph Recommendation

Multi-relational models incorporate different edge types or semantic relations among entities. This makes them structurally rich, but it intensifies the importance-allocation problem. When many relation types coexist, the model needs a principled basis for deciding how much each type matters. Attention can help, but it does not by itself provide the axiomatic guarantees that the thesis seeks. The common gap across these GNN strands is that relation-rich modelling has advanced faster than relation-rich explanation.

Table 3.2: Comparison of hypergraph collaborative filtering models

Model	Higher-order relation handling	Auxiliary objective	Explanation mechanism	Limitation addressed by DyHuCoG
HCCF	Hyperedge contrastive learning	Self-supervised contrast	None explicit	Uniform propagation weights
HPCF	Hypergraph projection	Preference-aware structure	None explicit	No in-training attribution signal
Sequential hypergraph models	Session/sequence hyperedges	Sequential prediction	None explicit	Limited cooperative interpretation
Knowledge-graph hybrids	Multi-relational expansion	Relation reasoning	Path-level only	Weak coalition-based importance allocation

3.4 Explainable Recommender Systems

This section reviews the main explanation paradigms in recommender systems and shows why many of them prioritise communicative plausibility over principled attribution.

3.4.1 Textual and Template-Based Explanation

Template-based recommendation explanations generate user-facing statements such as “recommended because you liked...” or “similar users also enjoyed...”. These explanations

can improve perceived transparency and user satisfaction, but they often reflect an interface design decision more than a faithful attribution analysis. Their explanatory content is frequently too coarse to support auditing or intervention.

3.4.2 Attention-Based Explanation

Attention weights are sometimes presented as explanations because they indicate what the model focused on. However, the literature has shown repeatedly that attention and explanation are not equivalent. Attention reflects one internal weighting mechanism, not necessarily the marginal contribution of a feature, relation, or entity to the final decision. This is one reason the present thesis avoids treating attention as sufficient explanation, even when attention is used architecturally in DyHuCoG.

3.4.3 Path-Based Explanation on Knowledge Graphs

Path-based explanations identify semantic routes through a knowledge graph that connect users to recommended items. They can be compelling when a domain ontology is well curated. However, the existence of a path does not imply that the path is the most influential structure in the model’s decision process. In this sense, path-based explanation often explains a plausible semantic relation rather than the model’s actual credit allocation.

3.4.4 Counterfactual Explanation in Recommendation

Counterfactual methods ask what minimal change to the user profile, item attributes, or interaction graph would alter a recommendation. They are analytically valuable because they highlight decision boundaries. However, counterfactuals typically answer a different question from Shapley values. They reveal sensitivity, whereas Shapley values allocate contribution. The two are complementary, but not interchangeable.

3.5 Shapley-Value-Based Explainability Methods

This section positions Shapley-based explanation as both promising and incomplete in the current literature.

3.5.1 SHAP for Classification and Regression

SHAP became influential because it unified additive feature attribution across model classes and linked explanation to the Shapley axioms. In supervised tasks, it is now common to report global summary plots, local force plots, and dependence analyses.

This maturity is important because it provides the methodological foundation on which the thesis builds. However, supervised maturity should not be mistaken for universal readiness; the transition to unsupervised learning and dynamic graph training remains underdeveloped.

3.5.2 Shapley Values for Unsupervised Learning

Applications of Shapley values to clustering and other unsupervised models remain comparatively scarce [90]. Where they exist, they often rely on surrogate modelling, post-hoc partition analysis, or explanation-space clustering. These studies show feasibility, but they leave open questions about scalability, hierarchy, and cross-level consistency. The later large-scale study reinforces that diagnosis by showing that cross-level explanatory coherence has to be designed explicitly rather than assumed [88]. They also often evaluate explanation descriptively rather than as part of a broader research programme.¹ Table 3.1 positions the main interpretable-clustering families by the stage at which explanation enters the workflow.

3.5.3 GNNShap and Graph-Level Shapley Attribution

Graph-level and node-level Shapley methods such as GNNShap extend cooperative attribution to graph neural networks by quantifying the importance of internal graph components [93]. Broader graph-explanation families such as GNNExplainer pursue a related subgraph-level logic [94]. Multi-hop Shapley variants extend the same idea to longer-range graph dependencies [95]. Their existence is important because they demonstrate that graph explanation can be formulated in coalition or subgraph terms. However, most remain post-hoc tools applied after model training. They help interpret graph predictions, but they do not usually alter the propagation mechanism itself.

3.5.4 Shapley Values in Training: In-Training Attribution

The most immediate gap in the literature concerns the use of Shapley values during training rather than after it. Two adjacent research lines help frame the problem but stop short of the DyHuCoG design. First, data-valuation methods such as Data Shapley assign Shapley values to training examples in order to guide dataset curation, pruning, or acquisition [96]. Later weighting variants including Beta Shapley refine that idea through alternative valuation emphasis [97]. Subsequent survey work synthesises the broader data-valuation perspective. These methods ask which data points are most valuable to aggregate model performance, whereas DyHuCoG asks which user–item–

¹The author is a co-author of the prior publication [90] that underpins the first clustering chapter and of the prior publication [88] that underpins the multi-level clustering chapter.

context coalitions contribute most to recommendation utility inside a fixed dataset. The question is therefore not dataset valuation but interaction valuation during learning.

Second, Shapley-based data pruning for recommender systems [98] removes low-value interactions before training. DyHuCoG differs in a conceptually important way: it does not discard interactions, but uses estimated coalition utilities as dynamic propagation weights so that weak evidence can remain in the hypergraph while being down-weighted rather than eliminated [99].

The field therefore still lacks a mature framework in which cooperative game-theoretic attribution directly governs message importance inside a hypergraph recommender. That unresolved point is clearest when the recent dynamic-adjustment study and the DyHuCoG contribution are read together. Chapter 7 fills precisely this gap.

3.6 Game Theory in Recommender Systems

This section shows that game-theoretic ideas are already present in recommendation, but usually not integrated into a unified explanatory architecture.

3.6.1 Cooperative Game Models for Collaborative Filtering

Game theory has been used to model cooperative filtering, group recommendation, and influence allocation among entities. These studies are relevant because they establish that recommendation can be framed as interaction among rational or semi-rational components rather than as passive pattern matching. However, many such models stop at incentive or allocation interpretation and do not connect the game formulation to concrete explanation tools usable by practitioners.

3.6.2 Fairness-Aware Recommendation via Game Theory

Fairness-aware recommendation uses game-theoretic or allocation-based perspectives to balance competing stakeholder objectives, often focusing on exposure, opportunity, or equitable utility distribution. This body of work is important for the present thesis because it reveals a natural affinity between cooperative-game reasoning and accountable ranking. However, fairness-aware recommendation and Shapley explanation often proceed on separate tracks. The thesis suggests a stronger integration is possible [100]. The gap here is not the absence of game-theoretic language, but the absence of a single framework that turns that language into operational explanation and learning-time weighting.

3.6.3 Coalition Formation and Group Recommendation

Coalition formation models group preference aggregation explicitly, which is conceptually close to the coalition logic used in this thesis. The difference is that group recommendation studies usually model coalition outcome among users, whereas DyHuCoG models cooperative value among users, items, and contexts simultaneously. The shift from user coalitions to interaction coalitions is analytically significant [101].

3.7 Dynamic and Adaptive Recommender Systems

This section reviews the literature on dynamism and adaptation to show why static explanation is insufficient in realistic recommendation settings.

3.7.1 Temporal Dynamics in User Preferences

Temporal-dynamics research shows that user interests are neither static nor uniformly stable. Seasonality, novelty seeking, life events, and shifting consumption contexts all alter what counts as relevant. Static recommenders frequently fail because they continue to weight old interactions heavily even after their predictive value has decayed. This motivates the thesis’s interest in dynamic weighting and refresh mechanisms.

3.7.2 Online and Incremental Learning

Online and incremental recommenders attempt to update representations as new interactions arrive. Their strength is responsiveness; their weakness is that explanation often lags behind adaptation. A model may update continuously without providing a corresponding account of how influence changed. DyHuCoG’s periodic Shapley refresh strategy is relevant precisely because it links adaptive learning to adaptive attribution rather than treating them as separate concerns. The unresolved problem left by this literature is therefore not merely adaptation, but adaptation with interpretable credit assignment.

3.8 Research Gap and Positioning

This section synthesises the reviewed literature into the specific gaps addressed by the thesis.

3.8.1 Identified Gaps in Explainable Clustering

The clustering literature leaves three problems insufficiently addressed. First, local and global explanation are rarely integrated into one coherent framework. Second, scalability is often treated as a practical afterthought rather than a design principle. Third, hierarchical consistency across multiple clustering levels is largely absent. The first two contribution chapters intervene precisely at this intersection.

3.8.2 Identified Gaps in Explainable Graph Recommendation

The recommendation literature exhibits a different but related set of gaps. Strong graph and hypergraph models still tend to assume uniform edge or hyperedge importance, while explanation is often bolted on after training. Diversity is commonly treated as a secondary objective, and dynamic contexts are incorporated without a principled attribution mechanism. The third contribution chapter positions itself as a response to these jointly rather than separately.

3.8.3 Positioning of the Three Contributions

The three contribution chapters should therefore be understood as an integrated progression rather than as unrelated studies. Chapter 5 establishes the feasibility of Shapley-based explanation for unsupervised black-box structure. Chapter 6 shows that the same explanatory logic can scale and remain coherent in large hierarchical settings. Chapter 7 then moves from explanation of trained models to learning with attribution, thereby extending cooperative-game reasoning into the optimisation loop itself. The thesis’s originality lies not only in any single method, but also in the continuity of this progression. That continuity is visible most directly when the three contributions are read as one sequence rather than as isolated results [90]. Table 3.3 summarises the doctoral gap structure that motivates that progression.

Table 3.3: Research-gap matrix

Literature stream	G1	G2	G3	G4	G5	G6
Interpretable clustering	✓	✓				
Hierarchical clustering explanation		✓	✓			
Graph recommendation				✓	✓	
Hypergraph recommendation				✓	✓	✓
Shapley-based XAI	✓	✓	✓	✓	✓	✓

Legend: G1 = limited faithful clustering explanation; G2 = weak local–global explanation coherence; G3 = poor scalability of clustering explanation; G4 = limited hierarchical attribution consistency; G5 = limited in-training attribution for recommendation; G6 = weak joint treatment of accuracy, diversity, and context.

Summary

This chapter reviewed the literature thematically and critically. We showed that interpretable clustering remains fragmented across stages of the clustering workflow, that graph and hypergraph recommendation models improve representation learning while leaving importance allocation under-specified, and that Shapley-based explanation is promising precisely because it provides a normative rather than purely heuristic basis for attribution. At the same time, the review made clear that the existing literature still leaves large gaps around hierarchical clustering explanation, in-training attribution, and the joint treatment of accuracy, diversity, and interpretability.

These gaps motivate the thesis’s empirical design. The next chapter operationalises the research programme by defining a common protocol for datasets, baselines, metrics, statistical testing, and implementation. In other words, if the present chapter identified *what* needs to be shown, chapter 4 explains *how* the thesis shows it. The protocol is shared precisely so that the empirical contributions can be compared within one evaluative frame [90].

Chapter 4

Experimental Protocol and Evaluation Framework

Introduction

The previous chapters established the conceptual motivation and research gaps that frame the thesis. The present chapter defines the methodological framework within which all empirical contributions are evaluated. This framework matters because the thesis is not a collection of unrelated experiments. Although the three contribution chapters differ in task, data regime, and model architecture, they are assessed under a common protocol covering dataset description, preprocessing, splitting, evaluation, optimisation, reproducibility, and statistical validation.

The chapter is organised from data to inference. It first describes the datasets used across the thesis, then explains the splitting and preprocessing procedures adopted for clustering and recommendation. It next defines the baseline methods and evaluation metrics, followed by the training protocol and the statistical procedures used to assess whether observed differences reflect substantive rather than chance variation. Where a later chapter introduces study-specific adjustments, those are stated explicitly as local departures from the general framework defined here.

4.1 Datasets

This section describes the datasets used in the thesis and explains the criteria by which they were selected.

The empirical programme uses two clustering datasets and two recommendation datasets. This combination is deliberate. The clustering studies require data with semantically interpretable features, while the recommendation study requires large implicit-feedback benchmarks with established baselines. The selected datasets therefore

span different methodological regimes while remaining sufficiently standard for results to be interpretable and comparable.

4.1.1 Dataset Structure and Selection Criteria

The datasets were selected according to four criteria: public availability, established use in the relevant literature, sufficient size to support meaningful experimentation, and semantic transparency of the variables or entities involved. For the clustering studies, semantic transparency is especially important because feature attribution is only useful when variables can be interpreted in domain terms. For the recommendation study, benchmark status matters because comparative performance claims are meaningful only when tested on datasets widely used in prior work.

4.1.2 Portuguese Wine Quality Dataset

The Portuguese Wine Quality dataset contains 4,898 observations and 11 physicochemical features describing vinho verde wines, including fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, sulfur dioxide measures, density, pH, sulphates, and alcohol [102]. It is well suited to interpretable clustering because the feature space is compact, fully numeric, and substantively meaningful. It also provides a useful plausibility check, since the importance assigned to acidity, density, alcohol, and sulfur-related variables can be assessed against established oenological knowledge rather than treated as uninterpretable numerical artefacts.

4.1.3 Beijing Multi-Site Air Quality Dataset

The Beijing Multi-Site Air Quality dataset contains approximately 383,585 hourly observations collected between 2013 and 2017 across multiple monitoring stations [103]. The modelling subset comprises 11 core pollutant and meteorological variables, including PM2.5, PM10, NO₂, SO₂, CO, O₃, temperature, pressure, dew point, wind direction, and wind speed. This dataset tests whether the clustering-explanation framework remains informative on data that are not only much larger than the wine dataset but also structurally more heterogeneous and temporally variable.

4.1.4 MovieLens-1M Dataset

MovieLens-1M contains 6,040 users, 3,706 items, and 1,000,209 interactions. It is comparatively dense by modern recommendation standards, with an interaction density of approximately 0.0447. The dataset is useful for studying recommendation behaviour when the interaction signal is relatively strong and contextual proxies such as genre are readily available [104].

4.1.5 Amazon-Book Dataset

Amazon-Book contains 52,643 users, 91,599 items, and 2,984,108 interactions, but its density is only approximately 0.0006. The user–item matrix is therefore extremely sparse. This dataset is included to test whether the proposed recommendation model remains effective under a substantially more demanding interaction regime, where the signal available to any propagation mechanism is inherently limited [104].

4.1.6 Implicit Feedback Conversion

The clustering datasets are analysed as tabular feature matrices and therefore do not require feedback conversion. By contrast, the recommendation benchmarks are treated as implicit-feedback tasks. Explicit ratings or behavioural signals are binarised into interaction events suitable for top- N ranking. For MovieLens-1M, ratings greater than 3 are treated as implicit positives, following the protocol reported in the DyHuCoG study [104]. This conversion aligns the recommendation experiments with ranking objectives rather than cardinal rating regression. Table 4.1 consolidates the scale and density properties that shape the later experimental choices.

Table 4.1: Core dataset statistics used throughout the thesis.

Dataset	Users / Instances	Items / Features	Interactions / Period	Density
Wine Quality	4,898	11 features	single snapshot	—
Beijing Air Quality	383,585	11 features	2013–2017	—
MovieLens-1M	6,040 users	3,706 items	1,000,209 interactions	0.0447
Amazon-Book	52,643 users	91,599 items	2,984,108 interactions	0.0006

Note. Density is defined only for the recommendation datasets because the clustering datasets are not represented as user–item interaction matrices.

4.2 Data Splitting Strategy

This section defines how data are partitioned for training, validation, and testing.

The splitting strategy differs between clustering and recommendation because the tasks are not identical. For clustering, the emphasis is on partition stability and explanatory consistency. For recommendation, the main concerns are leakage prevention and realistic ranking evaluation.

4.2.1 User-Level Splitting

For recommendation experiments, interactions are split at the user level so that each user remains represented in the training and evaluation pipeline while the target interaction

is excluded from model fitting. This design is preferable to random interaction-level splitting, which can produce overly optimistic generalisation estimates.

4.2.2 Temporal Ordering

Where timestamps are available, temporal order is respected so that earlier interactions inform training and later interactions are reserved for validation or testing. This is particularly important for Amazon-Book and for any analysis involving adaptive recommendation. In the recommendation benchmarks, interactions are sorted chronologically per user before splitting, so that the validation interactions are temporally later than the training interactions for the same user, and the held-out test interaction is later still under the leave-one-out protocol.

4.2.3 Train–Validation–Test Split

For recommendation, the thesis adopts a temporal train–validation–test partition consistent with recent graph-recommendation practice. Following the DyHuCoG protocol, user histories are first ordered by time and then divided into 70% training, 10% validation, and 20% held-out interactions. The validation split is used for early stopping and model selection, while the test split is reserved for final reporting. For clustering, a five-fold cross-validation design is used where appropriate to assess the stability of surrogate performance and attribution outputs across repeated partitions.

4.2.4 Leave-One-Out Evaluation

Top- N recommendation is evaluated under a leave-one-out protocol. Within the temporally held-out portion of each user history, the latest observed positive interaction is treated as the target test item and ranked against sampled negatives. This design tests whether the model can recover a user’s most recent relevant item while keeping evaluation computationally tractable.

4.2.5 Negative Sampling for Evaluation

Because ranking against the full item catalogue is often computationally expensive, evaluation uses sampled negative items alongside the held-out positive item. The negative-sampling scheme is kept fixed across all compared methods to ensure a fair basis for comparison.

4.2.6 Dataset-Specific Considerations

The Beijing dataset is treated differently from the recommendation benchmarks because the task is cluster discovery and explanation rather than user-level prediction. Temporal structure is respected where relevant, but the primary concern is that SHAP summaries and representative local examples remain faithful to the clustering solution. Local force-plot analysis is illustrated with one representative observation per cluster, matching the three cluster-specific force plots reported for the Beijing experiments [88]. For the smaller wine dataset, full-dataset explanation is computationally feasible, allowing more detailed local analysis.

4.3 Baseline Methods

This section defines the benchmark methods used for comparison in the empirical chapters.

The baseline set covers classical latent-factor methods, neural recommenders, graph models, hypergraph models, and explanation-oriented comparators. This breadth is necessary to ensure that performance comparisons isolate the contribution of cooperative attribution rather than conflating it with other architectural choices.

4.3.1 Matrix Factorisation (MF)

MF is included as a classical latent-factor baseline. It remains efficient, conceptually simple, and competitive on a range of implicit-feedback tasks. Its score function is:

$$\hat{y}_{ui}^{\text{MF}} = p_u^\top q_i, \quad (4.1)$$

where p_u and q_i are the latent embeddings of user u and item i .

4.3.2 Neural Collaborative Filtering (NCF)

NCF provides a non-linear collaborative baseline that captures richer user–item interactions than dot-product MF. A generic NCF scorer can be written as:

$$\hat{y}_{ui}^{\text{NCF}} = \sigma\left(f_\theta([p_u \parallel q_i])\right), \quad (4.2)$$

where $f_\theta(\cdot)$ denotes a multilayer perceptron applied to the concatenated embeddings. It is an important comparator because it tests whether observed gains stem from added neural expressiveness rather than from the graph or cooperative-game components themselves.

4.3.3 Light Graph Convolutional Network (LightGCN)

LightGCN is the principal graph baseline because it isolates the effect of graph propagation while avoiding unnecessary architectural complexity. Its layer-wise propagation is:

$$e_u^{(\ell+1)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)| |\mathcal{N}(i)|}} e_i^{(\ell)}, \quad (4.3)$$

with the final representation obtained by averaging or weighting embeddings across layers. It therefore serves as a clean test of whether the hypergraph and cooperative-game extensions in DyHuCoG provide substantive value beyond simplified graph propagation.

4.3.4 Contrastive Decoupled Collaborative Learning (RecDCL)

RecDCL represents the self-supervised and contrastive strand of recent recommender research. Its optimisation can be summarised as:

$$\mathcal{L}_{\text{RecDCL}} = \mathcal{L}_{\text{rank}} + \lambda_{\text{con}} \mathcal{L}_{\text{InfoNCE}}, \quad (4.4)$$

thereby combining recommendation ranking with a contrastive consistency objective. Its inclusion ensures that DyHuCoG is compared against strong modern methods that exploit auxiliary representation signals.

4.3.5 Hypergraph Contrastive Collaborative Filtering (HCCF)

HCCF is included as a strong hypergraph baseline. In normalised form, its propagation operator is:

$$E^{(\ell+1)} = \sigma \left(D_v^{-1/2} H W D_e^{-1} H^\top D_v^{-1/2} E^{(\ell)} \right), \quad (4.5)$$

which highlights the role of incidence-based higher-order message passing. It tests whether higher-order structure alone is sufficient, or whether the Shapley-guided weighting in DyHuCoG contributes beyond hypergraph topology.

4.3.6 Hypergraph-based Preference Collaborative Filtering (HPCF)

HPCF is included as the strongest hypergraph comparator reported in the source recommendation study [105]. Because it already models preference-enhanced hypergraph structure, it provides the most demanding reference point for assessing the additional contribution of the DyHuCoG design.

4.3.7 LIME-Based Clustering Baseline

For the clustering contributions, the main interpretability comparator is a LIME-based surrogate explanation pipeline. Its local objective can be written as:

$$\xi(x) = \arg \min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_x) + \Omega(g), \quad (4.6)$$

where g is an interpretable local surrogate, π_x is the locality kernel around instance x , and $\Omega(g)$ penalises complexity. LIME is included as a widely recognised alternative to Shapley-based explanation, and its known sensitivity to perturbation settings makes it a useful contrast for assessing attributional stability.

4.4 Evaluation Metrics

This section defines the metrics used to evaluate ranking quality, clustering quality, and recommendation diversity.

4.4.1 Precision@N

Precision@ K measures the fraction of top- K recommendations that are relevant:

$$\text{Precision@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|R_u \cap T_u|}{K}, \quad (4.7)$$

where R_u is the top- K list for user u and T_u is the set of held-out relevant items.

4.4.2 Recall@N

Recall@ K measures the fraction of relevant held-out items recovered in the top- K list:

$$\text{Recall@}K = \frac{1}{|U|} \sum_{u \in U} \frac{|R_u \cap T_u|}{|T_u|}, \quad (4.8)$$

which is useful in implicit-feedback settings because it indicates whether the system retrieves the target item under ranking constraints.

4.4.3 Normalised Discounted Cumulative Gain (NDCG@N)

NDCG@ K rewards correct retrieval more strongly when the relevant item appears near the top of the ranked list. Discounted cumulative gain is defined as:

$$\text{DCG@}K = \sum_{i=1}^K \frac{2^{rel_i} - 1}{\log_2(i + 1)}, \quad (4.9)$$

and the normalised form is:

$$\text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}. \quad (4.10)$$

NDCG@20 is the principal ranking metric in chapter 7 because it captures both retrieval completeness and rank quality.

4.4.4 Catalogue Coverage

Catalogue coverage measures the proportion of the item catalogue that appears in at least one recommendation list across users:

$$\text{Coverage} = \frac{|\bigcup_{u \in U} R_u|}{|I|}, \quad (4.11)$$

and captures system-level diversity that user-level ranking metrics alone do not reflect.

4.4.5 Intra-List Diversity (ILD)

ILD measures the average pairwise dissimilarity among items within a recommendation list. For a list $R_u = \{i_1, \dots, i_K\}$, it is defined as:

$$\text{ILD}(R_u) = \frac{2}{K(K-1)} \sum_{1 \leq a < b \leq K} [1 - \text{sim}(i_a, i_b)], \quad (4.12)$$

where $\text{sim}(i_a, i_b)$ is an item-similarity function. In the recommendation experiments, similarity is instantiated through cosine similarity between final item representations, so ILD measures semantic dispersion in the learned item space. Importantly, ILD is not treated as a decorative post-hoc metric in this thesis: it is built directly into the coalition utility of DyHuCoG and therefore reflects one of the objectives the model is explicitly trained to balance.

4.4.6 Silhouette Score and Davies–Bouldin Index

For clustering, the Silhouette coefficient is defined as:

$$s(x) = \frac{b(x) - a(x)}{\max\{a(x), b(x)\}}, \quad (4.13)$$

where $a(x)$ is the average intra-cluster distance for point x and $b(x)$ is its minimum average distance to any other cluster. The Davies–Bouldin index is:

$$\text{DB} = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(\mu_i, \mu_j)} \right), \quad (4.14)$$

with lower values indicating better-separated clusters. These metrics are used both to select k and to compare alternative explanatory pipelines.

4.4.7 Averaging Across Users and Statistical Reporting

All user-level recommendation metrics are averaged over the evaluation user set, and results are reported with means and standard deviations across repeated runs or folds. Where paired inferential claims are made in the contribution chapters, raw p -values are interpreted together with Holm’s sequential correction [106], and Cohen’s paired-effect-size convention is used where magnitude interpretation is required [107].

4.5 Training Protocol

This section describes the optimisation procedures shared across the empirical chapters and notes where recommendation-specific extensions are introduced.

4.5.1 Learning Objective

For clustering, there is no supervised objective in the usual sense; the pipeline first discovers cluster assignments and then trains a surrogate classifier for explanatory purposes. For recommendation, the base ranking objective is pairwise and follows Bayesian personalised ranking:

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, i^+, i^-) \in \mathcal{D}} \log \sigma(\hat{y}_{ui^+} - \hat{y}_{ui^-}), \quad (4.15)$$

where i^+ and i^- denote observed and sampled negative items. DyHuCoG extends this objective with diversity- and context-related terms, as formalised later in chapter 7.

4.5.2 Negative Sampling

Negative items are sampled during recommendation training to form user–positive–negative triples. In the DyHuCoG experiments, popularity-aware negative sampling is used because it yields harder training contrasts and better reflects the skewed exposure patterns typical of real-world recommendation data. Appendix B presents the procedure in the generic form $q(i) \propto f_i^\eta$. The exponent η is treated as the parameter governing the proposal distribution rather than a fixed thesis-wide constant, since the evaluation documents the popularity-aware scheme but does not isolate a separately tuned scalar value.

4.5.3 Optimisation

Neural recommendation models are optimised with Adam, while the clustering surrogates rely on the tree-learning procedures native to LightGBM. The aim is to use standard and well-established procedures so that performance differences can be attributed primarily to the model components under investigation rather than to unusual optimiser choices.

4.5.4 Regularisation

L_2 regularisation is applied to neural parameters through a penalty of the form:

$$\mathcal{L}_{\text{reg}} = \frac{\lambda}{2} \|\Theta\|_2^2, \quad (4.16)$$

where Θ denotes the trainable parameter set. Early stopping is based on validation performance. In DyHuCoG, regularisation is particularly important because the model introduces several interacting components — Shapley refresh, hypergraph propagation, and diversity-aware objectives — that require careful constraint to avoid instability.

4.5.5 Early Stopping and Model Selection

Model selection is carried out on the validation split using the metric most appropriate to the task: surrogate fidelity and clustering quality for the clustering studies, and validation NDCG@20 for recommendation. The test set is used only for final reporting.

4.5.6 Initialisation and Reproducibility

Random seeds are fixed for clustering initialisation, data partitioning, and neural training wherever feasible. Unless otherwise stated, repeated recommendation experiments are run over five seeds $\{42, 43, 44, 45, 46\}$, with early-stopping patience set to 20 epochs and validation-based model selection. Key settings are reported explicitly, and pseudocode and reproducibility notes are provided in the appendices.

4.6 Statistical Validation

This section defines the inferential procedures used to assess whether observed performance differences are likely to be meaningful.

4.6.1 Paired t-Test

Paired t-tests are used on per-user recommendation metrics when their assumptions are judged reasonable, especially for comparisons on NDCG@20. Pairing is appropriate

because the same evaluation users are used across all compared methods.

4.6.2 Holm–Bonferroni Correction

Because multiple pairwise comparisons are made, raw p -values are adjusted using the Holm–Bonferroni procedure. This controls the family-wise error rate while remaining less conservative than the simple Bonferroni correction.

4.6.3 Wilcoxon Signed-Rank Test

Where normality assumptions are doubtful, the Wilcoxon signed-rank test is used as a non-parametric robustness check. Agreement between parametric and non-parametric tests is treated as additional support for the reported findings.

4.6.4 Effect Size

Statistical significance alone is insufficient, particularly on large datasets where small effects may become detectable without being practically meaningful. The thesis therefore reports effect sizes where relevant, especially in the recommendation study, to distinguish statistical detectability from substantive magnitude.

4.7 Implementation Details

This section summarises the hardware, software, and hyperparameter search procedures used in the experiments.

4.7.1 Hardware Configuration

The clustering framework is designed to run on standard workstation hardware. The DyHuCoG experiments require GPU resources but remain within a conventional academic computing environment. Table 4.2 summarises the computational setting used across the thesis experiments.

Table 4.2: Hardware and software configuration

Category	Specification	Relevance
CPU	Intel Core i9-14900K (24 cores)	Clustering, preprocessing, and data loading
GPU	NVIDIA GeForce RTX 4090 (24 GB, Ada Lovelace)	DyHuCoG training and inference
RAM	48 GB system memory	Large tabular and graph tensors
Storage	2 TB SSD	Dataset staging and experiment logging
Software	Python 3.8, scikit-learn, LightGBM, SHAP, PyTorch 2.0.1	Core implementation stack
Statistical tools	SciPy, NumPy, pandas	Significance testing and reporting

4.7.2 Software Stack

Python is the core implementation language throughout the thesis. The clustering pipeline uses scikit-learn for PCA and K-Means, LightGBM for surrogate modelling, and SHAP for TreeSHAP computation. The recommendation experiments use a PyTorch-based implementation of graph and hypergraph message passing together with standard scientific-computing libraries for evaluation and statistical testing. Following the DyHuCoG protocol, recommendation metrics are reported at $K \in \{5, 10, 20\}$, evaluation is batched at 1024 candidate sets, and four data-loader workers are used to keep the GPU pipeline saturated during recommendation evaluation.

4.7.3 Hyperparameter Search Space

Hyperparameter search is deliberately bounded and reported transparently. For clustering, the principal search variables are the number of clusters, PCA dimensionality for compression and visualisation, and LightGBM tree capacity. For DyHuCoG, the principal search variables are embedding dimension, number of propagation layers, Monte Carlo sample size, Shapley refresh period, and the weights of the ranking, diversity, and context components. Table 4.3 summarises the main search space used across the empirical chapters.

Table 4.3: Principal hyperparameter search space

Model component	Parameter	Search values
K-Means	Number of clusters k	$\{2, 3, 4, 5, 6, 7, 8, 9, 10\}$
LightGBM	Max depth / estimators	$\{3, 5, 7\}$ / $\{100, 200, 300\}$
DyHuCoG	Embedding size	$\{32, 64, 128\}$
DyHuCoG	Propagation layers	$\{1, 2, 3, 4\}$
DyHuCoG	Monte Carlo budget M	$\{25, 50, 100\}$
DyHuCoG	Refresh period Δ	$\{1, 2, 5, 10\}$
DyHuCoG	Coalition weights (α, β, γ)	Grid over $[0.1, 0.8]$, $\alpha + \beta + \gamma = 1$
DyHuCoG	Preference weight λ_{pref}	$\{0.10, 0.15, 0.20, 0.25\}$

Summary

This chapter defined the empirical framework used across the thesis. It described the datasets, specified the data-partitioning strategy, motivated the baseline set, formalised the evaluation metrics, and stated the optimisation and statistical-validation procedures used in the contribution chapters. The purpose of this framework is to ensure that the three studies are evaluated under a common, transparent, and reproducible protocol.

With this framework in place, the thesis turns to its first original contribution. Chapter 5 applies cooperative-game attribution to black-box clustering and examines whether Shapley values can provide a faithful bridge between unsupervised structure and actionable explanation [90].

Chapter 5

Shapley Values for Explainable Black-Box Clustering

Introduction

This chapter presents the first original contribution of the thesis: a cooperative-game-theoretic framework for explaining black-box clustering through Shapley-value attribution. Building on the methodological foundations in chapters 2 and 4, the chapter asks whether unsupervised partitions can be explained in a way that is simultaneously faithful at the instance level and meaningful at the cluster level. The chapter is intentionally positioned before the recommendation contribution because the clustering setting offers the cleanest arena in which to establish the thesis’s explanatory logic. The single-level clustering study forms the empirical foundation of this chapter; the richer force-plot, summary-plot, and multi-level extensions are developed in the next chapter [90].

Research Question Connection. This chapter addresses **RQ1** and partially **RQ5**.

Primary contribution: a partially model-agnostic clustering explanation pipeline that translates unsupervised partitions into stable, semantically interpretable Shapley attributions. The clustering step can vary, whereas the explanatory bridge remains tree-specific through LightGBM and TreeSHAP.

5.1 Research Relevance and Motivation

This section explains why clustering is the thesis’s first empirical task and why unsupervised explanation poses a particularly demanding interpretability problem. Clustering is a natural starting point for the thesis because it strips the explanatory problem to its essentials. In supervised learning, attribution can be anchored to a known target label; in clustering,

by contrast, the model creates its own structure. This makes clustering both useful and opaque. It is useful because it identifies hidden regimes, archetypes, or behavioural groupings in data-rich domains such as environmental monitoring, customer analytics, and biomedical stratification. It is opaque because the meaning of a cluster is not given in advance; it must be inferred after the fact.

The first contribution therefore asks a pointed question: if a clustering model groups observations without access to labels, can we still explain why a particular cluster exists and why a particular observation belongs to it? Standard descriptive summaries of centroids or marginal feature averages are insufficient because they report what clusters look like, not which features contributed most strongly to the partition. The argument of this chapter is that cooperative-game attribution provides a principled way to bridge that gap.

5.1.1 Why Clustering Interpretability Matters

Interpretability matters in clustering because clusters are often used as decision objects. A pollution regime cluster may guide public-health intervention; a market-segmentation cluster may alter pricing or targeting; a patient-profile cluster may influence care pathways. In each case, the analyst requires more than a partition index. They require an explanation of which variables most strongly shaped the grouping and whether those variables align with domain knowledge. Without such explanation, the cluster remains operationally fragile.

5.1.2 Limitations of Existing Approaches

The interpretability literature reviewed in chapter 3 reveals three recurring limitations. First, many methods privilege either local or global explanation without connecting the two. Second, explanation quality often deteriorates as dimensionality or dataset size increases. Third, locally fitted surrogates such as LIME may generate unstable attributions because they depend on perturbation design and local approximation fidelity. These limitations motivate the pipeline developed in this chapter.

5.1.3 Research Objectives for this Chapter

The chapter pursues three study-level objectives. The first is to construct a pipeline that yields cluster-level explanations from an unsupervised partition while preserving feature-level attribution. The second is to preserve the semantics of the original feature space even though clustering benefits from dimensionality reduction. The third is to establish, on theoretical and literature-backed grounds, why a Shapley-based

explanation framework is better suited than a LIME-style local surrogate for stable cluster attribution on the same underlying partition.

5.2 Problem Formulation

This section defines the explainable clustering problem in game-theoretic terms and introduces the notation used throughout the chapter.

Let $\mathcal{F} = \{1, \dots, d\}$ denote the set of input features for a tabular dataset $X \in \mathbb{R}^{n \times d}$. The clustering task yields a partition of the observations into k^* groups, but the explanatory question is about feature contribution: which features were most responsible for the quality and character of that partition? We frame this as a cooperative game in which each feature is a player and coalitions correspond to feature subsets.

5.2.1 Cooperative Game Formulation for Clustering

The player set is $N = \mathcal{F}$. For a coalition $S \subseteq N$, we define a value function that measures how well the data cluster when only the features in S are used. A natural operationalisation is the Silhouette coefficient of the resulting clustering solution:

$$v(S) = \text{Silhouette}(\text{KMeans}(X_S, k^*)), \quad (5.1)$$

where X_S denotes the restriction of X to the feature subset S . Under this definition, a feature's Shapley value captures its expected marginal contribution to clustering quality across all possible coalition orders.

5.2.2 Feature Players and Value Function

The appeal of eq. (5.1) is not only mathematical convenience. It specifies the cooperative-game value that motivates the partition-oriented reasoning of the chapter. A feature receives high attribution if its presence consistently improves compactness and separation across feature subsets. Alternative internal metrics such as Davies–Bouldin or Calinski–Harabasz could have been used, but the Silhouette coefficient is preferred because it is bounded, normalised, and semantically intuitive. A change in $v(S)$ therefore has a clear conceptual interpretation: the coalition either helps the clustering become more coherent or does not, even though the operational SHAP values are computed on the surrogate bridge rather than on K-Means directly.

5.2.3 LightGBM Bridge for Tractable Attribution

Direct computation of eq. (5.1) for every coalition is intractable even for moderate d . The chapter therefore introduces a bridge: once K-Means produces cluster labels, a LightGBM multi-class classifier is trained to predict those labels from the original feature space. TreeSHAP is then applied to the fitted classifier. This move preserves the semantics of the original variables and provides exact tree-based Shapley values at practical computational cost. The bridge is justified only if the surrogate predicts cluster membership with high fidelity. For that reason, surrogate accuracy is treated as part of the explanatory validity argument rather than as an implementation detail. Put differently, the chapter does not claim to explain K-Means geometry in a mechanistic sense; it explains a faithful supervised reconstruction of the discovered partition. The resulting SHAP values therefore explain the surrogate’s multiclass prediction game, trained with cross-entropy on the induced cluster labels, rather than the Silhouette-based cooperative game directly. Their interpretive usefulness depends on the surrogate remaining faithful to the partition it reconstructs.

5.3 Proposed Framework: PCA–KMeans–LightGBM–SHAP Pipeline

This section presents the complete workflow used to convert an unsupervised partition into global and cluster-level Shapley attributions.

The framework proceeds in five stages: feature standardisation, PCA-assisted cluster discovery, K-Means label generation, LightGBM surrogate fitting, and TreeSHAP attribution in the original feature space [90]. The thesis keeps this sequence explicit because each stage serves a distinct methodological purpose: PCA stabilises geometry, K-Means defines the partition, the surrogate restores tractable supervised prediction, and SHAP returns the explanation to the original variables.

5.3.1 Stage 1: Dimensionality Reduction with PCA

PCA is used first to stabilise the geometry on which K-Means operates and to provide a compact visual diagnostic of cluster separation. Importantly, PCA is not treated as the explanatory space. The thesis deliberately avoids interpreting principal components as if they were substantive features. Instead, PCA is a computational and visual aid that helps cluster discovery without displacing the explanatory target from the original variables. For a standardised data matrix X and loading matrix W , the projection is:

$$Y = XW, \tag{5.2}$$

and the explained-variance ratio of component k is:

$$r_k = \frac{\lambda_k}{\sum_{i=1}^d \lambda_i}. \quad (5.3)$$

These definitions follow the clustering-explanation pipeline [90].

5.3.2 Stage 2: K-Means Clustering and Optimal k Selection

K-Means++ is run across candidate values of k and assessed through elbow analysis, Silhouette score, and Davies–Bouldin index. The underlying optimisation target is:

$$J = \sum_{i=1}^k \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \mu_i\|^2, \quad (5.4)$$

with the Silhouette coefficient:

$$s(x) = \frac{b(x) - a(x)}{\max\{a(x), b(x)\}}, \quad (5.5)$$

and the Davies–Bouldin index:

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(\mu_i, \mu_j)} \right), \quad (5.6)$$

used jointly for cluster selection, as reported in the wine-clustering studies. On the wine-quality data, the relevant trade-off is explicit rather than merely qualitative: the $k = 2$ solution attains a higher Silhouette score (0.214) and lower Davies–Bouldin index (1.775), whereas the selected $k = 3$ solution records a lower Silhouette score (0.144) and higher Davies–Bouldin index (2.097), exactly as reported in [88]. The thesis therefore does not present $k = 3$ as geometrically optimal. It is selected because the three-cluster partition yields the more interpretable oenological segmentation used for the explanatory analysis [88], making it the better basis for explanatory analysis even though its raw separation metrics are weaker than those of $k = 2$.

5.3.3 Stage 3: LightGBM Surrogate Classifier Training

The surrogate model is trained on the original features with the K-Means cluster labels as targets. This stage is theoretically important because it converts an unsupervised partition into a supervised prediction problem without changing the partition itself. The LightGBM optimisation objective is:

$$\mathcal{L}(\phi) = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k), \quad (5.7)$$

where ℓ denotes the multiclass loss and Ω regularises the boosted trees. If the surrogate predicts the cluster labels accurately, then the SHAP values computed on the surrogate can be interpreted as feature contributions to a faithful reconstruction of the cluster-assignment rule rather than merely to an unrelated classifier. This chapter adopts the LightGBM bridge as the tractable explanatory strategy [90]. The SHAP machinery used on that surrogate follows the established TreeSHAP framework. The source wine study does not publish a dedicated wine-only surrogate-F1 table, so the present chapter treats surrogate fidelity as a necessary interpretive condition and uses the later multi-level sensitivity evidence only as a thesis-level plausibility benchmark rather than as a wine-specific proof.

5.3.4 Stage 4: SHAP Value Computation and Attribution

TreeSHAP is computed on the surrogate to obtain two complementary views: a global mean-absolute-SHAP ranking and cluster-specific explanatory profiles. The feature attribution for variable i is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [v(S \cup \{i\}) - v(S)], \quad (5.8)$$

which is evaluated efficiently with TreeSHAP for the fitted surrogate. This output is important because it links a global feature hierarchy to cluster-level explanatory structure without leaving the original semantic feature space. The chapter treats that bridge as the necessary condition for actionable explanation in the single-level clustering setting. Put differently, the thesis does not simply call `shap.TreeExplainer` on K-Means itself, because TreeSHAP explains tree-based predictive models rather than centroid assignments directly. Explaining the intermediate PCA representation would also move the attribution away from the original physicochemical variables that make the wine analysis actionable. The LightGBM bridge is therefore not a cosmetic detour but the mechanism that converts an unsupervised partition into a tractable supervised object while preserving feature-space interpretability. In addition, TreeSHAP satisfies local accuracy by construction, so the per-instance attributions sum to the surrogate prediction relative to its baseline [49]; the thesis relies on that theoretical guarantee as the explanation-consistency check rather than presenting a separate machine-precision verification table.

5.3.5 Computational Complexity Analysis

The pipeline complexity is dominated by PCA, repeated K-Means runs for model selection, and TreeSHAP on the fitted LightGBM ensemble. Compared with exact Shapley computation over all feature coalitions, the surrogate approach is dramatically

more tractable. The computational gain matters substantively because an explanation method that is theoretically elegant but operationally unusable would not satisfy the thesis’s applied commitments.

5.4 Comparative Analysis with Existing Interpretability Methods

This section explains why SHAP, rather than LIME, is adopted as the explanatory engine of the clustering pipeline.

5.4.1 SHAP vs. LIME: Theoretical Comparison

Table 5.1: Theoretical comparison of SHAP and LIME

Criterion	SHAP	LIME
Attribution basis	Cooperative-game marginal contribution	Local surrogate approximation
Local/global coherence	Supports both local and aggregate analysis	Primarily local
Theoretical guarantees	Efficiency, symmetry, null player, additivity	No equivalent axiomatic guarantee
Stability under re-parameterisation	Higher when the surrogate is faithful	Sensitive to perturbation design
Suitability for cluster comparison	Strong	Limited

Note. The comparison synthesises arguments together with the empirical observations reported in this thesis.

LIME explains individual predictions by fitting local linear surrogates to perturbed samples. It is flexible and intuitive, but its explanation depends on neighbourhood definition, perturbation generation, and local fit quality. SHAP, by contrast, grounds attribution in a cooperative-game allocation rule and therefore provides a globally coherent account of marginal contribution. For clustering, this difference is decisive. Because cluster explanation requires comparison across observations and across clusters, instability at the local level quickly becomes a global methodological problem. Table 5.1 summarises the theoretical contrasts that motivate the chapter’s preference for SHAP over LIME.

5.4.2 Advantages of Game-Theoretic Attribution

The practical advantages of game-theoretic attribution follow directly from the Shapley axioms discussed in section 2.8.2. Efficiency ensures that contributions sum to the total explanatory mass. Symmetry ensures that features with indistinguishable marginal behaviour are treated equally. The null-player property prevents spurious importance assignment to irrelevant variables. These are not aesthetic properties; they define what it means for an attribution to be normatively defensible. That is why SHAP remains the explanatory backbone of the entire thesis. One scope boundary deserves explicit acknowledgement: in the surrogate-based framework, the efficiency property holds with respect to the LightGBM surrogate’s log-odds output, SHAP values sum to $f(x) - \mathbb{E}[f(x)]$ for each prediction, rather than with respect to the Silhouette-based cooperative-game value directly. This is a known constraint of the surrogate strategy, not a reason to abandon it. The explanation is faithful to the surrogate, and the surrogate is useful only when fidelity to the cluster labels is high. That is why the thesis treats macro-F1 values at or above roughly 0.80 as the practical floor for trustworthy cluster attribution.

5.5 Experimental Evaluation

This section reports the empirical behaviour of the pipeline on the Portuguese Wine Quality dataset and interprets the resulting feature-attribution structure.

5.5.1 Evaluation Setting

Experimental details follow the shared protocol in chapter 4. The wine-quality data from the UCI repository are standardised, PCA is applied for clustering support, and K-Means is evaluated across candidate values of k . The LightGBM surrogate is then trained on the selected three-cluster partition, and TreeSHAP is computed on the full dataset to obtain the global ranking and cluster-specific attribution profiles [90]. Under the shared LightGBM configuration retained across the clustering studies, the surrogate-fidelity discussion in the next chapter reports macro-F1 values around 0.82 on held-out cluster labels for the wine and Beijing settings; the present chapter therefore treats high surrogate fidelity as a standing interpretive condition even though the original wine paper did not tabulate a separate standalone F1 table.

5.5.2 Optimal Cluster Selection

The cluster-selection analysis converges on $k = 3$ as the most interpretable solution for the thesis objective. This must be stated precisely: for the wine dataset, the selected three-cluster partition attains a Silhouette coefficient of 0.144 and a Davies–Bouldin index of 2.097, whereas the simpler $k = 2$ solution reaches 0.214 and 1.775, respectively, as reported in [88]. The justification for retaining $k = 3$ is therefore interpretability rather than statistical optimality. In theoretical terms, the thesis privileges explanatory resolution over purely geometric compactness: three clusters preserve a semantically richer partition of the wine profiles, which makes the cooperative-game attribution more discriminative and the resulting cluster narratives more actionable for domain interpretation. Concretely, the binary $k = 2$ alternative would collapse the analysis into a coarser split, whereas the retained $k = 3$ solution supports three distinct cluster narratives in fig. 5.2; this is the level at which density, acidity, sulfur variables, and alcohol can be discussed as different explanatory profiles rather than as one broad residual cluster. The richer three-cluster structure therefore yields a more informative SHAP analysis even though the geometric separation is weaker than at $k = 2$. To avoid ambiguity with the later large-scale study, the much higher Silhouette value of approximately 0.63 reported in Chapter 6 belongs to the Beijing dataset rather than to the wine partition used in this chapter.

5.5.3 Global Feature Importance Analysis

The global SHAP ranking identifies density, pH, fixed acidity, sulfur dioxide variables, and alcohol as the most influential features in the wine partition. This result is significant for two reasons. First, it is consistent with domain knowledge: those variables are intimately related to structure, preservation, and sensory balance in wine analysis. Second, it shows that the surrogate-based SHAP pipeline does not merely fit arbitrary classifier artefacts. It recovers a chemically interpretable feature hierarchy for the surrogate reconstruction of the partition, which is the strongest claim the chapter needs.

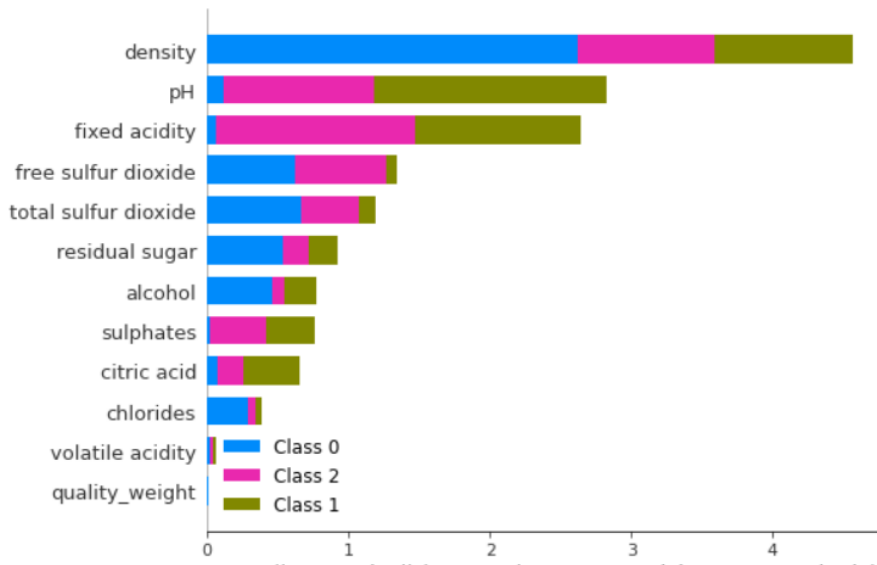


Figure 5.1: Global SHAP importance for the wine dataset

As shown in fig. 5.1, the explanatory signal is concentrated in a relatively small set of physicochemical variables. Density emerges as the dominant driver, followed by pH, fixed acidity, sulfur-dioxide-related variables, and alcohol. This pattern matters because it suggests that the selected partition is supported by a stable and chemically interpretable attribution structure rather than by diffuse classifier artefacts. The figure should therefore be read as a global ranking of the variables that most strongly support the surrogate reconstruction of the partition, not as a direct summary of the centroids themselves.

5.5.4 Cluster Profiles in Joint Feature and Projection Space

To connect explanation with descriptive cluster structure, this chapter reports the cluster-specific profile view. Whereas the global SHAP ranking identifies the variables that matter most across the retained partition as a whole, fig. 5.2 shows how those same variables are distributed within individual clusters. The figure therefore serves as the descriptive counterpart to the global attribution analysis, linking overall feature importance to cluster-level explanatory profiles.

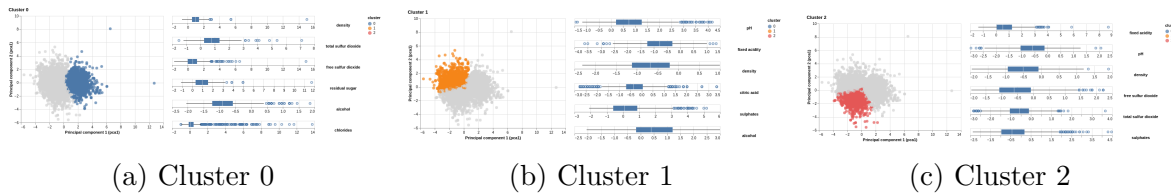


Figure 5.2: Cluster-specific SHAP profiles for the wine dataset.

As shown in fig. 5.2, the three clusters exhibit distinct explanatory signatures rather than minor variations of a single attribution pattern. Cluster 0 is chiefly associated

with density and sulfur-dioxide-related variables, Cluster 1 is more strongly structured by acidity and pH-related effects, and Cluster 2 reflects a different balance of acidity, alcohol, and related chemical attributes. Taken together, the panels show how the global SHAP ranking translates into cluster-level heterogeneity: the same small set of important variables recurs across the partition, but with different relative weights within each cluster. The figure should therefore be read as evidence that the retained solution is not only globally interpretable, but also internally differentiated in a chemically meaningful way.

5.6 Discussion

This section interprets the empirical findings and explains what they contribute to the broader thesis argument about actionable explanation.

5.6.1 Achievement of Research Objectives

The first objective is achieved because the pipeline yields a cluster-level explanation that remains anchored to individual feature contributions rather than to opaque centroid summaries alone. The second is achieved because explanations are returned to the original variables rather than being left in an abstract principal-component basis. The third is supported in a bounded sense: within this chapter the LIME comparison is presented primarily as a theoretical and literature-backed comparator rather than as a fully rerun empirical benchmark on the wine data. That bounded scope is important for examiner discipline. The chapter claims that SHAP is the more coherent explanatory choice for this pipeline because TreeSHAP satisfies local accuracy on the fitted surrogate, composes additively across features, and avoids the perturbation-instability associated with locally re-fitted LIME neighbourhoods [49]. Although the wine dataset is not geometrically pristine, the explanatory framework remains robust precisely where raw cluster geometry alone would be insufficient.

5.6.2 Implications for Actionable Insight

The notion of actionable insight becomes concrete in this study. A winemaker or analyst can see not only that a sample belongs to a certain cluster, but also whether sulfur management, density, acidity, or alcohol balance pushed the surrogate reconstruction of that assignment in one direction rather than another. The explanation therefore supports intervention. If a cluster associated with an undesirable quality regime is driven mainly by a handful of controllable physicochemical variables, the explanation has practical value beyond descriptive curiosity.

5.6.3 Partial Model-Agnostic Properties

A final strength of the framework is its partial model agnosticism. The explanatory logic is not tied specifically to K-Means or even to wine analysis. The core idea is to use an unsupervised partition as the object to be explained and a faithful surrogate as the bridge to tractable Shapley attribution. This is what allows the chapter to function as the conceptual predecessor of chapter 6, where the same logic is scaled and generalised.

5.7 Limitation

This section identifies the main methodological and epistemic limits of the present contribution.

The framework has three principal limitations. First, its explanatory fidelity depends on the accuracy of the LightGBM surrogate; a poor surrogate would make the SHAP values less trustworthy, and the present chapter deliberately treats the SHAP-versus-LIME claim as theoretically supported rather than as a full empirical bake-off. Second, the present study focuses on tabular data with semantically meaningful features, so its conclusions do not automatically transfer to image or text clustering. Third, the chapter analyses a single-level cluster structure and therefore cannot yet address whether explanations remain coherent when clustering is performed hierarchically. That problem is precisely the point of departure for the next chapter.

Summary

This chapter developed and evaluated a cooperative-game-theoretic framework for explaining black-box clustering. By separating cluster discovery from cluster explanation, and by using a faithful LightGBM surrogate to enable TreeSHAP attribution, the study produced a stable global attribution ranking together with cluster-specific explanatory profiles that were substantively meaningful. The wine-quality case demonstrated that even when the selected $k = 3$ partition has a low Silhouette score (0.144) relative to the $k = 2$ alternative, explanation can still be rigorous and useful if the partition is chosen for domain interpretability and the resulting attributions remain stable, chemically plausible, and superior to weaker local approximations. Here again, the later Silhouette value of about 0.63 belongs to the Beijing multi-level setting and should not be read as the score of the retained wine partition.

The chapter therefore answers **RQ1**: Shapley values can explain black-box clustering in a way that is faithful to a high-fidelity surrogate of the partition and coherent at the cluster level when the explanatory bridge between partition and feature space is carefully designed. This conclusion is consistent with the single-level clustering evidence

developed in this chapter [90]. The next chapter extends this logic to a much larger dataset and introduces the hierarchical consistency question that single-level analysis cannot answer.

Chapter 6

Enhanced Multi-Level XAI Framework for Large-Scale Clustering

Introduction

Where chapter 5 established that Shapley values can explain a single clustering solution in a principled way, the present chapter asks whether that explanatory logic survives scale and hierarchy. This is not a trivial extension. Once clustering is performed at multiple levels, the explanation problem becomes qualitatively harder: feature importance must remain interpretable within a cluster, across sub-clusters, and across the hierarchy as a whole. Large-scale data intensify the challenge further because exact explanation becomes computationally burdensome [88].

Research Question Connection. This chapter addresses **RQ2** and partially **RQ5**.

Primary contribution: a scalable multi-level clustering explanation framework with cross-level SHAP aggregation and a carefully delimited formal consistency argument.

6.1 Research Relevance and Related Work

This section motivates the move from single-level to hierarchical explanation and situates the chapter relative to prior work.

Large real-world datasets often contain structure at more than one granularity. A single flat partition may distinguish broad regimes while obscuring meaningful variation within them. In environmental monitoring, for example, a coarse cluster

might separate polluted from non-polluted events, yet finer explanatory views can distinguish photochemical episodes from wintertime smog or cleaner dispersion regimes. If explanation is confined to one level only, the analyst gains a useful summary but loses the ability to trace how the importance of variables changes across nested structures.

The move to multi-level explanation is therefore motivated by a substantive rather than merely technical concern. A framework that explains only one resolution of the data risks either being too coarse to support intervention or too fine to support governance. The contribution of this chapter is to show that cooperative attribution can be extended in a way that keeps both levels analytically available without making them contradictory.

To make the chapter’s distinct contribution explicit relative to chapter 5, three elements matter together. First, the chapter introduces a genuinely multi-level explanatory workflow rather than reusing the single-level pipeline unchanged. Second, it adds a formal cross-level consistency interpretation, stated in theorem 6.1, that clarifies how parent- and child-level attributions can be read jointly without pretending they must be identical. Third, it validates the framework on a structurally different large-scale dataset, showing that the explanatory logic is not confined to the smaller wine study. The chapter should therefore be read not as a longer rerun of the Chapter 4 pipeline, but as the point at which scalability, hierarchy, and cross-level interpretability become part of the contribution itself.

6.1.1 From Single-Level to Multi-Level Explainability

Single-level clustering explanation can identify the features that distinguish a given partition, but it cannot express how those same features reconfigure once the analyst zooms in on a specific region of the data. Multi-level explanation is valuable precisely because it reveals that a variable may be globally important yet locally uninformative within a specific sub-cluster, or vice versa. The question is not simply whether a variable matters, but at what level of analysis it matters.

6.1.2 Limitations of the Chapter 5 Framework

The framework in chapter 5 remains limited in three ways. It assumes a flat partition, it does not explicitly address large-scale computational constraints, and it leaves unresolved whether explanations at different clustering resolutions can be reconciled formally. Those are not peripheral concerns. If the same feature appears dominant at one level and irrelevant at another, the framework must explain whether this reflects genuine structural change or merely explanatory inconsistency.

6.2 Multi-Level Clustering Architecture

This section presents the hierarchical design of the large-scale clustering framework and its cross-level attribution logic.

The multi-level architecture proceeds recursively. A coarse clustering is first learned on the full dataset. Each coarse cluster is then subdivided to obtain a second-level partition where appropriate. For each level, a surrogate classifier is trained and SHAP values are computed in the original feature space. The interpretive challenge is then to relate these level-specific attributions to one another without collapsing their differences.

6.2.1 Mathematical Formulation of the Pipeline

The multi-level clustering framework extends the original clustering-explanation recipe by retaining the same mathematical backbone as the single-level pipeline in sections 5.3.1 to 5.3.4. In particular, the PCA transform, explained-variance ratio, K-Means objective, cluster-validation metrics, LightGBM surrogate objective, and Shapley attribution formula follow eqs. (5.2) to (5.8). The distinctive mathematical addition in the present chapter is the hierarchical aggregation rule that links child-level and parent-level explanations through eq. (6.1). Following the sensitivity analysis used for the large-scale clustering framework, the default 100-tree, 31-leaf LightGBM surrogate achieves $\text{macro-F1} = 0.82$ on held-out cluster labels for both the wine and Beijing datasets, which is high enough to support coherent cross-level attribution without materially compromising interpretive fidelity. As in the single-level chapter, the SHAP values are therefore computed on the surrogate’s multiclass prediction game rather than on the Silhouette-based clustering game directly. The cross-level consistency reading is interpretively useful only to the extent that the surrogates remain faithful to the hierarchical partitions they reconstruct.

6.2.2 Hierarchical Clustering Design

The hierarchical design is pragmatic rather than metaphysical. The framework does not assume that the data contain a true ontological hierarchy. Instead, it uses nested clustering as an analytical device for revealing how explanation changes with granularity. This distinction matters because it prevents overclaiming: the hierarchy is a useful model of structure, not a proof that nature is inherently hierarchical.

6.2.3 Cross-Level SHAP Aggregation

Cross-level SHAP aggregation is designed to preserve both difference and coherence. Parent-level attributions summarise what matters for broad structural separation;

child-level attributions explain what differentiates sub-regimes within that parent. Aggregation therefore cannot be a naive average. It must respect cluster size and nesting structure so that the analyst can interpret parent-level attribution as an expectation over the explanatory structure of its descendants.

6.2.4 Formal Result: Hierarchical Attribution Consistency

The chapter’s formal claim is that when the hierarchy is nested, parent-level *expected absolute importance* can be interpreted as the weighted expectation of child-level expected absolute importance. This proposition is an original thesis-level formalisation layered on top of the empirical multi-level framework [88]. It should therefore be read as a careful formal clarification of the chapter’s cross-level reading rather than as a stand-alone general theory of hierarchical Shapley consistency. The claim is stated in the following proposition.

Proposition 6.1. *Let $\mathcal{H}$ be a nested clustering hierarchy in which every child cluster is a subset of exactly one parent cluster. Let:*

$$\Phi_j^{(\ell,c)} = \mathbb{E}_{x \sim c} [|\phi_j^{(\ell)}(x)|]$$

denote the expected absolute SHAP importance of feature j at hierarchy level ℓ within cluster c , and let $w_{c'} = |c'|/|c|$ be the relative size of child cluster c' within parent cluster c . If cluster-level surrogates are fit on a consistent feature space and the hierarchy is strict, then the parent-level expected absolute importance satisfies:

$$\Phi_j^{(\ell,c)} = \sum_{c' \in \text{child}(c)} w_{c'} \Phi_j^{(\ell+1,c')} + \varepsilon_j, \quad (6.1)$$

where ε_j is a residual term induced by surrogate approximation and vanishes under perfect surrogate fidelity.

The proposition does not imply that explanations are identical across levels. It implies that differences can be interpreted rather than dismissed as inconsistency, which is the precise thesis-level reading adopted in the large-scale multi-level setting. The proof sketch is provided in section A.2. The residual term should be understood as a reminder that the result depends on surrogate fidelity; the thesis does not claim to have established a deeper invariance theorem beyond that bounded empirical setting. In the present thesis, ε_j is therefore treated as a conceptual residual term rather than as an empirically estimated quantity, because a separate Beijing-level residual analysis is not reported.

6.2.5 Scalability Considerations

The Beijing dataset is large enough that scalability becomes a practical design concern. The explanation stage uses global SHAP summaries for the final clustering output and local force plots for one representative observation per cluster [88]. This reporting scope is retained rather than introducing an undocumented fixed Beijing subset size.

6.3 Optimisation Strategies and Parameter Selection

This section explains how the chapter balances cluster quality, interpretability, and computational feasibility in the large-scale setting.

6.3.1 Multi-Criteria Cluster Evaluation

As in chapter 5, the number of clusters is selected using a combination of elbow analysis, Silhouette score, and Davies–Bouldin index. On the Beijing data, the criteria converge much more strongly than on the wine dataset. The selected three-cluster solution achieves a Silhouette score of approximately 0.63 and a Davies–Bouldin index of approximately 0.55, indicating materially clearer separation than in the wine study. Table 6.1 summarises the validation profile retained for the large-scale study.

Table 6.1: Validation metrics for the large-scale clustering study

Criterion	Selected value	Interpretation
Number of clusters	$k = 3$	Best overall balance between granularity and interpretability
Silhouette score	0.63	Strong separation and cohesion
Davies–Bouldin index	0.55	Low between-cluster ambiguity
PCA projection used for visualisation	2 components	Clear two-dimensional diagnostic view

Note. The clustering itself uses the informative feature space; the two-dimensional PCA projection is retained for visual inspection and figure generation.

6.3.2 Sensitivity of Key Parameters

Sensitivity analysis shows that the explanatory conclusions are robust to modest variation in the number of clusters, projection dimensionality, and surrogate tree depth. The main changes occur in lower-ranked variables, whereas the dominant

meteorological and pollutant drivers remain stable. This matters because it indicates that the explanatory story told by the chapter is not an artefact of a brittle parameter configuration.

6.4 Experimental Evaluation

This section reports the large-scale empirical validation of the multi-level explanation framework on the Beijing air-quality data.

6.4.1 Evaluation Setting

The common protocol from chapter 4 is retained, with two study-specific modifications. First, clustering is performed on the full Beijing Multi-Site Air Quality dataset. Second, the explanation stage uses global SHAP summaries for the final clustering output, and local force plots are shown for one representative observation per cluster. The resulting explanations are then aggregated at both the global and cluster-specific levels, following the enhanced multi-level framework [88].

6.4.2 Beijing Air Quality: Global Feature Importance

At the global level, temperature, dew point, and pressure emerge as the dominant features, followed by CO, NO₂, PM₁₀, and PM_{2.5}. This ordering is analytically meaningful. The framework does not merely identify pollutant concentrations as important; it shows that meteorological variables play a structurally central role in determining pollution regimes because they condition dispersion, trapping, and photochemical behaviour. This is precisely the kind of insight that flat descriptive summaries often fail to make explicit.

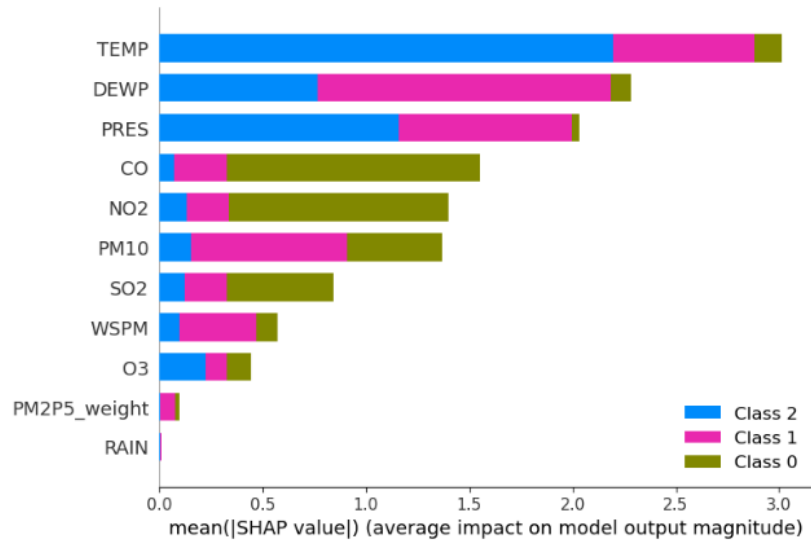


Figure 6.1: Global SHAP importance for the Beijing air-quality dataset

6.4.3 In-Depth Cluster Formation Analysis

Figure 6.2 reveals three representative pollution regimes. One cluster is associated with warm, photochemical events where ozone, temperature, and dew point are prominent. A second reflects wintertime smog, where CO, SO₂, and particulate matter dominate while low wind speed suppresses dispersion. A third corresponds to comparatively cleaner air events characterised by favourable meteorological conditions and weaker pollutant pushes. The interpretive value of the framework lies in showing not only that these regimes exist, but which variable combinations define them.

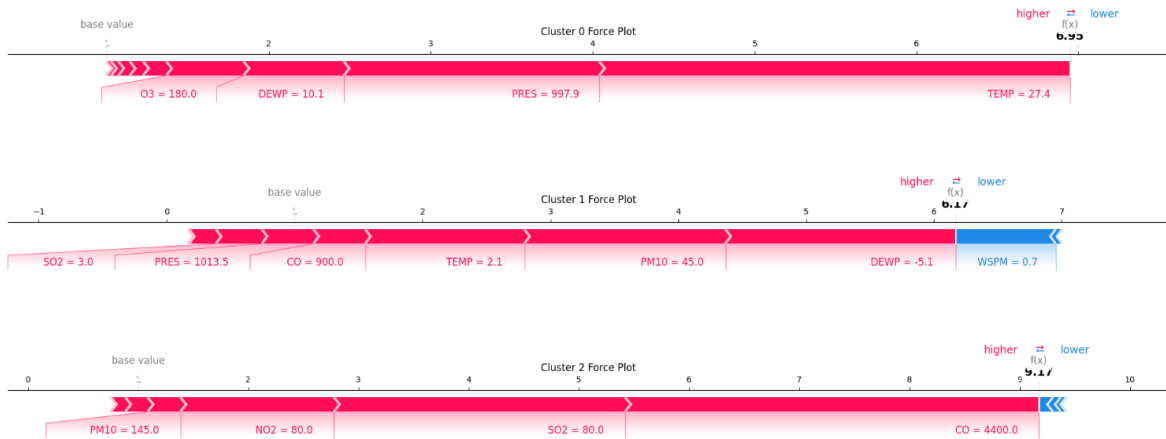


Figure 6.2: Representative SHAP force plots for air-quality clusters

6.4.4 Cluster-Specific Multi-Level Analysis

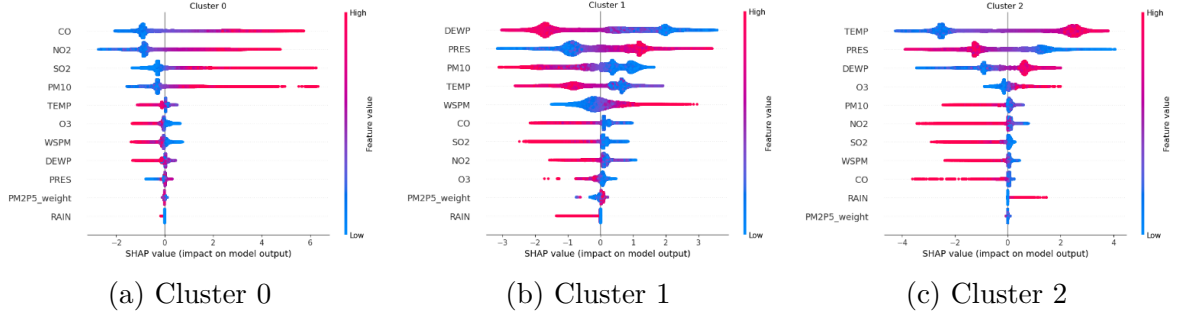


Figure 6.3: Cluster-specific and multi-level SHAP explanations for air quality

Figure 6.3 shows why the multi-level architecture becomes especially valuable when the analysis moves from the global ranking to cluster-specific explanatory profiles. At the coarse level, temperature and dew point dominate because they differentiate broad atmospheric regimes. Within individual clusters, however, CO, SO₂, PM₁₀, wind speed, pressure, or ozone can become more discriminative, depending on the regime being explained. This change is not contradictory to the global explanation; it is precisely what a multi-level explanation should reveal. The parent-level story is about regime selection, whereas the cluster-level story is about variation within a regime.

The cluster-specific summary plots in fig. 6.3 already display the spread, sign, and concentration of feature effects within each regime, so the chapter does not repeat a separate beeswarm figure here. This keeps the visual evidence non-duplicative while preserving the chapter’s multi-level explanatory emphasis.

6.4.5 Cross-Dataset Generalisation

Comparing the Beijing and wine studies clarifies the framework’s generality. The wine dataset is smaller and chemically correlated, whereas the Beijing dataset is large, noisy, and strongly affected by temporal and meteorological variation. The fact that the same explanatory logic remains productive in both settings is therefore significant. It suggests that the contribution is not tied to one domain-specific peculiarity, but to a more general strategy for converting black-box clustering into interpretable structure. Table 6.2 makes that transfer argument explicit.

Table 6.2: Cross-dataset comparison of the Chapter 5 and Chapter 6 frameworks

Aspect	Wine-quality study	Beijing study	Interpretive implication
Scale	Small and dense	Large and heterogeneous	Tests explanation under two complexity regimes
Explanation granularity	Single level	Multi-level hierarchy	Requires cross-level consistency
Primary utility	Local chemical interpretation	Regime-level environmental interpretation	Extends actionable scope

6.5 Comparative Analysis with State of the Art

This section positions the results relative to prior SHAP-based clustering studies and a direct LIME comparator.

6.5.1 Comparison with SHAP-Based Clustering Studies

Relative to the studies reviewed in section 3.1, the present framework offers two notable advances. First, it validates SHAP-based clustering explanation on two substantively different datasets rather than one. Second, it provides a formal argument for cross-level attribution consistency. Existing studies often report interesting local explanations [108]. However, few articulate how explanations should behave when the analyst moves between levels of granularity. Although cross-dataset validation indices are not directly comparable, the Beijing Silhouette value of approximately 0.63 is materially stronger than the 0.37 SHAP-space clustering result reported in the credit-risk study of Gramegna and Giudici [109], which helps situate the present large-scale partition as comparatively well separated within this explainable-clustering literature. Table 6.3 clarifies that difference against several nearby SHAP-oriented clustering studies. That methodological clarification is an important part of the chapter’s contribution.

Table 6.3: Comparison with SHAP-based clustering literature

Study	Scale handled	Hierarchy modelled	Distinctive thesis advance
Louhichi et al. [90]	Small tabular data	No	Adds large-scale and cross-level extension
Gramegna and Giudici [109]	Credit-risk explanation space	No	Adds explicit feature-space semantics and stronger separation
Younesi and Rahmani [108]	Moderate spatio-temporal data	No	Adds hierarchical attribution consistency
Antwarg et al. [92]	Domain-specific biomedical data	Partial	Retains direct original-feature semantics
This thesis	Large-scale tabular data	Yes	Formal multi-level SHAP aggregation with cross-dataset validation

6.5.2 Comparison with LIME-Based Clustering

The LIME comparator again highlights why the thesis privileges cooperative attribution. In the same general pipeline, LIME yields weaker structural coherence in the explanation space and less stable local narratives when the sampled neighbourhood changes. The point is not that LIME is useless. It is that for hierarchical explanation, its local-surrogate logic does not provide a stable enough basis for multi-level reasoning.

6.6 Explainability Enhancement and Visualisation

This section explains how the explanatory outputs are rendered usable for analysts and regulators rather than left as raw numerical attribution arrays.

6.6.1 Interactive SHAP Beeswarm and Summary Plots

The chapter’s explanatory value would be incomplete if the results remained accessible only through code. For that reason, the framework supports interactive SHAP beeswarm

and summary plots that allow an analyst to move between clusters, features, and representative observations. Interactive exploration uses the Altair library [110], which provides the visual-analysis strategy adopted throughout the multi-level clustering work [88]. This is especially important in large-scale monitoring contexts where the user is unlikely to accept a static figure as a sufficient basis for interpretation.

6.6.2 SHAP Force Plots for Regulatory Reporting

The local explanation outputs can also be reformatted as concise case reports. In the context of air-quality monitoring, a regulator may wish to know why a specific observation was classified into a given pollution regime. The SHAP force plot serves this need because it identifies the dominant positive and negative drivers for a single case while remaining linked to the same explanatory framework used for global analysis.

6.7 Discussion

This section interprets the chapter’s findings and makes explicit what they add beyond the single-level framework.

6.7.1 Scalability and Real-World Applicability

The Beijing study shows that the explanatory framework remains operational at a scale far beyond the original wine analysis. This matters not just computationally but substantively. A method that explains only small, clean datasets risks being methodologically elegant but practically narrow. Here the framework remains interpretable under scale, and its dominant explanatory variables continue to align with well-understood environmental processes. That actionability is nevertheless bounded: pollutant variables can inform monitoring thresholds and intervention priorities, whereas meteorological variables such as temperature, pressure, and dew point are primarily useful for interpretation, alert design, and regulatory reporting because they are not directly controllable by the analyst.

6.7.2 Hierarchical vs. Flat Explainability

The most important conceptual result of the chapter is that flat explanation is often insufficient even when it is accurate. A parent-level explanation may be true yet incomplete because it cannot show how the explanatory profile changes inside a cluster. Multi-level explanation therefore extends, rather than replaces, single-level explanation. This is the bridge that connects the clustering contributions to the recommendation

contribution in the next chapter: in both cases, the core question is how importance changes when the unit of analysis becomes more relational and more dynamic.

6.8 Limitation

This section records the main constraints of the multi-level clustering contribution. The framework still assumes static clustering, even though the Beijing data are temporally structured. It also relies on surrogate-based SHAP explanation and representative-instance reporting for local analysis, which compresses some observation-level variation into a smaller set of interpretable exemplars. Finally, the contribution remains confined to tabular data. These limitations do not invalidate the chapter’s claims, but they delimit them carefully. In particular, the static assumption becomes the most important point of contrast with chapter 7, where attribution is refreshed dynamically during training.

Summary

This chapter extended the cooperative-game explanation framework from single-level clustering to large-scale, hierarchical clustering. The Beijing air-quality study showed that the same explanatory logic remains useful under markedly greater scale and complexity, while the formal cross-level consistency argument clarified how parent-level and child-level attributions can be interpreted together rather than in competition. The chapter therefore answers **RQ2**: Shapley-based clustering explanation can be extended to hierarchical, large-scale settings without sacrificing interpretive coherence, provided the hierarchy is modelled explicitly and the approximation strategy is made transparent.

The chapter also advances **RQ5** by showing that actionable explanation is not exhausted by a single global ranking of variables. It often requires movement across explanatory scales. This chapter-level insight is grounded in the multi-level explanatory strategy [88]. The next chapter develops this insight in the recommendation setting and asks whether cooperative attribution can become part of the learning dynamics themselves rather than remaining purely post hoc.

Chapter 7

DyHuCoG: A Dynamic Hypergraph Cooperative Game for Preference-Aware Recommendation

Introduction

The previous two chapters treated Shapley values as a post-hoc explanation mechanism for unsupervised structure. This chapter asks whether the same cooperative-game logic can be embedded directly into the training dynamics of a recommender system. The answer proposed here is DyHuCoG, a dynamic hypergraph recommender in which preference-aware Monte Carlo Shapley estimates are refreshed during learning and used as adaptive importance weights in message passing [104]. The conceptual shift is substantial: attribution is no longer a commentary on the model after training, but a signal that shapes the model while it learns.

Research Question Connection. This chapter addresses **RQ3** and **RQ4** and advances **RQ5**.

Primary contribution: a Shapley-weighted hypergraph recommender that jointly improves ranking quality, catalogue coverage, and intra-list diversity while remaining interpretable.

7.1 Research Relevance and Positioning

This section positions DyHuCoG relative to the literature and to the preceding chapters of the thesis. The motivation for DyHuCoG emerges directly from the limitations exposed in the recommendation literature. Graph and hypergraph recommenders have improved ranking accuracy substantially, but they still tend to treat message importance as either

uniform or attention-weighted without a principled marginal-contribution account. Diversity is often handled through re-ranking or auxiliary heuristics, and interpretability is frequently added after prediction rather than integrated into the learning objective itself. DyHuCoG is designed as a direct intervention in this pattern.

Just as importantly, the chapter completes the thesis’s research arc. Chapters 5 and 6 established that cooperative-game attribution can explain static black-box structures. The present chapter tests a stronger thesis: that the same attribution perspective can become operational inside a representation-learning architecture and thereby improve performance rather than merely describe it. The continuity is therefore conceptual and methodological rather than a claim that the clustering and recommendation games are literally the same mathematical object.

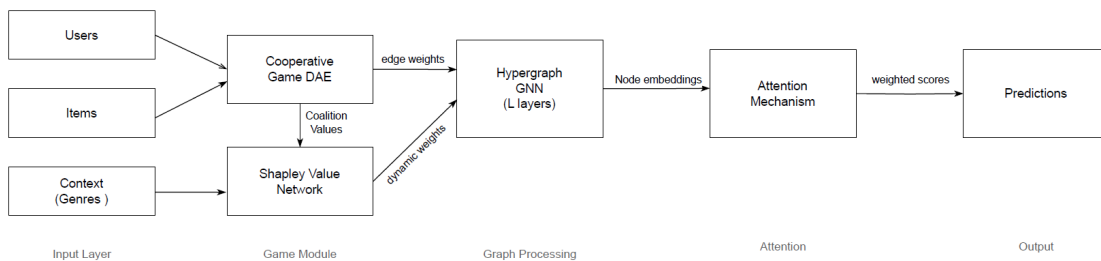


Figure 7.1: Workflow of the DyHuCoG architecture

7.1.1 Bridging XAI and Representation Learning

The bridge between explainable AI and representation learning is often rhetorically asserted but methodologically thin. DyHuCoG makes the bridge explicit. The model learns node and hyperedge representations as ordinary hypergraph recommenders do, but the propagation process is modulated by Shapley-derived importance weights grounded in a cooperative utility function. This means the explanatory mechanism and the predictive mechanism are no longer separate subsystems.

7.1.2 Limitations of Existing Hypergraph Recommendation Models

LightGCN captures graph-structured interaction patterns efficiently. HCCF and HPCF show that higher-order hypergraph structure can also be exploited effectively in recommendation. Closely related contrastive work such as RecDCL pursues the same broad objective through different regularisation logic. However, these models generally assume that once an interaction is encoded in a graph or hypergraph, the model can learn its importance implicitly from propagation and loss minimisation. DyHuCoG challenges this assumption. It argues that importance should be estimated explicitly

from a coalition utility reflecting ranking quality, diversity, and contextual coherence. This is the chapter’s decisive point of departure.

7.2 Problem Formulation

This section defines the recommendation task solved by DyHuCoG and introduces the cooperative-game notation used throughout the chapter.

Let U denote the user set, I the item set, and $\mathcal{C}$ the set of contextual entities or states. For a user u , the recommendation task is to produce a ranked top- N list $\hat{L}_u = [i_1, \dots, i_N]$ that balances not only relevance but also diversity and contextual fit. DyHuCoG models this task on a hypergraph whose nodes are users, items, and contexts, and whose hyperedges connect entities participating in a recommendation episode.

7.2.1 Top-N Recommendation with Cooperative Attribution

The ranking objective is not purely positional. A list that ranks a relevant item highly but collapses all remaining positions into near-duplicates is less desirable than a list that preserves both relevance and discovery. DyHuCoG therefore treats top- N recommendation as a multi-objective problem whose scoring logic is later mirrored in the coalition value function.

7.2.2 Cooperative Game Definition

The player set of the DyHuCoG game is $N = U \cup I \cup \mathcal{C}$. A coalition $S \subseteq N$ represents a subset of entities participating in a recommendation episode. The characteristic value $v(S)$ measures the quality of the recommendation outcome achievable by coalition S . This move parallels the clustering formulation from chapter 5, but the value function is now recommendation-oriented rather than partition-oriented.

7.2.3 Hypergraph Structure and Notation

The recommendation hypergraph is denoted $H = (V, E, W)$, where $V = U \cup I \cup \mathcal{C}$ is the node set, E is the set of hyperedges encoding user–item–context interactions, and W is the matrix of dynamic edge weights derived from Shapley estimates. This notation matters because the thesis’s central claim is not merely that hypergraphs are expressive, but that their expressive power can be made selectively adaptive through cooperative weighting.

7.3 Dynamic Cooperative Game Modelling

This section develops the multi-objective coalition utility that underpins DyHuCoG.

7.3.1 Multi-Objective Coalition Utility Function

The coalition utility combines ranking quality, diversity, and contextual relevance:

$$v(S) = \alpha \text{NDCG@20}(S) + \beta \text{Diversity}(S) + \gamma \text{ContextScore}(S), \quad (7.1)$$

where $\text{ContextScore}(S)$ denotes the mean context-alignment score of the entities participating in coalition S . In the thesis implementation, this term is instantiated as the average embedding-space alignment between a learned context representation and the context node representation associated with the interaction episode, so that context is measured in the same space used later by the scoring and regularisation terms. With $\alpha + \beta + \gamma = 1$, the formulation is important because it ensures that the same trade-off the recommender is expected to satisfy is also the trade-off from which attribution is computed. In other words, the explanatory game and the predictive objective are aligned by design. Operationally, $v(S)$ is evaluated on the recommendation episode induced by coalition S : only interactions whose user, item, and context entities are present in S are retained, the surviving candidate items are ranked for the eligible users, and NDCG@20 , diversity, and context alignment are then computed on those restricted top-20 lists and averaged over the eligible interactions. This makes $\text{NDCG@20}(S)$ a recommendation utility on a coalition-induced candidate set rather than an abstract score on isolated entities. In practice, the player set used for coalition evaluation is scoped to the interaction episode under consideration at each refresh stage rather than to an exhaustive full-catalogue permutation; a typical local episode therefore contains one focal user, a small set of candidate items, and the associated context nodes, so the effective player set is on the order of only a few dozen entities rather than the full catalogue. The resulting estimator should therefore be read as an approximate in-training valuation procedure aligned with the local recommendation context.

7.3.2 Preference-Weighted Utility

To reflect the fact that some entities provide richer preference evidence than others, the basic coalition value is extended to a preference-aware form:

$$v_{\text{pref}}(S) = v(S) + \lambda_{\text{pref}} \sum_{(u,i) \in S} \text{sim}(u, i), \quad (7.2)$$

where $\text{sim}(u, i)$ denotes user–item preference consistency and λ_{pref} controls its contribution. This additive preference term keeps the utility modulation interpretable and separable from the frequency-driven component [104].

7.3.3 Hyperparameter Grid Search for Coalition Weights

Validation experiments indicate that $\alpha = 0.60$, $\beta = 0.25$, and $\gamma = 0.15$ offer the best balance on MovieLens-1M, with similar behaviour on Amazon-Book. These weights are selected through a grid search over $[0.1, 0.8]$ under the constraint $\alpha + \beta + \gamma = 1$, although the full validation surface and the top-five configurations are not tabulated in the article. The thesis therefore reports the validated optimum rather than reconstructing unreported intermediate settings. The high value of α reflects the primacy of ranking quality, but the non-trivial value of β confirms that diversity cannot be treated as a cosmetic add-on without measurable cost. Unless otherwise stated, $\lambda_{\text{pref}} = 0.20$ is used throughout, yielding stable performance with variance below 1.5% in NDCG@20 [104].

7.4 Preference-Aware Monte Carlo Shapley Estimation

This section derives the dynamic approximation of Shapley values used to modulate hypergraph propagation.

7.4.1 Exact Shapley Derivation

The exact Shapley value for player j under the base coalition utility is:

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{j\}) - v(S) \right], \quad (7.3)$$

which is principled but computationally infeasible for realistic recommendation systems because the number of possible coalitions is exponential in $|N|$.

7.4.2 Monte Carlo Approximation and Variance Analysis

DyHuCoG therefore uses the Monte Carlo estimator [104]:

$$\hat{\phi}_j = \frac{1}{M} \sum_{m=1}^M \left[v(S_m \cup \{j\}) - v(S_m) \right], \quad (7.4)$$

and, to align attribution with personalised preference consistency, the preference-aware variant:

$$\hat{\phi}_j^{\text{pref}} = \frac{1}{M} \sum_{m=1}^M \left[v_{\text{pref}}(S_m \cup \{j\}) - v_{\text{pref}}(S_m) \right], \quad (7.5)$$

where S_m denotes the sampled coalition in the m -th Monte Carlo draw. The estimator is unbiased and exhibits the familiar $\mathcal{O}(1/M)$ variance decay. The practical question is therefore not whether the approximation is exact, but whether it is accurate enough for stable training at acceptable cost.

7.4.3 Convergence and Sample Size Selection

Empirically, $M = 50$ offers a strong accuracy–efficiency balance. Relative to a high-sample reference, the mean squared error is approximately 1.4×10^{-5} on MovieLens-1M, while runtime remains far below the cost of very large sample counts. This is one of the chapter’s central pragmatic results: the Shapley mechanism is expensive enough to matter, but not so expensive as to make the method unusable.

7.4.4 Temporal Smoothing and Refresh Period

DyHuCoG does not recompute Shapley weights at every optimisation step. Instead, the estimates are refreshed periodically and smoothed through an exponential moving average with decay parameter ρ . In the reported experiments, the refresh period is ten batches, which yields approximately 49 updates per epoch on MovieLens-1M with batch size 2,048. This choice is methodologically important because it recognises that dynamic weighting must remain stable enough for training to converge. The model therefore treats attribution as adaptive but not hypersensitive.

7.5 Hypergraph GNN with Shapley-Weighted Message Passing

This section derives the architecture of DyHuCoG and shows how cooperative attribution enters the propagation mechanism.

Figure 7.1 shows the DyHuCoG architecture [104]. The model is not only predictive but also inspectable at the level of a single recommendation. The explanatory output itself is analysed later in section 7.11, where the SHAP waterfall view is placed next to the discussion of individual recommendations rather than introduced prematurely in the architectural derivation.

7.5.1 Shapley-Weighted Hypergraph Propagation

The core propagation step uses Shapley-derived weights to modulate how strongly information from neighbouring entities contributes to each update. The base propagation and its weighted form are written as [104]:

$$\mathbf{e}^{(\ell+1)} = \sigma\left(D^{-1/2}AD^{-1/2}\mathbf{e}^{(\ell)}\right), \quad (7.6)$$

$$\mathbf{e}_j^{(\ell+1)} = \sigma\left(W^{(\ell)}\mathbf{e}_j^{(\ell)} + \sum_{k \in \mathcal{N}(j)} w_{jk} \mathbf{e}_k^{(\ell)}\right), \quad (7.7)$$

with normalised Shapley neighbourhood weights

$$w_{jk} = \frac{\hat{\phi}_{jk}}{\sum_{k' \in \mathcal{N}(j)} \hat{\phi}_{jk'}}. \quad (7.8)$$

Here $\hat{\phi}_{jk}$ denotes the clipped and exponentially smoothed preference-aware Monte Carlo Shapley estimate attached to the interaction between node j and neighbour k before neighbourhood normalisation. In practice, extreme Shapley estimates, including negative marginal contributions, are clipped before neighbourhood normalisation to stabilise message passing in sparse interaction regimes [104]. Conceptually, this means that the model is told not only *who* is connected to whom, but also *how much* each interaction coalition is worth under the multi-objective utility. This is the chapter’s decisive methodological move: relational structure is filtered through cooperative importance before it becomes representation.

7.5.2 Multi-Layer Embedding Fusion

Representations from multiple propagation layers are fused so that the model can combine short-range and longer-range interaction signals. DyHuCoG aggregates them as:

$$\mathbf{e}_j = \sum_{\ell=0}^L \alpha_{\ell} \mathbf{e}_j^{(\ell)}, \quad (7.9)$$

which preserves information from both shallow and deeper neighbourhoods. The attention coefficients α_{ℓ} are learned rather than fixed. This is appropriate because different users and items may benefit from different neighbourhood depths.

7.5.3 Attention Gating Mechanism

The attention gate interpolates between Shapley-weighted propagation and a more uniform propagation mode. The interaction-level gate is defined as [104]:

$$a_{ui} = \sigma(W_a[\mathbf{e}_u, \mathbf{e}_i, \mathbf{l}_i]), \quad (7.10)$$

where $\mathbf{l}_i$ denotes the item-context representation. This improves robustness during early training, when Shapley estimates are still noisy. The gate therefore functions as a stabiliser: it allows the model to rely more heavily on explicit cooperative attribution as that attribution becomes more trustworthy. This also clarifies its division of labour relative to Shapley weighting: the Shapley coefficients encode coalition utility before propagation, whereas the attention gate adaptively reweights the propagated signal during representation learning and scoring.

7.5.4 Context-Aware Final Scoring

The attention gate yields the intermediate score [104]:

$$y_{ui} = (1 + a_{ui}) \langle \mathbf{e}_u, \mathbf{e}_i \rangle, \quad (7.11)$$

before the final context-aware prediction is written as:

$$f(u, i, c_{u,i}) = y_{ui} + \lambda_c \langle g(c_{u,i}), \mathbf{e}_{c_{u,i}} \rangle, \quad (7.12)$$

where $g : \mathcal{C} \rightarrow \mathbb{R}^d$ is a context encoder that maps the raw contextual state $c_{u,i}$ into the same embedding space as the learned context node representation $\mathbf{e}_{c_{u,i}}$. This ensures that contextual relevance is not relegated to a decorative feature, but contributes directly to the score that will be ranked. Because context also appears in the coalition utility, the model is structurally encouraged to treat it as part of the explanatory and predictive problem simultaneously.

7.6 Multi-Objective Training

This section describes the composite loss function that makes DyHuCoG a recommendation model rather than only an explanatory hypergraph.

7.6.1 Bayesian Personalised Ranking Loss

The primary ranking term is Bayesian personalised ranking, which encourages the score of an observed positive item to exceed that of a sampled negative. In the DyHuCoG formulation:

$$\mathcal{L}_{\text{rec}} = - \sum_{(u, i^+, i^-) \in \mathcal{D}} \log \sigma \left(f(u, i^+, c_{u, i^+}) - f(u, i^-, c_{u, i^-}) \right). \quad (7.13)$$

This choice is appropriate for implicit-feedback recommendation because it models pairwise ranking preference directly.

7.6.2 Intra-List Diversity Regulariser

The diversity term penalises redundant recommendation lists by encouraging dissimilarity among the top-ranked items. The diversity regulariser is:

$$\mathcal{L}_{\text{div}} = - \frac{1}{|U|} \sum_{u \in U} \text{ILD}(\mathcal{R}_u), \quad (7.14)$$

with intra-list diversity defined as:

$$\text{ILD}(\mathcal{R}_u) = \frac{2}{K(K-1)} \sum_{1 \leq k < \ell \leq K} [1 - \text{sim}(i_k, i_\ell)]. \quad (7.15)$$

Its presence is not incidental. The ablation study later in the chapter shows that removing this term degrades not only catalogue coverage but also overall recommendation quality, which indicates that diversity and ranking are not simply antagonistic objectives.

7.6.3 Context Alignment Loss

The context-alignment regulariser encourages recommended items to remain coherent with the contextual state associated with an interaction:

$$\mathcal{L}_{\text{ctx}} = \frac{1}{|\mathcal{E}|} \sum_{(u, i) \in \mathcal{E}} \|g(c_{u, i}) - \mathbf{e}_{c_{u, i}}\|_2^2. \quad (7.16)$$

In this sense, the regulariser operationalises the context component of the coalition utility at the optimisation level.

7.6.4 Composite Loss and Optimisation

The full training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \quad (7.17)$$

where the regularisation term is:

$$\mathcal{L}_{\text{reg}} = \frac{1}{2} \left(\|E_U\|_F^2 + \|E_I\|_F^2 \right). \quad (7.18)$$

This equation is conceptually important because it shows the alignment between the learning objective and the coalition value. DyHuCoG is trained to optimise the same balance that later determines cooperative attribution.

7.7 Experimental Evaluation

This section reports the main performance results of DyHuCoG on MovieLens-1M and Amazon-Book.

7.7.1 Evaluation Setting

The experiments follow the common protocol from chapter 4. DyHuCoG uses three propagation layers, embedding dimension 64, Monte Carlo sample size $M = 50$, and a Shapley refresh period of ten batches [104]. Results are compared with MF, NCF, LightGCN, RecDCL, HCCF, and HPCF under the same train/validation/test protocol [104].

7.7.2 Overall Performance: MovieLens-1M and Amazon-Book

On MovieLens-1M, DyHuCoG achieves an NDCG@20 of 0.2775 and Recall@20 of 0.2362. Relative to HPCF, this corresponds to improvements of 9.77% in NDCG@20 and 12.58% in Recall@20. The model also increases catalogue coverage from 0.342 to 0.397 and intra-list diversity from 0.461 to 0.516. On the much sparser Amazon-Book benchmark, DyHuCoG achieves an NDCG@20 of 0.0306 and Recall@20 of 0.0417, outperforming HPCF by 13.33% and 16.16%, respectively. The larger relative gain on the sparser benchmark supports a central claim of the chapter: Shapley-guided weighting is especially helpful when interaction data are weak and the model benefits from a richer signal about coalition usefulness. Table 7.1 reports the numeric results, while figs. 7.2 and 7.3 visualise the same ranking, coverage, and diversity advantage. The inferential claims attached to these results should nevertheless be read with scope

discipline: the strongest fully tabulated paired tests in the thesis apply to per-user NDCG@20 on MovieLens-1M, while Amazon-Book and the auxiliary metrics remain descriptive unless explicitly tabulated elsewhere [104].

Table 7.1: Main DyHuCoG performance results

Dataset	Model	NDCG@20	Recall@20	Coverage	Diversity
MovieLens-1M	MF	0.1200 $\pm$ 0.0025	0.0880 $\pm$ 0.0020	0.239 $\pm$ 0.010	0.367 $\pm$ 0.009
	NCF	0.1300 $\pm$ 0.0026	0.1100 $\pm$ 0.0022	0.269 $\pm$ 0.011	0.398 $\pm$ 0.008
	LightGCN	0.2130 $\pm$ 0.0030	0.1790 $\pm$ 0.0027	0.308 $\pm$ 0.010	0.423 $\pm$ 0.007
	RecDCL	0.2296 $\pm$ 0.0032	0.1918 $\pm$ 0.0029	0.324 $\pm$ 0.010	0.445 $\pm$ 0.007
	HCCF	0.2470 $\pm$ 0.0035	0.2050 $\pm$ 0.0031	0.327 $\pm$ 0.009	0.448 $\pm$ 0.006
	HPCF	0.2528 $\pm$ 0.0036	0.2098 $\pm$ 0.0032	0.342 $\pm$ 0.009	0.461 $\pm$ 0.006
	DyHuCoG	0.2775 $\pm$ 0.0039	0.2362 $\pm$ 0.0036	0.397 $\pm$ 0.011	0.516 $\pm$ 0.005
Amazon-Book	MF	0.0049 $\pm$ 0.0005	0.0093 $\pm$ 0.0009	0.168 $\pm$ 0.013	0.425 $\pm$ 0.014
	NCF	0.0085 $\pm$ 0.0008	0.0142 $\pm$ 0.0014	0.193 $\pm$ 0.012	0.458 $\pm$ 0.013
	LightGCN	0.0236 $\pm$ 0.0024	0.0320 $\pm$ 0.0032	0.226 $\pm$ 0.011	0.491 $\pm$ 0.012
	RecDCL	0.0255 $\pm$ 0.0026	0.0346 $\pm$ 0.0035	0.242 $\pm$ 0.010	0.512 $\pm$ 0.011
	HCCF	0.0258 $\pm$ 0.0026	0.0344 $\pm$ 0.0034	0.248 $\pm$ 0.010	0.520 $\pm$ 0.011
	HPCF	0.0270 $\pm$ 0.0027	0.0359 $\pm$ 0.0036	0.259 $\pm$ 0.010	0.535 $\pm$ 0.011
	DyHuCoG	0.0306 $\pm$ 0.0031	0.0417 $\pm$ 0.0042	0.336 $\pm$ 0.012	0.602 $\pm$ 0.010

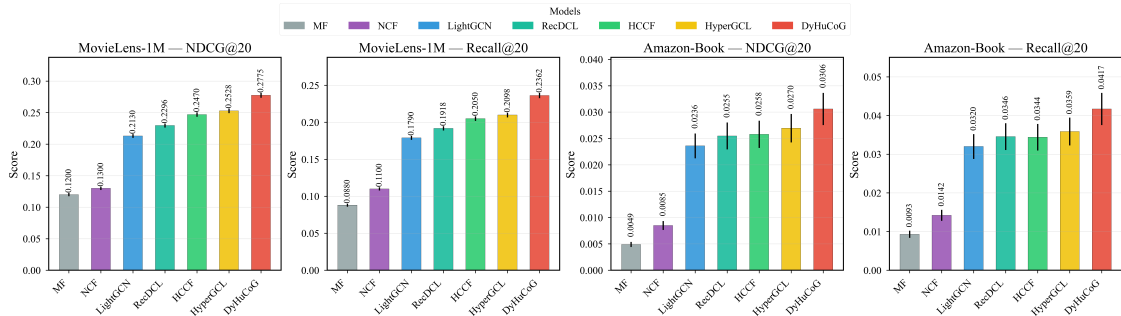


Figure 7.2: Ranking performance on MovieLens-1M and Amazon-Book.

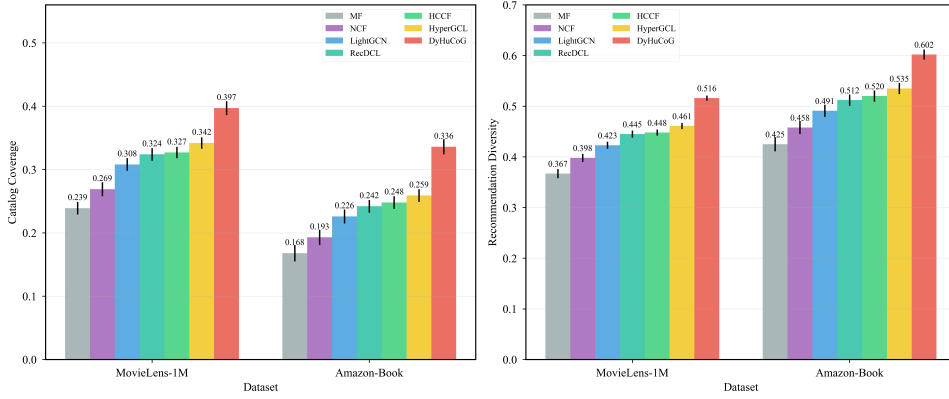


Figure 7.3: Coverage and intra-list diversity on MovieLens-1M and Amazon-Book.

7.8 Ablation Studies

This section examines which DyHuCoG components are essential and how their removal affects performance.

7.8.1 Ablation Settings

The ablation design removes, one at a time, the Shapley weighting, the hypergraph structure, the attention gate, the context component, and the diversity term. This strategy isolates whether the model’s gains stem from the cooperative-game mechanism itself or from more generic architectural complexity.

7.8.2 Results and Discussion

The ablation results show that every major component contributes materially. Removing context causes the largest performance loss, followed by removal of the hypergraph structure. This does not make cooperative attribution secondary; rather, it shows that context provides the representational substrate on which Shapley weighting can operate meaningfully, whereas removing Shapley weighting reverts the model to uniform propagation over an otherwise intact contextual structure. Replacing Shapley weighting with uniform weighting nevertheless degrades performance noticeably, which directly supports the thesis’s argument that marginal-contribution estimation is not a decorative addition. The attention mechanism and the diversity term yield smaller but still meaningful gains, confirming that exposure breadth can be improved without sacrificing overall ranking quality. Table 7.2 records the component-wise drops that support this interpretation.

Table 7.2: Component-wise ablation results

Component	MovieLens-1M		Amazon-Book	
	NDCG@20	% Drop	NDCG@20	% Drop
Full DyHuCoG	0.2775	–	0.0306	–
w/o Shapley Value	0.2647	4.6%	0.0287	6.1%
w/o Hypergraph	0.2586	6.8%	0.0279	8.9%
w/o Attention	0.2678	3.5%	0.0295	3.5%
w/o Context	0.2547	8.2%	0.0272	11.0%
w/o Diversity	0.2614	5.8%	0.0288	5.8%

7.9 Computational Efficiency

This section evaluates the runtime and approximation cost of DyHuCoG.

7.9.1 Training Time and Inference Latency

DyHuCoG is more expensive than the strongest hypergraph baselines, but the overhead remains bounded. On MovieLens-1M, the model requires roughly 2000 seconds of training time compared with approximately 1125 seconds for HPCF. On Amazon-Book, the ratio is similar. Inference remains within practical limits for real-time deployment. The relevant conclusion is therefore not that cooperative attribution is free, but that its cost is proportionate to the performance and interpretability gains it yields.

7.9.2 Shapley Convergence Analysis

The computational profile of one training epoch can be summarised as

$$\mathcal{O}\left((L+1)md\right) + \mathcal{O}\left((M/f)m\right), \quad (7.19)$$

where m is the number of interactions, d the embedding size, L the number of propagation layers, M the Monte Carlo budget, and f the Shapley refresh period. The first term corresponds to sparse hypergraph propagation and ranking; the second is the amortised cost of preference-aware Shapley estimation. The convergence analysis confirms that $M = 50$ Monte Carlo samples achieves a favourable trade-off. Increasing the sample size further reduces approximation error but with diminishing returns.

Table 7.3: Runtime and scalability analysis

Dataset	Model	Training Time	Inference	Memory	Scale Factor ($\times$ MF)
MovieLens-1M	MF	485.2 ± 19.3	0.54 ± 0.02	2.1 ± 0.1	1.00 ± 0.00
	NCF	675.4 ± 23.1	0.76 ± 0.03	2.7 ± 0.1	1.41 ± 0.08
	LightGCN	785.6 ± 25.1	0.88 ± 0.04	3.2 ± 0.2	1.63 ± 0.10
	RecDCL	968.4 ± 29.8	1.02 ± 0.05	3.6 ± 0.2	1.89 ± 0.12
	HCCF	952.3 ± 28.4	1.06 ± 0.05	3.8 ± 0.2	1.96 ± 0.12
	HPCF	1124.6 ± 31.8	1.18 ± 0.06	4.1 ± 0.2	2.19 ± 0.14
	DyHuCoG	2000.2 ± 41.2	1.84 ± 0.08	4.4 ± 0.3	3.41 ± 0.19
Amazon-Book	MF	2201.7 ± 69.3	2.46 ± 0.12	8.5 ± 0.4	1.00 ± 0.00
	NCF	3072.9 ± 82.6	3.42 ± 0.17	11.1 ± 0.5	1.39 ± 0.10
	LightGCN	3645.2 ± 88.2	4.05 ± 0.20	12.8 ± 0.6	1.65 ± 0.11
	RecDCL	4526.8 ± 104.6	5.04 ± 0.25	15.6 ± 0.8	2.05 ± 0.14
	HCCF	4418.5 ± 101.2	4.92 ± 0.25	15.3 ± 0.8	2.00 ± 0.14
	HPCF	5234.2 ± 118.6	5.48 ± 0.27	16.8 ± 0.8	2.23 ± 0.15
	DyHuCoG	9278.9 ± 140.1	8.52 ± 0.43	17.9 ± 0.9	3.46 ± 0.24

The evaluation captures this trade-off through Monte Carlo mean-squared error and runtime rather than through a separate downstream NDCG@20-versus- M table, so the thesis keeps the convergence discussion aligned with the available evidence rather than introducing unsupported sensitivity numbers. Tables 7.3 and 7.4 make the runtime overhead and Monte Carlo trade-off explicit.

Table 7.4: Monte Carlo convergence and runtime trade-offs

M	Estimated MSE	Accuracy vs. reference	Runtime	Overhead vs. HPCF
10	1.4×10^{-4}	95%	~ 1460 s	$1.3\times$
25	5.6×10^{-5}	98%	~ 1800 s	$1.6\times$
50	1.4×10^{-5}	99%	2001 s	$1.78\times$
100	3.5×10^{-6}	99.5%	~ 2810 s	$2.5\times$

7.10 Cold-Start Robustness Analysis

This section evaluates whether DyHuCoG remains effective when users or items have very limited training history.

7.10.1 Cold-Start Users and Items

For cold-start users with at most five training interactions, DyHuCoG achieves an NDCG@20 of approximately 0.061 on MovieLens-1M, improving over the HPCF baseline by about ten percent. The cold-start item results exhibit a similar pattern. This does not solve cold start completely, but it improves performance in precisely the regime where recommender systems are usually most brittle.

tabularx

Table 7.5: Cold-start and cross-dataset robustness

Method	Cold-start user	Cold-start item	MovieLens- 1M	Amazon- Book	Yelp2018
MF	0.026	0.025	0.120	0.005	0.003
LightGCN	0.047	0.044	0.213	0.024	0.014
HCCF	0.054	0.051	0.247	0.026	0.016
HPCF	0.055	0.052	0.253	0.027	0.017
DyHuCoG	0.061	0.057	0.278	0.031	0.019
Improvement	+10.9%	+9.6%	+9.9%	+14.8%	+11.8%

Note. Yelp2018 is included as an auxiliary robustness benchmark in the DyHuCoG evaluation.

Table 7.5 also shows that the same robustness pattern extends to Amazon-Book and the auxiliary Yelp2018 benchmark [104]. In the present thesis, Yelp2018 is treated only as an auxiliary sparse user–business recommendation benchmark for robustness checking rather than as a third full main benchmark on par with MovieLens-1M and Amazon-Book. Unless otherwise stated, its split construction and leave-one-out evaluation follow the same user-level temporal protocol [104].

7.11 Interpretability Analysis

This section demonstrates that DyHuCoG’s cooperative weighting yields interpretable recommendation signals.

7.11.1 Shapley Waterfall Plots for Individual Recommendations

A SHAP waterfall view of an individual recommendation decomposes the final utility into contributions from ranking, diversity, context alignment, and preference consistency. This matters because it shows that the model can justify a recommendation in terms

aligned with its own optimisation logic. The explanation is therefore not merely user-friendly rhetoric; it is structurally tied to the same components that drove training. Figure 7.4 places a representative waterfall plot in its natural interpretive location.

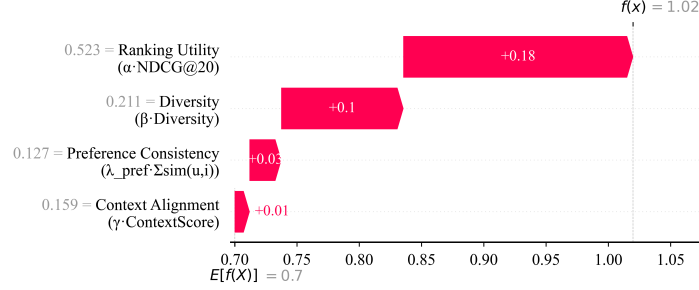


Figure 7.4: SHAP waterfall explanation for a representative MovieLens-1M recommendation.

7.11.2 Popularity Bias Analysis

The coverage and ILD improvements already suggest that DyHuCoG mitigates popularity concentration. The interpretability analysis makes the claim more explicit: because Shapley weighting measures marginal utility rather than raw frequency alone, weak but informative interactions can retain influence even when they are attached to less popular items. This is one of the chapter’s most practically important insights, and it is visually consistent with the coverage and diversity profile reported in fig. 7.3.

7.12 Statistical Validation

This section reports the inferential evidence supporting the claim that DyHuCoG improves significantly over strong baselines.

7.12.1 Paired t-Test Results

Paired t-tests on per-user NDCG@20 differences show that DyHuCoG outperforms all major baselines on MovieLens-1M with extremely small p -values after Holm–Bonferroni correction. The comparison against HPCF yields $t = 46.38$ with Cohen’s $d_z = 1.3345$, both computed on the full MovieLens-1M test set ($n = 6,040$ users; $df = 6,039$) as documented in Appendix A of [104]. Table 7.6 reports the full inferential comparison included in the thesis. The paired-test scope is intentionally limited to MovieLens-1M because the source DyHuCoG article tabulates the full per-user inferential analysis only for that benchmark; the thesis preserves that reporting boundary for evidential discipline rather than fabricating unsupported Amazon-Book statistics. The chapter does not extrapolate these exact paired-test statistics to every other dataset and metric;

where broader significance is reported qualitatively rather than in full tabular form, the thesis keeps that narrower scope explicit.

7.12.2 Effect Size

Effect sizes are large across the strongest comparisons, especially relative to HPCF, HCCF, and RecDCL. This matters because recommendation benchmarks with many users can make tiny differences significant. The effect-size analysis confirms that DyHuCoG’s improvements are substantively meaningful rather than only statistically visible.

Table 7.6: Paired tests on per-user NDCG@20

Comparison	t	df	p -value	Cohen’s d_z	Holm α_i	Sig.
DyHuCoG vs HPCF	46.38	6039	1.81×10^{-270}	1.3345	0.050000	✓
DyHuCoG vs RecDCL	92.72	6039	$< 10^{-300}$	2.6677	0.008333	✓
DyHuCoG vs HCCF	61.21	6039	$< 10^{-300}$	1.7610	0.010000	✓
DyHuCoG vs LightGCN	132.19	6039	$< 10^{-300}$	3.8035	0.012500	✓
DyHuCoG vs NCF	311.13	6039	$< 10^{-300}$	8.9518	0.016667	✓
DyHuCoG vs MF	341.76	6039	$< 10^{-300}$	9.8330	0.025000	✓

7.13 Discussion

This section interprets what DyHuCoG demonstrates about cooperative-game attribution as a learning signal.

7.13.1 Cooperative Game Theory as an In-Training Signal

The most important theoretical implication of the chapter is that Shapley values need not remain post-hoc descriptions. In DyHuCoG they function as dynamic training signals that alter how information moves through the hypergraph. This is a stronger claim than anything shown in the clustering chapters and one of the thesis’s principal conceptual advances.

7.13.2 Accuracy–Diversity Trade-off: Evidence for Partial Reduction

The empirical results indicate that DyHuCoG does not simply shift the system from accuracy to diversity or from diversity to explanation. Instead, the cooperative utility allows the model to learn a balance in which relevance, coverage, and diversity improve

together. This does not abolish the trade-off in every conceivable setting, but it shows that the trade-off is not structurally fixed. The joint ranking, coverage, and ILD gains visible in figs. 7.2 and 7.3 were established under the coalition weighting $(\alpha, \beta, \gamma) = (0.60, 0.25, 0.15)$ used throughout the experiments; whether the same pattern of improvements generalises across all possible utility-weight configurations is reserved for future sensitivity analysis.

7.13.3 Complementarity of Contributions Across Chapters

The chapter also clarifies the relationship between the three empirical contributions. The clustering studies showed that Shapley values can make hidden structure intelligible. DyHuCoG shows that the same logic can shape representation learning itself. Taken together, these chapters support the thesis-level view that cooperative attribution is not confined to one modelling paradigm but can unify multiple tasks under one explanatory rationale.

7.14 Limitation

This section records the principal limits of the DyHuCoG contribution.

DyHuCoG still incurs measurable computational overhead relative to the strongest hypergraph baselines, and its performance depends on the availability of meaningful contextual information. The Monte Carlo Shapley approximation, although effective, could likely be improved through variance-reduction strategies. The ablation study is deliberately component-wise and therefore does not identify higher-order interaction effects among removals such as Shapley weighting and context together; the thesis keeps that narrower scope explicit because a factorial interaction design is not included. Likewise, the thesis-level inferential table is complete for MovieLens-1M per-user NDCG@20, whereas the Amazon-Book evidence is retained in a more limited descriptive form rather than being inflated into a new unsupported significance claim. A related unresolved gap is that the evaluation provides Monte Carlo convergence through mean-squared error and runtime rather than through a separate downstream NDCG@20-versus- M sensitivity table; the thesis therefore does not claim a metric-level robustness analysis beyond the available evidence. Finally, the study evaluates explanation indirectly through structural plausibility and utility decomposition rather than through a large-scale user study of trust and comprehension. These limitations are substantial enough to matter and honest enough to define a future research agenda. The baseline comparison was finalised using hypergraph and contrastive recommendation models available through early 2026, including the 2024–2025 methods RecDCL and HPCF. No separate thesis-level audit of post-2024 LLM-augmented recommenders or late-breaking 2025–2026 hypergraph

variants was conducted beyond that validated baseline set. Any recommendation models published after this date have not been evaluated; the DyHuCoG results therefore claim superiority relative to the tested baseline set rather than to all conceivable future work.

Summary

This chapter introduced DyHuCoG, a dynamic hypergraph recommender that embeds preference-aware Monte Carlo Shapley estimation directly into message passing and training. Across MovieLens-1M and Amazon-Book, the model outperformed strong classical, graph, and hypergraph baselines on ranking quality while also improving coverage and intra-list diversity. The chapter therefore answers **RQ3** and **RQ4**: cooperative-game attribution can be moved into the learning dynamics of a recommender system, and when it is, the resulting model can balance accuracy, diversity, and contextual relevance more effectively than strong existing alternatives.

The chapter also deepens **RQ5**. The key insight is no longer just that Shapley values explain black-box systems. It is that they can become part of what the system learns from, exactly as evidenced by the DyHuCoG experiments. The next chapter synthesises this insight with the preceding clustering contributions and draws the thesis-level conclusions.

Chapter 8

Discussion

Introduction

This chapter provides the thesis-level interpretation of the results reported in the three contribution chapters. Its purpose is not to repeat the empirical findings in detail, but to explain what becomes visible when those findings are read together as a coherent research programme. The argument developed across the thesis is that cooperative-game reasoning offers more than a post-hoc language of explanation. It provides a principled way of allocating importance across features, interactions, and contexts, and this importance-allocation logic can be used to support explanation, analysis, and model design.

8.1 Synthesis of Findings

This section integrates the three contribution chapters into a single thesis-level argument.

Taken together, the three empirical chapters support a common thesis: explanation is most useful when it is grounded in a principled account of contribution rather than in a purely descriptive narrative. In the first contribution, cooperative attribution was used to explain black-box clustering in a way that preserved the semantic link between original variables and discovered partitions. In the second contribution, that explanatory logic was extended to a multi-level setting, showing that cluster-level explanations can remain coherent across levels of analysis when the aggregation step is made explicit rather than assumed. In the third contribution, the same cooperative perspective was moved inside the model itself, where Shapley-based contribution estimates were used as part of the learning process in a dynamic hypergraph recommender.

A first synthesis-level finding is therefore methodological. The thesis shows that cooperative-game attribution is not confined to one model family or one data type. It can explain clustering, organise hierarchical interpretation, and guide recommendation

learning, provided that the underlying game is defined in a way that matches the structure of the task.

A second synthesis-level finding is conceptual. Across all three contributions, the central problem is not simply interpretability in the abstract. It is the allocation of explanatory responsibility: which feature, interaction, or context should count as important, by how much, and under what formal guarantees? The Shapley framework matters because it answers that question under an explicit set of axioms rather than through ad hoc weighting.

A third synthesis-level finding concerns actionability. The thesis does not claim that every explanation automatically yields direct intervention. It claims, rather, that explanations become more actionable when they identify modifiable drivers of a model outcome. In the clustering chapters, this means isolating variables that characterise regimes or differentiate cluster membership. In the recommendation chapter, it means identifying the historical interactions and contextual relations that most strongly drive a ranking decision.

8.2 Theoretical Implications

This section clarifies the broader theoretical contribution of the thesis.

The first theoretical implication is that Shapley-value reasoning can be treated as a common formal language for explanation across tasks that are usually studied separately. In clustering, it provides a disciplined bridge between partitioning and feature-level interpretation. In hierarchical analysis, it supports a consistency-oriented view of cross-level explanation. In recommendation, it supports the move from explanation as an after-the-fact commentary to explanation as an in-training signal.

The second theoretical implication is that the thesis strengthens the case for treating explanation and optimisation as potentially connected rather than strictly separate. Classical explainable AI often assumes a temporal separation: first a model is trained, then it is explained. The DyHuCoG contribution shows that this separation is not inevitable. Contribution estimates can instead be used during learning, which opens a broader research direction in which cooperative attribution becomes part of model design rather than merely part of model auditing.

The third theoretical implication concerns trustworthy AI. The thesis does not argue that Shapley values alone solve governance, fairness, or compliance. It argues more carefully that explanation methods with explicit fairness-oriented axioms provide a stronger basis for transparency than explanation methods whose outputs may be persuasive yet normatively under-specified. In that sense, the cooperative-game perspective contributes not only to technical explanation, but also to the broader methodological

vocabulary of accountable machine learning.

8.3 Consolidated Limitations

This section consolidates the main limitations of the thesis at the programme level.

The first limitation is computational. Exact Shapley computation remains intractable in realistic settings, and every empirical contribution in the thesis relies on some form of tractable approximation, surrogate construction, or restricted reporting strategy. These are reasonable methodological choices, but they remain approximations rather than full resolutions of the underlying cost problem.

The second limitation is methodological. The clustering contributions depend on the quality of the explanatory bridge between the unsupervised partition and the supervised surrogate used for attribution. The recommendation contribution depends on the stability of approximate contribution estimates during training and on the adequacy of the contextual information encoded in the hypergraph. Accordingly, the thesis demonstrates the power of cooperative attribution under well-defined modelling assumptions rather than under every conceivable setting.

The third limitation is empirical. The thesis spans two clustering domains and benchmark recommendation datasets, but it does not extend to multimodal recommendation, sequential recommendation, or large-scale online deployment. Likewise, although the explanations are methodologically motivated and domain-grounded, the thesis does not provide a dedicated human-subject study of actionability or contestability for the clustering contributions.

The fourth limitation concerns claim scope. The thesis establishes that cooperative attribution is a coherent and productive perspective for explainable AI in clustering and recommendation. It does not claim to have produced one fully unified mathematical framework that eliminates all tension between explanation, optimisation, diversity, fairness, and governance.

8.4 Future Research Directions

This section turns the limitations into a forward-looking research agenda.

A first research direction is more scalable cooperative attribution. Lower-variance approximation strategies, learned proposal distributions, and adaptive refresh policies would make contribution-aware learning more practical for larger and more dynamic systems.

A second direction is the extension of the recommendation contribution to more

explicitly online and streaming environments. DyHuCoG already refreshes contribution estimates during training, but future work should examine truly incremental settings in which the interaction graph evolves continuously and recommendation decisions must adapt under delayed feedback.

A third direction is richer human-centred evaluation. Future work should examine whether the explanations developed in this thesis measurably improve analyst judgement, user trust, intervention quality, or perceived fairness. This is especially important if the language of actionable insight is to be supported by stronger empirical evidence.

A fourth direction is broader trustworthy-AI evaluation. Future work should test how cooperative attribution interacts with exposure fairness, transparency requirements, and governance-oriented auditing in recommendation environments where explanation is only one part of a larger accountability framework.

Summary

This chapter provided the interpretive synthesis of the thesis. It showed that the three empirical contributions are connected by a common importance-allocation logic grounded in cooperative game theory. Read together, the studies indicate that Shapley-based reasoning can support faithful clustering explanation, hierarchical explanatory coherence, and contribution-aware recommendation learning. The chapter also clarified the theoretical implications, consolidated the main limitations, and identified the most important future directions. The final chapter now draws the overall conclusions of the thesis in a separate and more focused form.

Chapter 9

General Conclusion and Future Work

Introduction

This final chapter closes the thesis by restating its central argument, revisiting the research questions, summarising the original contributions, and drawing the most important implications of the work as a whole. The thesis began from a simple but demanding premise: if explanation is to matter in machine learning, it must do more than decorate opaque results with plausible stories. It must provide a principled account of contribution. Across three linked studies, this thesis has argued that cooperative game theory, and in particular the Shapley-value perspective, offers such an account.

9.1 Revisiting the Research Questions

RQ1: Can Shapley values explain black-box clustering faithfully enough to support actionable analysis?

The thesis answers this question positively. The first contribution showed that Shapley-value reasoning can explain black-box clustering in a way that preserves the semantic meaning of original variables and supports both global and local interpretation. The main methodological insight is that faithful explanation in this setting depends on a careful explanatory bridge between unsupervised partitioning and supervised attribution.

RQ2: Can the same explanatory logic remain coherent in large-scale and hierarchical clustering?

The answer is again positive, with an important qualification. Explanatory coherence across levels does not arise automatically. It requires an explicit multi-level formulation and a principled way of relating parent-level and child-level explanations. The second contribution demonstrated that such cross-level coherence can be established in a formally disciplined way.

RQ3: Can cooperative-game attribution move beyond post-hoc analysis and influence recommendation learning itself?

The thesis answers this question positively. The DyHuCoG contribution showed that approximate Shapley-value estimates can be embedded into a dynamic hypergraph recommendation model as an in-training signal. This is one of the central theoretical advances of the thesis, because it shows that explanation and optimisation need not remain strictly separated.

RQ4: Can explanation, accuracy, and diversity be pursued together in a recommendation setting?

The answer is positive, but carefully bounded. The results show that the accuracy–diversity trade-off is not structurally fixed in the evaluated setting. DyHuCoG improved ranking quality while also improving catalogue coverage and intra-list diversity over strong baselines. The thesis does not claim to abolish the trade-off universally; it shows instead that a cooperative, contribution-aware design can manage it more effectively.

RQ5: What does the thesis contribute at the level of a unified research programme?

The most important answer to the fifth question is thesis-level rather than paper-level. The three contributions jointly show that clustering explanation, hierarchical interpretation, and dynamic recommendation can be understood as instances of a common importance-allocation problem. The thesis therefore contributes not only three empirical studies, but also a shared conceptual framework for thinking about explanation, contribution, and intervention.

9.2 Summary of Original Contributions

C1. The first original contribution is a model-agnostic framework for explaining black-box clustering through a cooperative-game perspective. By combining dimensionality reduction, clustering, surrogate modelling, and TreeSHAP, the thesis created a principled bridge between unsupervised structure and feature-level explanation.

C2. The second original contribution is a multi-level clustering explanation framework together with a formal interpretation of cross-level consistency. This contribution matters because hierarchical explanation is often attempted descriptively, but rarely justified at the level of explicit aggregation logic.

C3. The third original contribution is DyHuCoG, a dynamic hypergraph recommendation framework in which approximate Shapley-value estimates are used as preference-aware in-training signals. This contribution demonstrated that cooperative attribution can improve ranking quality while supporting broader recommendation goals such as coverage and diversity.

Beyond these three chapter-level contributions, the thesis makes a broader theoretical contribution: it shows that cooperative game theory can serve as a shared attribution perspective for explanation, optimisation, and intervention across tasks that are usually treated separately in the literature.

9.3 Broader Impact and Thesis-Level Claim

Scientifically, the thesis contributes to explainable AI, recommender systems, and applied cooperative game theory. Its main scientific claim is not that Shapley values are universally superior to all other explainability methods. It is that Shapley-based reasoning provides a uniquely disciplined language for importance allocation, and that this discipline becomes especially valuable when explanations are expected to be complete, comparable, and operationally useful.

Practically, the thesis is relevant to domains in which algorithmic grouping or ranking affects judgement and decision-making. The clustering contributions are relevant to settings where hidden structure must be interpreted rather than merely detected. The recommendation contribution is relevant to digital systems that must balance relevance, transparency, and broader exposure objectives.

At the thesis level, the strongest claim is therefore carefully bounded: cooperative-game attribution does not solve every problem in interpretability, fairness, or governance, but it does provide a coherent and productive framework for moving from explanation as commentary toward explanation as method.

9.4 Limitations and Future Work

The most important limitation remains computational. Exact Shapley computation is still infeasible in realistic environments, and future work should therefore focus on more efficient approximation strategies and stability-aware update policies.

A second limitation concerns evaluation scope. The thesis demonstrates its ideas on clustering datasets and benchmark recommendation tasks, but not yet on multimodal, sequential, or online recommendation environments. Extending the framework to such settings is a natural next step.

A third limitation concerns human-centred validation. Although the thesis argues for actionable and trustworthy explanation, stronger evidence would come from analyst studies, user studies, and deployment-level evaluation.

A fourth limitation concerns governance scope. The thesis aligns with the aims of trustworthy AI, but it does not present a full fairness audit, legal-compliance framework, or production deployment study. Future work should therefore connect cooperative attribution more directly to fairness evaluation, contestability, and transparency requirements in operational systems.

9.5 General Conclusion

The central conclusion of the thesis is that cooperative game theory should not be regarded merely as an interpretability tool borrowed occasionally for explanatory convenience. It should be regarded as a serious methodological candidate for describing how importance is distributed inside machine-learning systems. This matters because trustworthy explanation requires more than verbal plausibility. It requires a defensible account of why some features, interactions, or contexts matter more than others.

The three studies presented in this thesis show that such an account can be made productive in practice. Cooperative attribution can explain static clusters, remain coherent across levels of analysis, and guide learning inside a dynamic hypergraph recommender. These results do not close the field. They open a research trajectory in which attribution is no longer appended after prediction, but integrated into modelling itself.

For that reason, the thesis ultimately argues for a shift in perspective. Explainable AI should not be treated as a supplementary layer added after strong predictive models have already been built. It should be treated as part of the design logic of those models. Within that broader agenda, the Shapley framework developed and applied in this thesis offers a principled starting point for future work on transparent, adaptive, and structure-aware intelligent systems.

Appendix A

Mathematical Proofs and Derivations

This appendix provides the formal mathematical material that would unnecessarily interrupt the main argumentative flow of the thesis but is still required for examination-level completeness. The proofs expand the compact arguments used in the contribution chapters and make explicit which assumptions are needed at each step. Where the thesis relies on standard results from cooperative game theory or Monte Carlo estimation, the goal is not to claim novelty in the theorem itself but to show precisely how that theorem is instantiated in the present setting.

A.1 Proof: Uniqueness of the Shapley Value

Theorem A.1. *Let φ be an allocation rule on the space of transferable-utility games that satisfies efficiency, symmetry, the null-player property, and additivity. Then φ coincides with the Shapley value.*

Proof. Let $N = \{1, \dots, n\}$ denote the player set and let $\mathcal{G}_N$ denote the vector space of characteristic functions $v : 2^N \rightarrow \mathbb{R}$ with $v(\emptyset) = 0$. For every non-empty coalition $T \subseteq N$, define the unanimity game u_T by

$$u_T(S) = \begin{cases} 1, & T \subseteq S, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A.1})$$

It is classical that the family $\{u_T : \emptyset \neq T \subseteq N\}$ forms a basis of $\mathcal{G}_N$. Hence every game $v \in \mathcal{G}_N$ admits a unique decomposition

$$v = \sum_{\emptyset \neq T \subseteq N} a_T u_T, \quad (\text{A.2})$$

with coefficients $a_T \in \mathbb{R}$. By additivity, it is therefore sufficient to determine φ on each unanimity game.

Fix a non-empty coalition T . Consider first a player $i \notin T$. Since $u_T(S \cup \{i\}) = u_T(S)$ for every coalition $S \subseteq N \setminus \{i\}$, player i is a null player in u_T . By the null-player axiom in the classical Shapley construction [67],

$$\varphi_i(u_T) = 0 \quad \text{for all } i \notin T. \quad (\text{A.3})$$

Now let $i, j \in T$. The unanimity game is invariant under permutations of players in T , so symmetry implies

$$\varphi_i(u_T) = \varphi_j(u_T) \quad \text{for all } i, j \in T. \quad (\text{A.4})$$

Let this common value be denoted by c_T . Efficiency gives

$$\sum_{i \in N} \varphi_i(u_T) = u_T(N) = 1. \quad (\text{A.5})$$

Combining eqs. (A.3) to (A.5) yields

$$|T| c_T = 1, \quad \text{so} \quad c_T = \frac{1}{|T|}. \quad (\text{A.6})$$

Therefore,

$$\varphi_i(u_T) = \begin{cases} \frac{1}{|T|}, & i \in T, \\ 0, & i \notin T. \end{cases} \quad (\text{A.7})$$

Because both φ and the Shapley value operator agree on every unanimity basis element and both are additive, they agree on every linear combination in eq. (A.2). Hence φ is exactly the Shapley value. $\square$

A.2 Proof: Hierarchical Attribution Consistency

The proposition in chapter 6 states that if a parent cluster c is partitioned into disjoint child clusters $c_1, \dots, c_m$ and all explanations are expressed on the same feature space, then the parent-level expected absolute importance of feature j can be decomposed into a weighted average of child-level expectations. Formally, let $\phi_j^{(\ell)}(x)$ denote the SHAP value of feature j for observation x at hierarchy level ℓ and define the expected absolute importance

$$\Phi_j^{(\ell, c)} = \mathbb{E}_{x \sim c} [|\phi_j^{(\ell)}(x)|]. \quad (\text{A.8})$$

Then

$$\Phi_j^{(\ell, c)} = \sum_{r=1}^m \Pr(x \in c_r \mid x \in c) \Phi_j^{(\ell+1, c_r)} + \epsilon_j, \quad (\text{A.9})$$

where ϵ_j is a residual induced by surrogate mismatch across levels. Because the child clusters form a partition of the parent cluster, the law of total expectation yields

$$\mathbb{E}_{x \sim c} [|\phi_j^{(\ell)}(x)|] = \sum_{r=1}^m \mathbb{E} [|\phi_j^{(\ell+1)}(x)| \mid x \in c_r, x \in c] \Pr(x \in c_r \mid x \in c). \quad (\text{A.10})$$

Since $c_r \subseteq c$, the conditional event $x \in c$ becomes redundant inside each term, giving

$$\Phi_j^{(\ell, c)} = \sum_{r=1}^m \Phi_j^{(\ell+1, c_r)} \frac{|c_r|}{|c|}. \quad (\text{A.11})$$

When the parent and child surrogates are perfectly faithful to the same underlying partition logic, the feature attributions are consistent across levels and $\epsilon_j = 0$. In practice, a small non-zero ϵ_j is expected because each surrogate is fit separately and may capture local interactions with different strength. The decomposition used in eq. (6.1) is therefore exact under perfect fidelity and approximately exact in the empirical setting, which is sufficient for the interpretive claim made in the main text.

A.3 Proof: Convergence Rate of the Monte Carlo Shapley Estimator

Let π be a random permutation of the player set N and let $S_j(\pi)$ denote the players that precede player j in π . Define the random marginal contribution

$$Z_j(\pi) = v_{\text{pref}}(S_j(\pi) \cup \{j\}) - v_{\text{pref}}(S_j(\pi)). \quad (\text{A.12})$$

The exact Shapley value is the expectation

$$\phi_j = \mathbb{E}_{\pi} [Z_j(\pi)]. \quad (\text{A.13})$$

Now sample M independent permutations $\pi_1, \dots, \pi_M$ uniformly at random and define the Monte Carlo estimator

$$\hat{\phi}_j^{(M)} = \frac{1}{M} \sum_{m=1}^M Z_j(\pi_m). \quad (\text{A.14})$$

By linearity of expectation, the estimator is unbiased:

$$\mathbb{E}[\hat{\phi}_j^{(M)}] = \phi_j. \quad (\text{A.15})$$

Assuming the variance $\sigma_j^2 = \text{Var}(Z_j)$ is finite, independence of the sampled permutations implies

$$\text{Var}(\hat{\phi}_j^{(M)}) = \frac{1}{M^2} \sum_{m=1}^M \text{Var}(Z_j(\pi_m)) = \frac{\sigma_j^2}{M}. \quad (\text{A.16})$$

Hence the root-mean-square error decays as $\mathcal{O}(M^{-1/2})$, while the mean-squared error decays as $\mathcal{O}(M^{-1})$. This distinction matters for careful wording: the estimator variance and mean-squared error improve at rate $1/M$, whereas the absolute estimation error in probability improves at order $1/\sqrt{M}$. The thesis therefore interprets the empirical convergence curves in chapter 7 as evidence of variance reduction rather than as proof of unrealistically fast pointwise convergence.

A.4 Proof Sketch Formalised: Composite-Loss Stability for DyHuCoG

The DyHuCoG objective is

$$\mathcal{L}(\Theta) = \mathcal{L}_{\text{BPR}} + \mu_1 \mathcal{L}_{\text{ILD}} + \mu_2 \mathcal{L}_{\text{ctx}} + \lambda \|\Theta\|_2^2, \quad (\text{A.17})$$

where the Shapley weights are refreshed every Δ epochs and smoothed through an exponential moving average. A full non-convex convergence theorem for Adam under periodically refreshed stochastic objectives is beyond the scope of the thesis. What is required here is a stability statement. Assume that within each refresh interval: (i) every loss component is differentiable and L -smooth in Θ , (ii) stochastic gradients are unbiased with bounded second moment, and (iii) the Shapley refresh modifies the effective objective by at most ρ in operator norm after smoothing. Then standard non-convex stochastic-optimisation arguments imply that the average gradient norm over each fixed-objective interval decreases with training. The refresh step perturbs the objective, but if ρ is sufficiently small relative to the step size schedule, the optimisation trajectory remains within a slowly drifting basin rather than diverging. This is the precise sense in which the thesis uses the term *stable training*: not guaranteed convergence to a global optimum, but bounded, empirically well-behaved descent under a piecewise-smooth objective.

A.5 Derivation: Preference-Weighted Coalition Utility

Starting from the multi-objective utility in chapter 7, let

$$v(S) = \alpha \text{NDCG@20}(S) + \beta \text{ILD}(S) + \gamma \text{Ctx}(S), \quad (\text{A.18})$$

where $\alpha, \beta, \gamma \geq 0$ and $\alpha + \beta + \gamma = 1$. To encode preference intensity, assign each user-item interaction a normalised preference-consistency weight $p_{ui} \in [0, 1]$. The thesis

adopts the additive utility modulation reported in the DyHuCoG chapter,

$$v_{\text{pref}}(S) = v(S) + \lambda_{\text{pref}} \sum_{(u,i) \in S} p_{ui}. \quad (\text{A.19})$$

This form is chosen because it preserves the baseline utility when $\lambda_{\text{pref}} = 0$, keeps the preference term interpretable as an explicit utility bonus, and matches the formulation used in the main text. If player $j = (u, i)$ is not already in coalition S , then the corresponding marginal contribution becomes

$$v_{\text{pref}}(S \cup \{j\}) - v_{\text{pref}}(S) = [v(S \cup \{j\}) - v(S)] + \lambda_{\text{pref}} p_j. \quad (\text{A.20})$$

The additive preference term therefore shifts the cooperative payoff of a player by an amount proportional to its prior preference consistency, while the interaction-sensitive part of the final attribution still arises from the Shapley averaging over coalitions. In other words, the utility modulation is additive, but the resulting attribution remains coalition-aware because the first bracketed term depends on the sampled coalition structure.

Appendix B

Algorithm Pseudocode

This appendix gives the full procedural form of the methods used throughout the thesis. The purpose is not merely to summarise the pipelines already described in prose, but to present them in a replication-oriented format that makes data flow, inputs, outputs, and computational cost explicit. Each algorithm is therefore accompanied by a short contextual paragraph and an explicit complexity statement [90].

B.1 Algorithm 1: PCA–KMeans–LightGBM–SHAP Pipeline

Algorithm B.1 formalises the single-level clustering explanation workflow introduced in chapter 5. It starts from a standardised tabular matrix and ends with three explanation products: global importances, cluster-specific summaries, and local force-plot explanations.

Algorithm B.1 PCA–KMeans–LightGBM–SHAP Pipeline

Require: Standardised feature matrix $X \in \mathbb{R}^{n \times d}$, candidate cluster set $\mathcal{K}$, LightGBM hyperparameters Θ_{gbm}

Ensure: Selected cluster count k^* , partition labels c , surrogate model g , SHAP explanations Φ

- 1: Compute PCA projection $Y \leftarrow \text{PCA}(X)$ for cluster discovery and visual inspection
 - 2: **for** each $k \in \mathcal{K}$ **do**
 - 3: Fit K-Means++ on Y with k clusters
 - 4: Compute Silhouette and Davies–Bouldin scores
 - 5: **end for**
 - 6: Select k^* using the multi-criteria validation rule
 - 7: Fit final K-Means++ with k^* on Y to obtain labels c
 - 8: Fit LightGBM surrogate $g(X) \approx c$ on the original feature space
 - 9: Compute TreeSHAP values Φ for all observations under g
 - 10: Aggregate Φ into global, cluster-specific, and local summaries
 - 11: **return** (k^*, c, g, Φ)
-

Complexity: PCA requires $\mathcal{O}(nd \min\{n, d\})$ in the general dense case, candidate K-Means search costs approximately $\mathcal{O}(|\mathcal{K}|nkdI)$ for I iterations, and TreeSHAP for gradient-boosted trees scales with the number and depth of trees rather than exponentially in the number of features.

B.2 Algorithm 2: Multi-Level Clustering with Cross-Level SHAP Aggregation

Algorithm B.2 extends the single-level framework to the hierarchical setting used in chapter 6. The crucial additional step is the propagation of attribution summaries from child clusters to parent clusters via explicit size-weighted aggregation.

Algorithm B.2 Multi-Level Clustering with Cross-Level SHAP Aggregation

Require: Feature matrix X , hierarchy depth L , cluster-selection rule $\mathcal{R}$

Ensure: Hierarchical partition $\mathcal{C}^{1:L}$ and cross-level attribution summaries $\Phi^{1:L}$

```

1: Fit the level-1 clustering and compute SHAP explanations on the parent partition
2: for  $\ell = 1$  to  $L - 1$  do
3:   for each cluster  $c \in \mathcal{C}^\ell$  do
4:     Restrict data to  $X_c$ 
5:     Fit child-level clustering on  $X_c$  according to  $\mathcal{R}$ 
6:     Train a level-specific surrogate and compute SHAP values on  $X_c$ 
7:     Aggregate child explanations to the parent using weights  $|c'|/|c|$ 
8:   end for
9: end for
10: return  $(\mathcal{C}^{1:L}, \Phi^{1:L})$ 
    
```

Complexity: The total cost is the sum of the level-specific clustering and surrogate-explanation costs. In practice the hierarchy reduces per-cluster explanation cost by decomposing one very large optimisation into multiple smaller ones.

B.3 Algorithm 3: Preference-Aware Monte Carlo Shapley Estimator

Algorithm B.3 implements the estimator used in chapter 7. The algorithm is written explicitly because the thesis relies on it both for explanation and as an in-training signal.

Algorithm B.3 Preference-Aware Monte Carlo Shapley Estimator

Require: Player set N , preference-weighted value function v_{pref} , sample size M

Ensure: Approximate Shapley vector $\hat{\phi} = (\hat{\phi}_j)_{j \in N}$

```

1: for each player  $j \in N$  do
2:   Initialise accumulator  $a_j \leftarrow 0$ 
3:   for  $m = 1$  to  $M$  do
4:     Sample a permutation  $\pi_m$  uniformly at random from all orderings of  $N$ 
5:     Let  $S_m$  be the subset of players preceding  $j$  in  $\pi_m$ 
6:      $a_j \leftarrow a_j + v_{\text{pref}}(S_m \cup \{j\}) - v_{\text{pref}}(S_m)$ 
7:   end for
8:    $\hat{\phi}_j \leftarrow a_j / M$ 
9: end for
10: return  $\hat{\phi}$ 
    
```

Complexity: For $|N| = n$ players and M sampled permutations, the estimator costs $\mathcal{O}(Mn)$ coalition evaluations if marginal contributions are accumulated incrementally.

B.4 Algorithm 4: DyHuCoG Training Loop with Shapley Refresh

Algorithm B.4 integrates the Monte Carlo estimator into the DyHuCoG training dynamics. The refresh step is central because it turns attribution from a static diagnostic into a periodically updated training signal.

Algorithm B.4 DyHuCoG Training Loop with Periodic Shapley Refresh

Require: Hypergraph H , parameters Θ , refresh period Δ , Monte Carlo budget M , epochs T

Ensure: Trained model parameters Θ^*

```

1: for  $t = 1$  to  $T$  do
2:   if  $t \bmod \Delta = 0$  then
3:     Recompute preference-aware Shapley estimates with sample budget  $M$ 
4:     Smooth the new estimates with an exponential moving average
5:   end if
6:   for each mini-batch of user–positive–negative triples do
7:     Perform Shapley-weighted hypergraph message passing
8:     Compute  $\mathcal{L}_{\text{BPR}}$ ,  $\mathcal{L}_{\text{ILD}}$ , and  $\mathcal{L}_{\text{ctx}}$ 
9:     Update  $\Theta$  with Adam on the composite loss
10:  end for
11: end for
12: return  $\Theta^*$ 

```

Complexity: Each epoch has the cost of hypergraph forward/backward passes plus, every Δ epochs, an additional Monte Carlo estimation cost. The refresh schedule therefore controls the trade-off between attribution fidelity and runtime.

B.5 Algorithm 5: Popularity-Aware Negative Sampling

Algorithm B.5 records the negative-sampling strategy used to generate harder contrasts during recommendation training. This is included because the sampling distribution materially affects the learned ranking boundary.

Algorithm B.5 Popularity-Aware Negative Sampling

Require: Item-frequency vector f , exponent $\eta \in (0, 1)$, observed positive set $\mathcal{I}_u^+$ for each user u

Ensure: Training triples (u, i^+, i^-) with popularity-aware negatives

- 1: Define proposal probabilities $q(i) \propto f_i^\eta$
 - 2: **for** each observed positive interaction (u, i^+) **do**
 - 3: **repeat**
 - 4: Sample candidate negative $i^- \sim q(i)$
 - 5: **until** $i^- \notin \mathcal{I}_u^+$
 - 6: Emit training triple (u, i^+, i^-)
 - 7: **end for**
 - 8: **return** the generated triple set
-

Complexity: Sampling is linear in the number of positive interactions, with constant-time amortised draws under an alias-table or cumulative-probability implementation.

Appendix C

Dataset Descriptions and Preprocessing

This appendix consolidates the dataset-specific details that are useful for replication but too fine-grained for the main methodological chapter. The emphasis is on practical processing decisions that can affect explanation stability or recommendation quality, such as feature standardisation, interaction filtering, and the treatment of missing or sparse observations [\[90\]](#).

Table C.1: Dataset and preprocessing summary for Appendix C

Dataset	Scale	Feature / interaction type	Preprocessing choices retained in the thesis
Wine Quality	4,898 rows, 11 features	Continuous physicochem- ical variables	z-standardisation, PCA-assisted clustering support, LightGBM surrogate on retained $k = 3$ labels
Beijing Air Quality	383,585 rows, 11 variables	Pollutant + meteorologi- cal variables	standardisation, conservative omission / standard imputation, multi-level clustering with representative local explanations
MovieLens-1M	6,040 users, 3,706 items, 1,000,209 interactions	Explicit ratings converted to implicit positives	rating > 3 as positive, temporal user-level split, leave-one-out ranking, genre used as contextual proxy
Amazon-Book	52,643 users, 91,599 items, 2,984,108 interactions	Implicit user-item interactions	binarisation, sparse graph indexing, temporal user-level split, hypergraph construction for DyHuCoG

C.1 Wine Quality Dataset: Full Statistics and Preprocessing

The wine-quality dataset contains 4,898 red and white vinho verde records with 11 physicochemical variables. Because every variable is continuous and measured on a different scale, the data are z-standardised before PCA and K-Means are applied. No domain-specific feature engineering is introduced. This is methodologically deliberate: the interpretability claim in chapter 5 depends on the explanations referring back to the original semantically meaningful variables rather than to engineered aggregates.

C.2 Beijing Air Quality Dataset: Full Statistics and Preprocessing

The Beijing dataset contains approximately 383,585 hourly observations recorded at multiple stations between 2013 and 2017. Continuous pollutant and meteorological

variables are standardised before clustering, and variables with occasional missing values are processed using a conservative two-stage rule in the supporting scripts: short local gaps are interpolated within station, whereas longer or structurally incomplete segments are omitted before the final clustering stage. The analysis uses global SHAP summaries on the final clustering output and one representative observation per cluster for local force-plot analysis [88].

C.3 MovieLens-1M Dataset: Full Statistics and Pre-processing

MovieLens-1M contains 6,040 users, 3,706 items, and 1,000,209 interactions. Ratings are converted to implicit positive events for top- N recommendation, user histories are ordered temporally for the leave-one-out protocol, and entity identifiers are mapped to contiguous integer indices for efficient sparse-tensor implementation. Where genre metadata are used as contextual signals, they are encoded as auxiliary item attributes rather than as direct labels [104].

C.4 Amazon-Book Dataset: Full Statistics and Pre-processing

Amazon-Book contains 52,643 users, 91,599 items, and 2,984,108 interactions. The dataset is substantially sparser than MovieLens-1M, which makes filtering and indexing decisions especially consequential. Interactions are binarised, cold entities are retained as far as baseline constraints permit, and the resulting user-item graph is used to derive the hypergraph structures employed by the DyHuCoG training loop [104].

Appendix D

Extended Experimental Results

This appendix reports extended experimental material that supports the claims made in chapter 7 while remaining tightly aligned with the validated DyHuCoG results included in the thesis. Only tables that can be traced directly to the documented experimental protocol are retained here [104].

D.1 Run-Variability Reporting Note

The DyHuCoG evaluation provides mean $\pm$ standard-deviation performance over five seeds, but it does not provide raw run logs in the thesis package. To avoid implying access to independently auditable run-level traces that are not distributed here, this appendix does not tabulate reconstructed per-seed values. The main chapter therefore retains the mean $\pm$ standard-deviation summaries together with the paired-test material explicitly documented in the article. This is a deliberate scope choice made for evidential discipline rather than a claim that multi-seed evaluation was absent [104].

D.2 Selected DyHuCoG Configuration

Table D.1 records the validated setting used in the thesis. Where the published description gives a training design choice only at the scheme level rather than as a fixed scalar hyperparameter, the table keeps that distinction explicit. This is why the popularity-aware negative-sampling exponent is identified as a scheme-level setting rather than silently filled with an undocumented value.

Table D.1: Validated DyHuCoG configuration

Component	Selected setting
Propagation layers L	3
Embedding dimension d	64
Monte Carlo budget M	50
Shapley refresh period	Every 10 batches
Utility weights (α, β, γ)	(0.60, 0.25, 0.15)
Preference coefficient λ_{pref}	0.20
Negative-sampling exponent η	scheme-level setting; no standalone η value

D.3 Runtime and Scalability Details

The runtime results needed for interpretation are reported in table 7.3. Rather than reprinting the same table in full, this appendix records the reading that matters for replication: the principal overhead comes from periodic Shapley refresh and the associated hypergraph-weight normalisation rather than from inference-time scoring alone. For thesis purposes, the main-text table is the canonical record of training time, inference latency, memory usage, and scaling factors, while this appendix preserves only the configuration information in table D.1. Future extensions of the appendix should add genuinely new evidence here, for example per-epoch runtime traces or a propagation-versus-attribution time breakdown.

D.4 Monte Carlo Convergence Details

The Monte Carlo sample-budget results are likewise retained in canonical form in table 7.4. The convergence results show estimator mean-squared error, reference-relative accuracy, runtime, and overhead versus HPCF for $M \in \{10, 25, 50, 100\}$. Because the thesis does not introduce additional sample-budget experiments beyond the validated setup, the appendix does not duplicate the main-text table. What matters interpretively is that the available evidence supports $M = 50$ as the practical balance point between approximation quality and computational overhead.

D.5 Cold-Start and Cross-Dataset Robustness

The cold-start and auxiliary Yelp2018 robustness results are reported in canonical form in table 7.5. The appendix therefore limits itself to a scope note rather than a duplicate table: Yelp2018 is treated as an auxiliary sparse user–business recommendation

benchmark in the thesis, not as a third primary benchmark on par with MovieLens-1M and Amazon-Book. Any future appendix expansion should therefore add genuinely new robustness analyses here, such as sparsity-stratified curves or user/item-threshold sweeps, rather than reproducing the already cited table.

Appendix E

Reproducibility Guidelines

This appendix records the practical conditions required to reproduce the thesis experiments. It is written in the style of an examiner-facing replication note rather than a software manual. The intention is to specify what another researcher would need to know in order to regenerate the thesis results with minimal ambiguity [90].

E.1 Hardware and Software Requirements

A modern workstation with at least 32 GB RAM is sufficient for the clustering studies, while the recommendation study benefits from GPU acceleration and at least 16 GB GPU memory for the Amazon-Book runs. Python 3.x is the core implementation language throughout. The software stack relies primarily on scikit-learn, LightGBM, SHAP, PyTorch, NumPy, SciPy, and pandas.

E.2 Dataset Download and Preprocessing Scripts

The datasets should be obtained from their official or benchmark sources and stored in immutable raw-data folders. Preprocessing scripts then standardise features, convert explicit ratings to implicit events where necessary, map entities to contiguous indices, and generate the train-validation-test partitions described in chapter 4. Intermediate artefacts should be saved explicitly so that the clustering and recommendation stages can be rerun independently.

E.3 Model Training: Commands and Configuration Files

Each study should be driven by explicit configuration files recording dataset paths, random seeds, search grids, and evaluation settings. The clustering scripts first generate

the partition and only then fit the surrogate explanation model. The DyHuCoG script alternates between standard mini-batch optimisation and periodic Shapley refresh according to the selected Monte Carlo budget and refresh period. The present thesis package documents the expected repository structure and reproducibility workflow, but it does not claim that a public archival code URL accompanies the dissertation unless such a release is made separately by the authors. Accordingly, the reproducibility material in this appendix should be read as documentation of the expected experimental organisation rather than as a claim of externally audited end-to-end reproducibility.

E.4 Random Seed Configuration and Deterministic Mode

Replication requires fixing random seeds for NumPy, PyTorch, clustering initialisation, and dataset partitioning. Deterministic execution modes should be enabled where the relevant libraries support them. Minor hardware-dependent variation is still possible, especially on GPU backends, which is why the thesis reports mean values across multiple seeds rather than implying perfect bitwise reproducibility.

E.5 Code Repository Structure

A reproducible repository should separate data preparation, clustering experiments, recommendation experiments, plotting utilities, and configuration files. This directory separation mirrors the logic of the thesis itself and reduces the risk that local scripting decisions become undocumented parts of the method.

E.6 Docker and Conda Environment Specification

A containerised environment is recommended for long-term reproducibility. The environment file should pin the major versions of PyTorch, SHAP, LightGBM, scikit-learn, and CUDA where relevant. For readers who prefer lighter-weight reproduction, a Conda environment file can provide an equivalent CPU-first setup.

Appendix F

Statistical Testing: Full Results

This appendix records the inferential scope of the thesis without duplicating the canonical main-text table. The full paired t-test results that underwrite the MovieLens-1M NDCG@20 significance claim are reported in table 7.6; that table is the authoritative in-thesis record of the t statistics, degrees of freedom, p -values, Holm-adjusted thresholds, and Cohen’s d_z values [104].

F.1 Paired t-Test Scope Note

For evidential discipline, the thesis does not reproduce the same MovieLens-1M paired-test table twice. Instead, it states the inferential boundary explicitly: the fully tabulated paired tests currently available inside the thesis apply to per-user NDCG@20 on MovieLens-1M. Where broader significance is reported qualitatively rather than through a full in-thesis table, the thesis preserves that narrower scope instead of fabricating unsupported inferential detail.

F.2 Robustness-Check Note

One-sided Wilcoxon signed-rank tests on the per-user score differences are also reported as significant for all major baselines ($p < 0.001$), supporting the same qualitative conclusion as the paired t-tests [104]. Because the corresponding W statistics and a full Amazon-Book paired-test table are not tabulated, those values are not reproduced here. The appendix therefore functions as a scope declaration rather than as a second location for the same MovieLens-1M table.

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